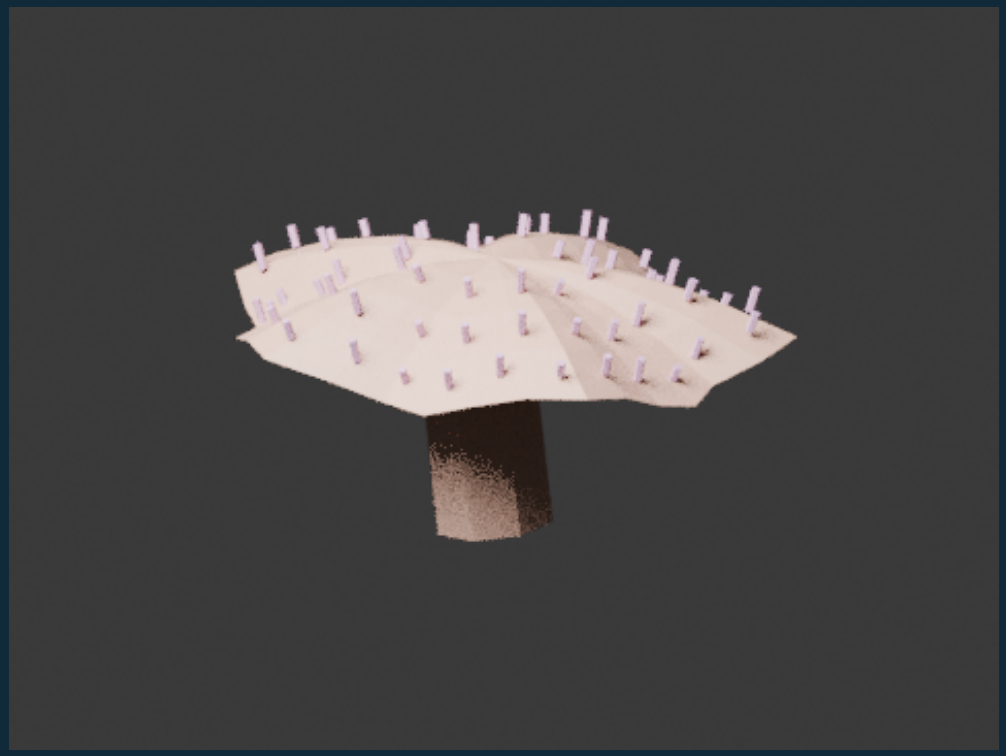
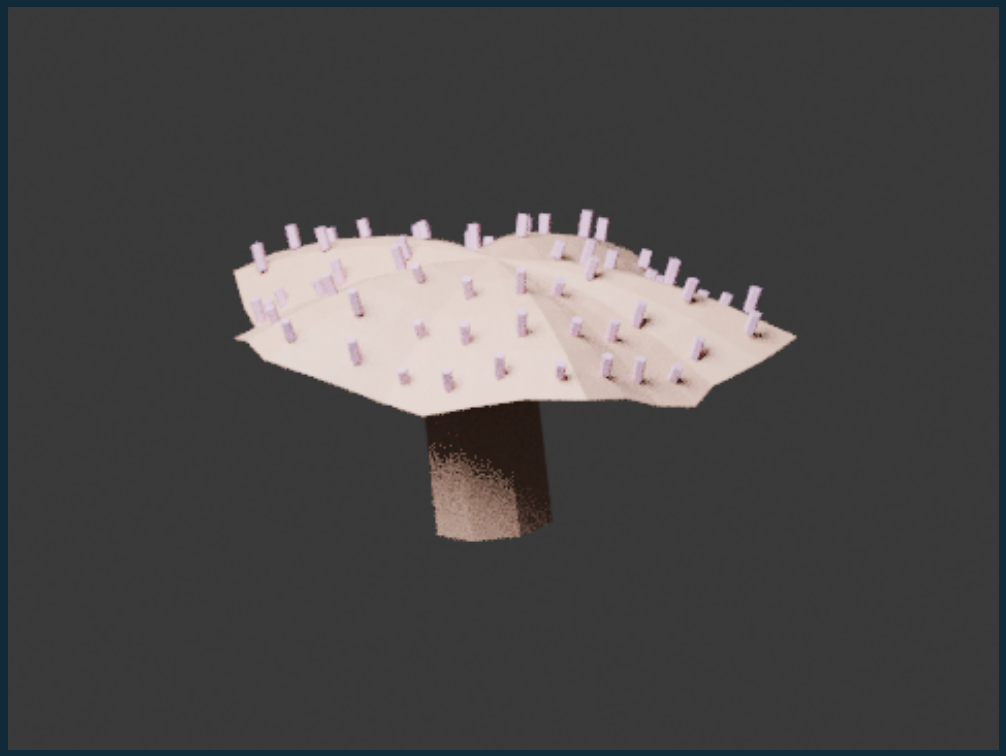
# Leather coral: export survival



## BEFORE



## AFTER



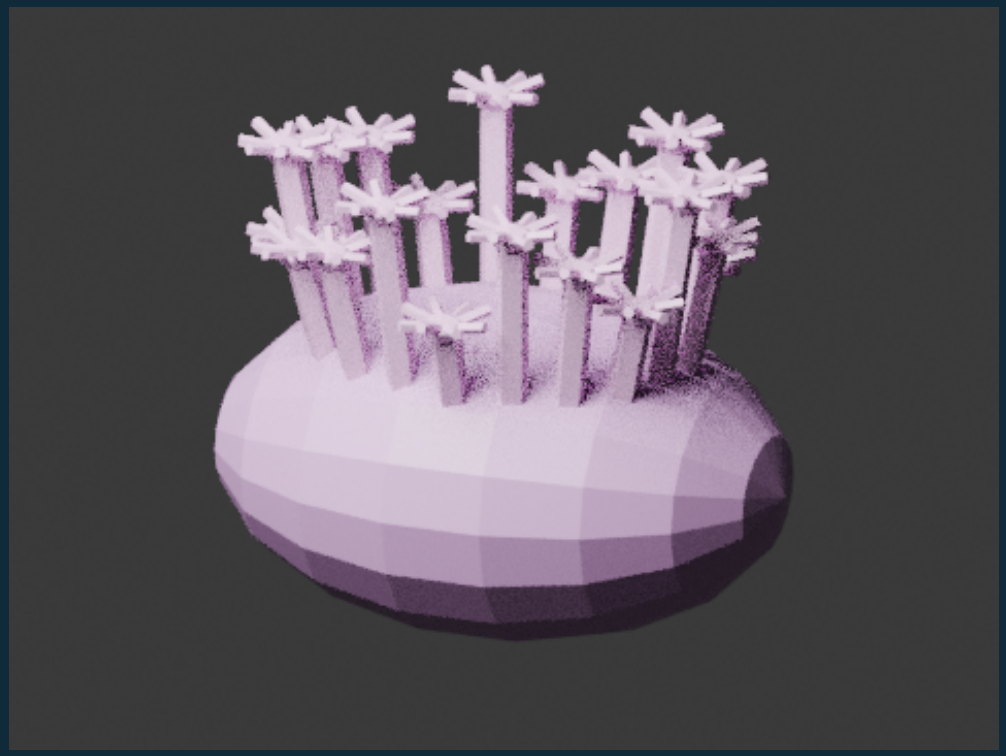
Thickened polyps to preserve their end caps during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

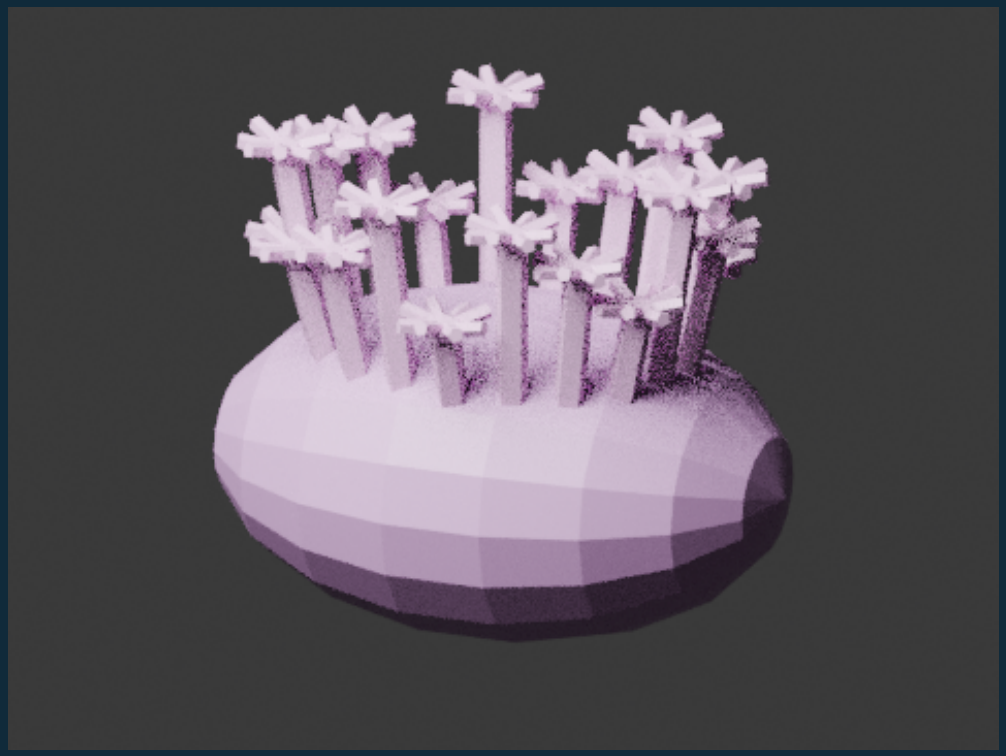
# Flowerpot coral: export survival



**BEFORE**



**AFTER**



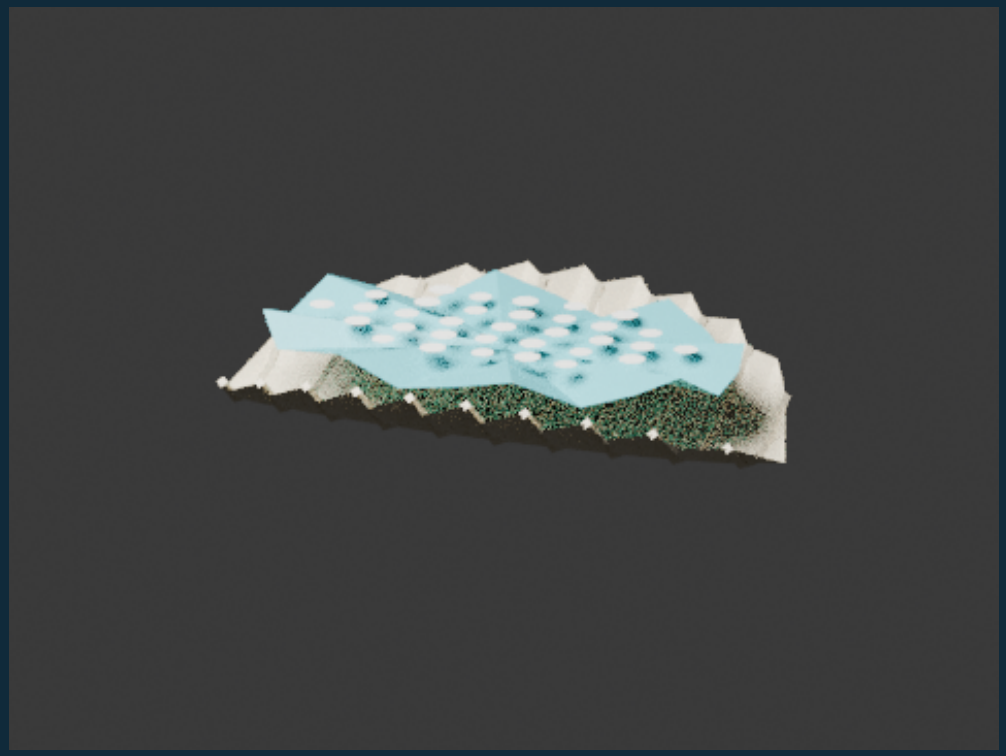
Thickened crown tentacles to retain their end caps.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

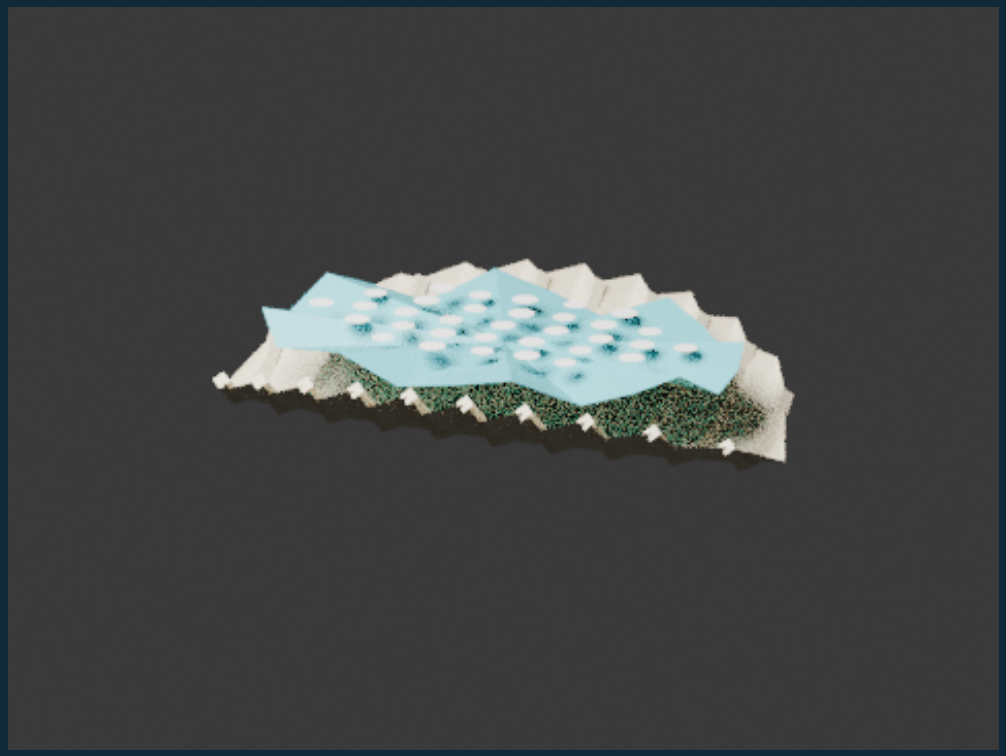
# Giant clam: shell ribs



## BEFORE



## AFTER



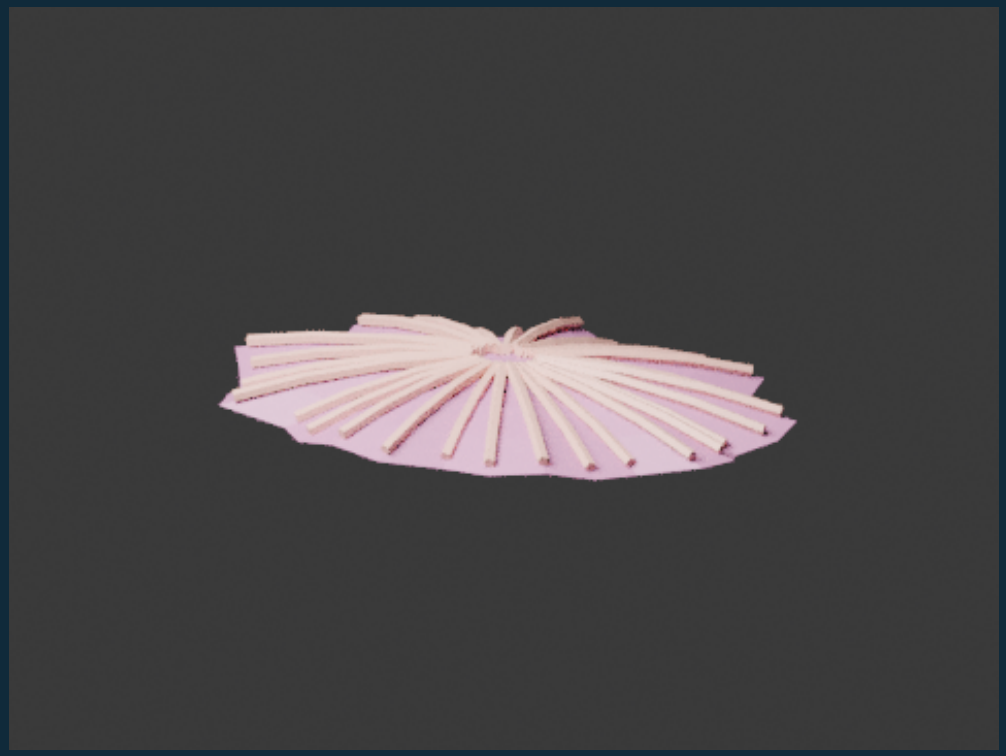
Thickened shell ribs so more of their faces survive export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

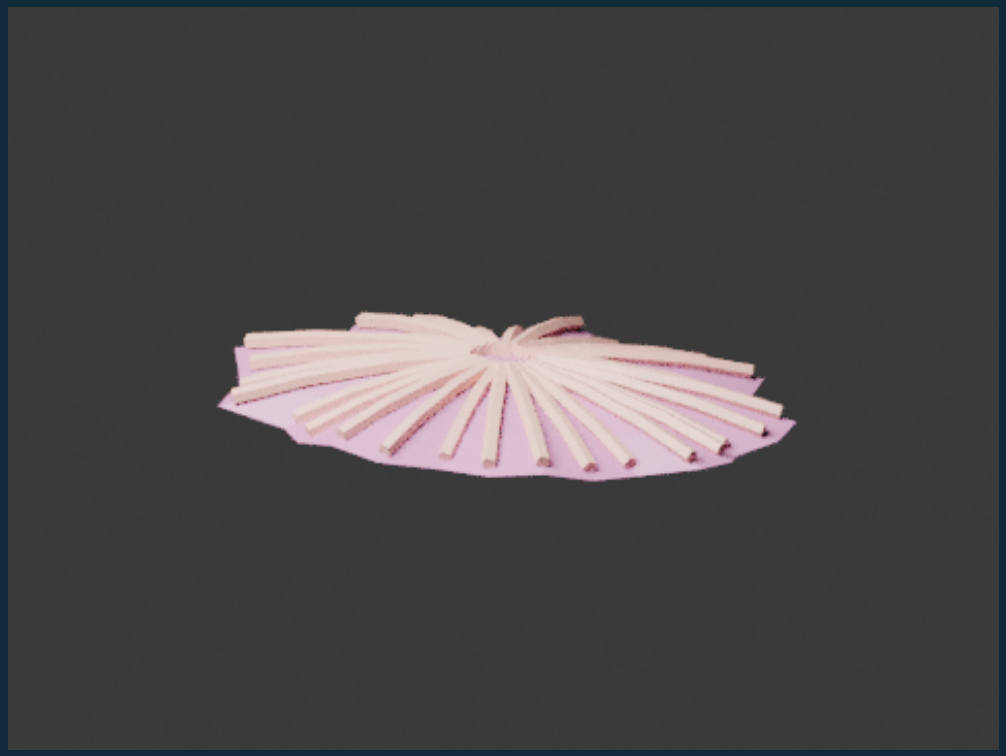
# Mushroom coral: export survival



## BEFORE



## AFTER



Thickened the ridges to retain their faces during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Cup coral: export survival



**BEFORE**

**AFTER**

Thickened polyp tentacles to retain their faces during export.

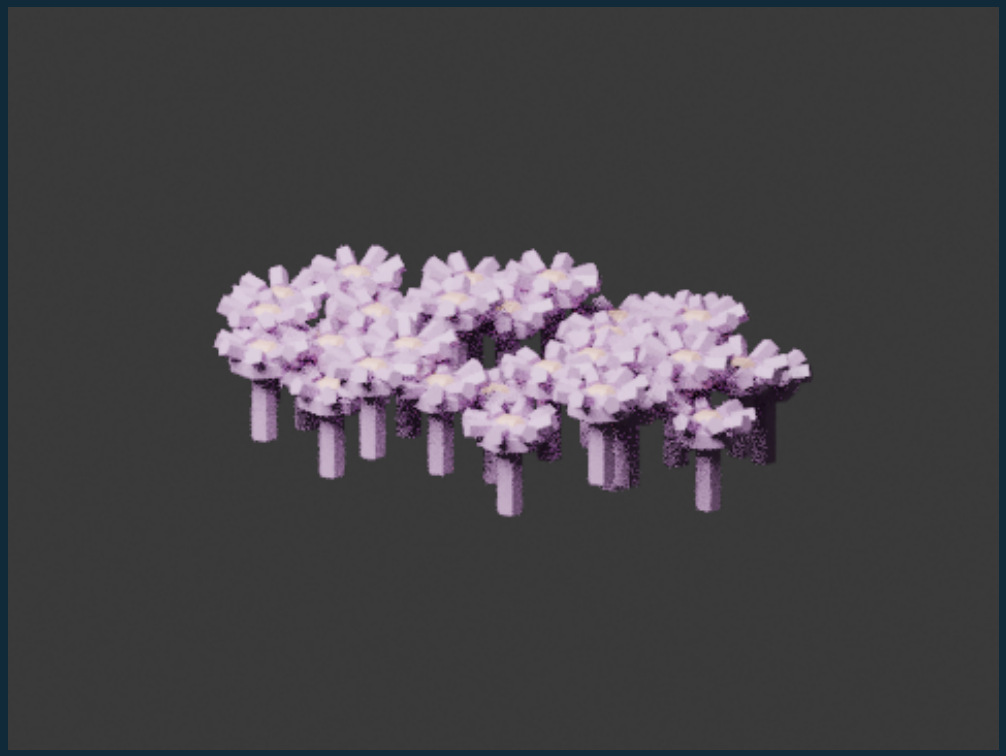
Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Zoanthids: export survival



## BEFORE

## AFTER



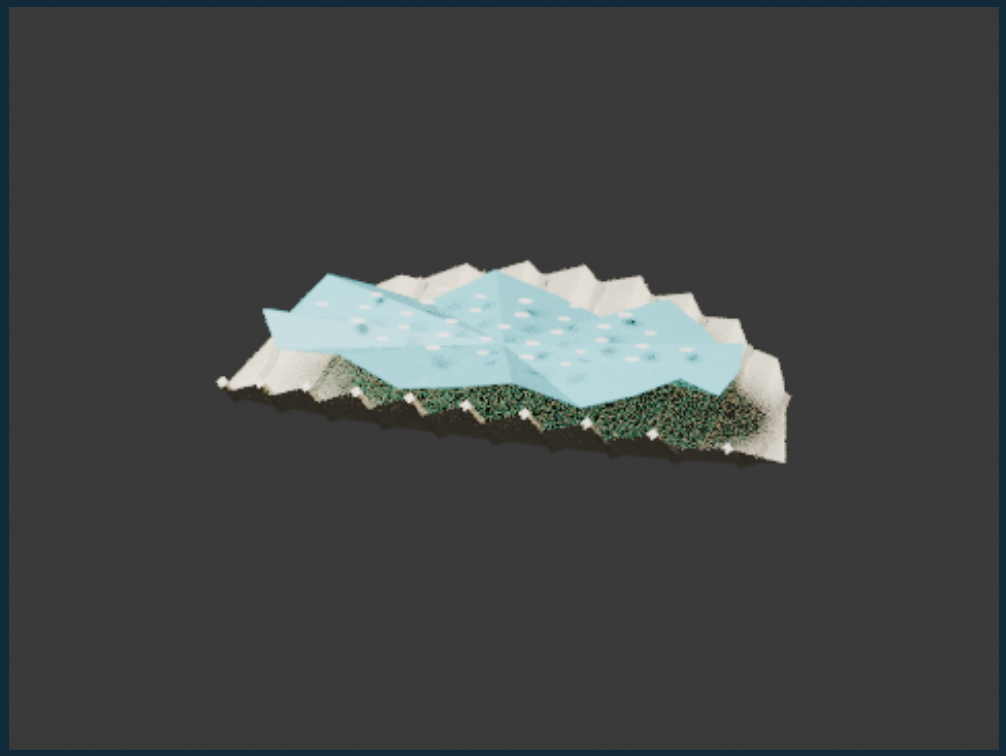
Thickened the tentacle tips so their faces survive export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

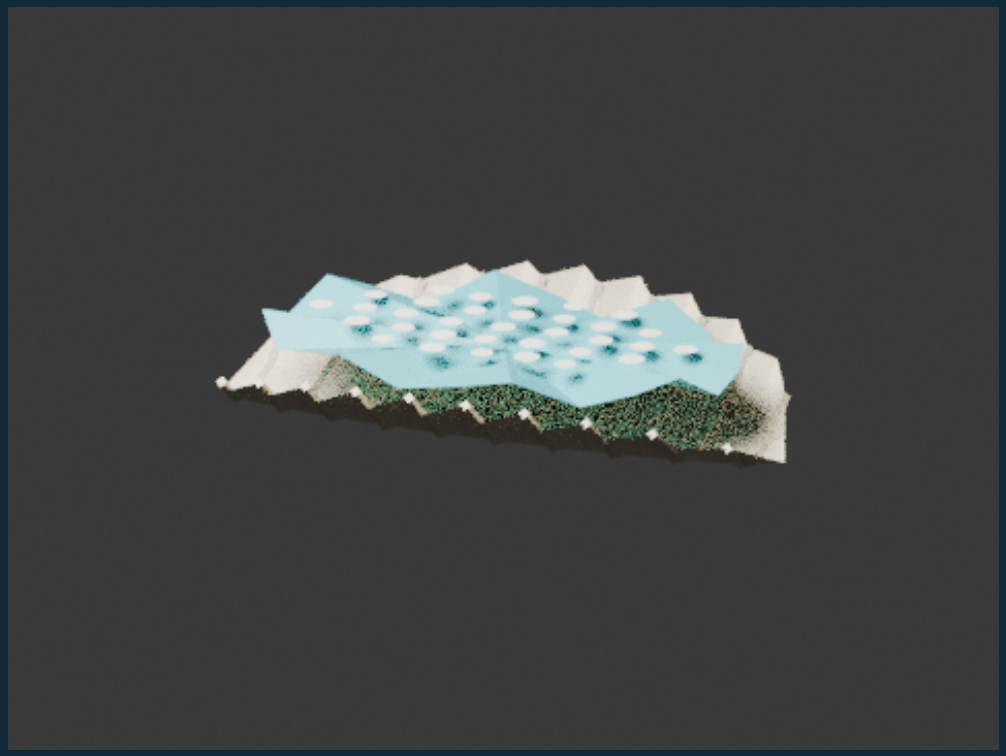
# Giant clam: export survival



## BEFORE



## AFTER



Reshaped the mantle markings so their faces survive export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Boxfish: export survival



## BEFORE

## AFTER

Reshaped small markings to retain their faces during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Humbug: export survival



## BEFORE

## AFTER

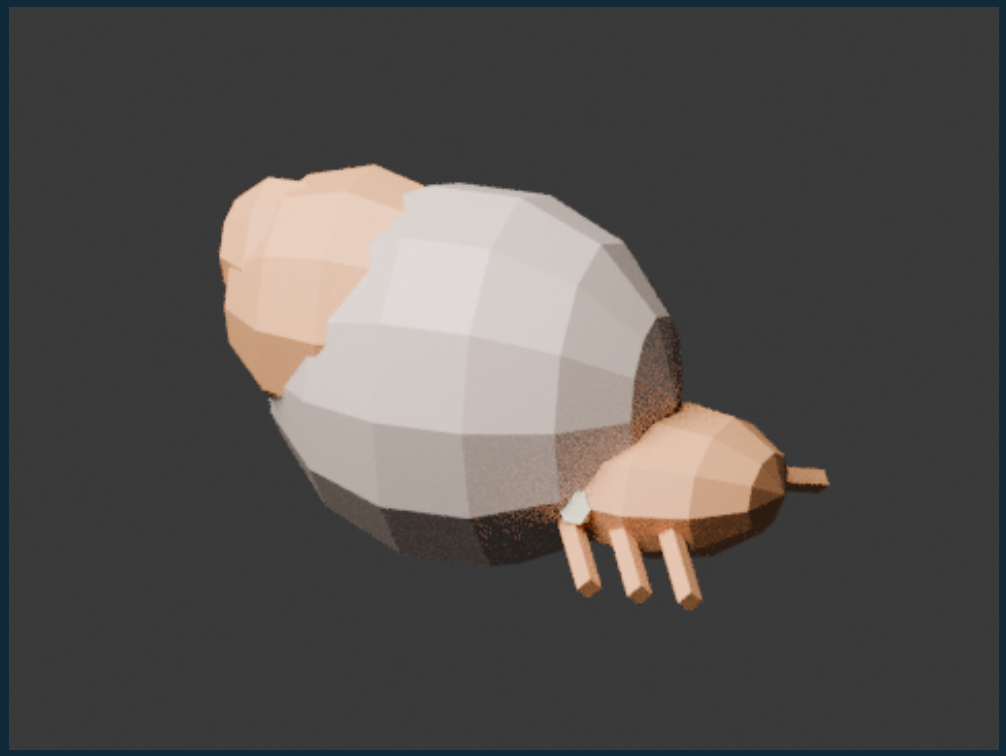
Reshaped small markings to retain their faces during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

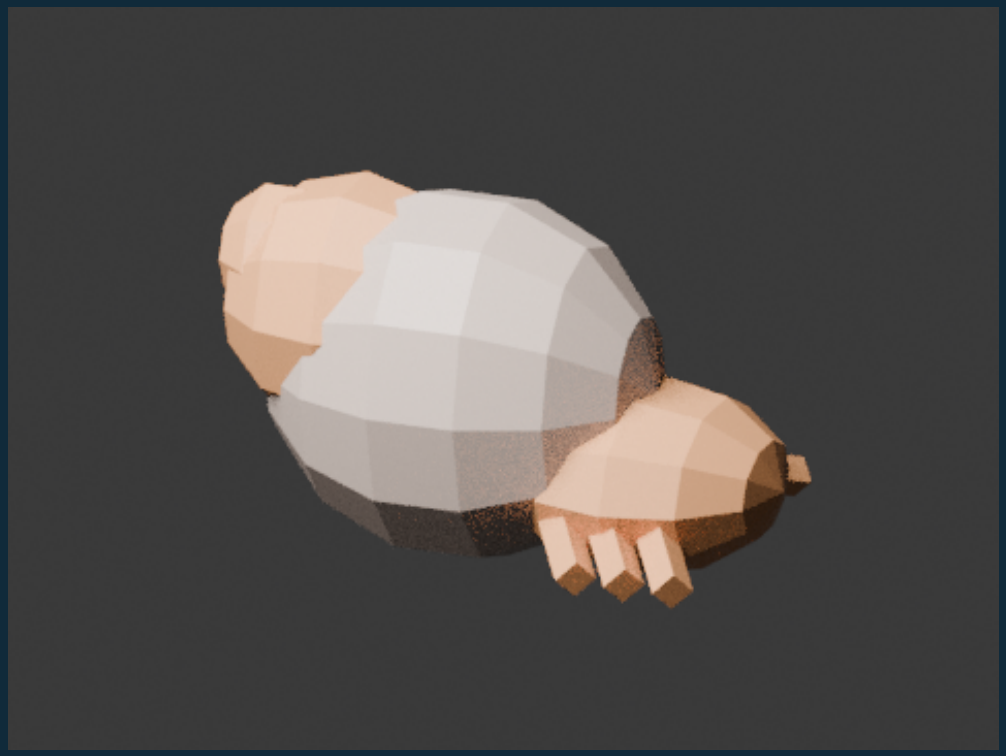
# Hermit crab: export survival



**BEFORE**



**AFTER**



Adjusted the body, shell lip and legs to retain their faces during export.

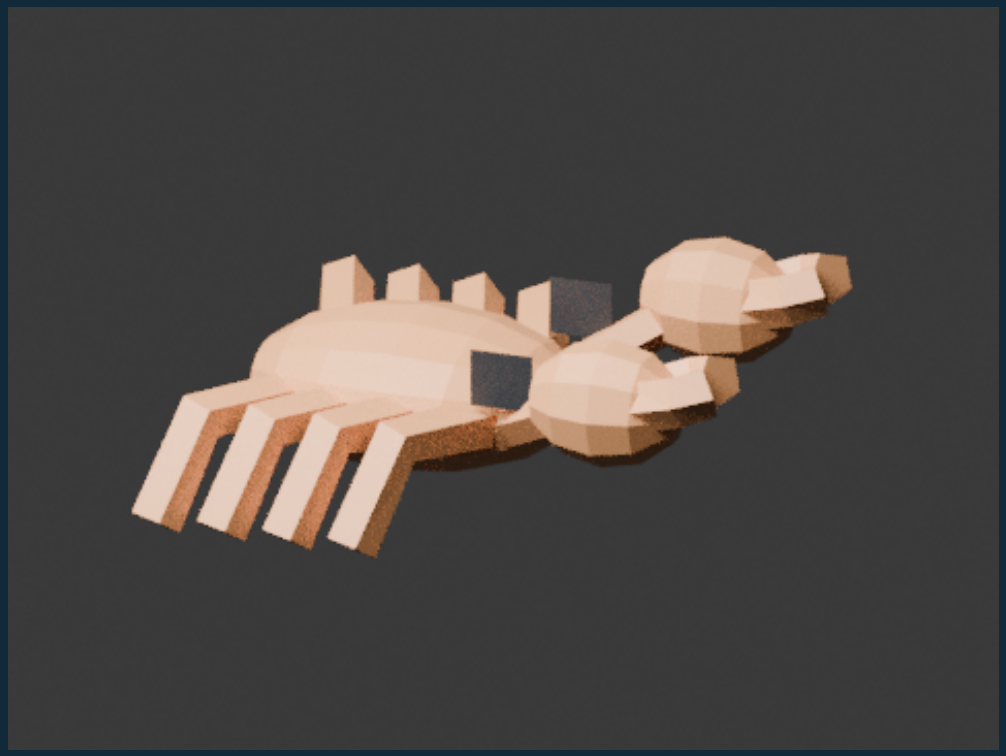
Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Guard crab: claws



## BEFORE

## AFTER



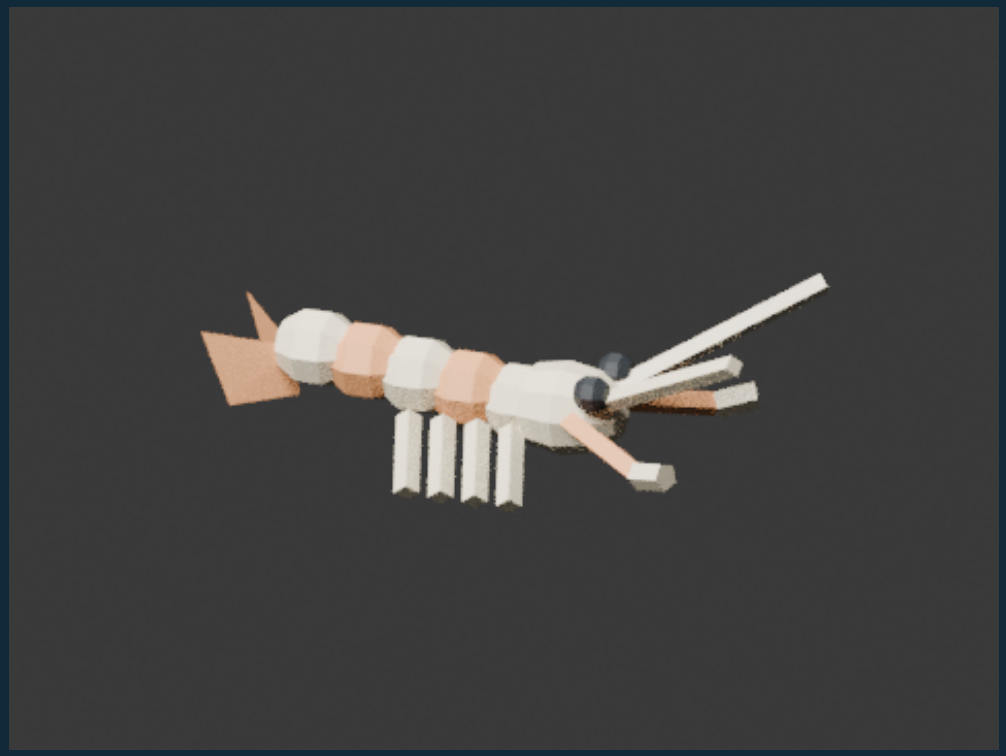
Adjusted claw proportions to retain their faces during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

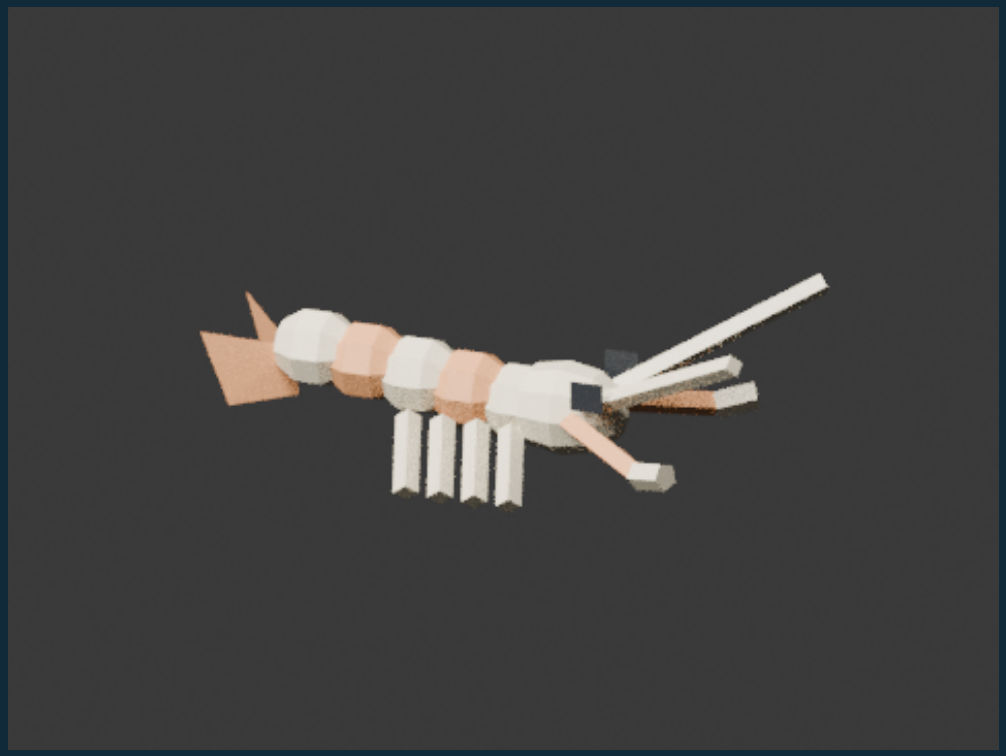
# Shrimp: eyes



## BEFORE



## AFTER



Replaced tiny subdivided eyes with surfaces that survive export.

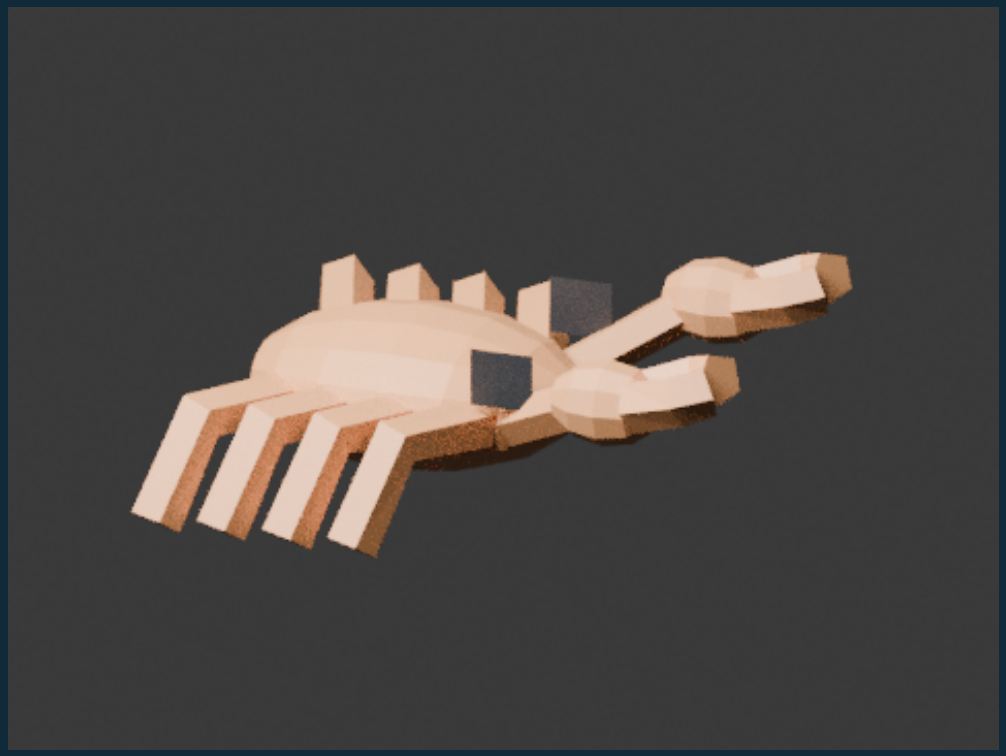
Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Guard crab: eyes



## BEFORE

## AFTER



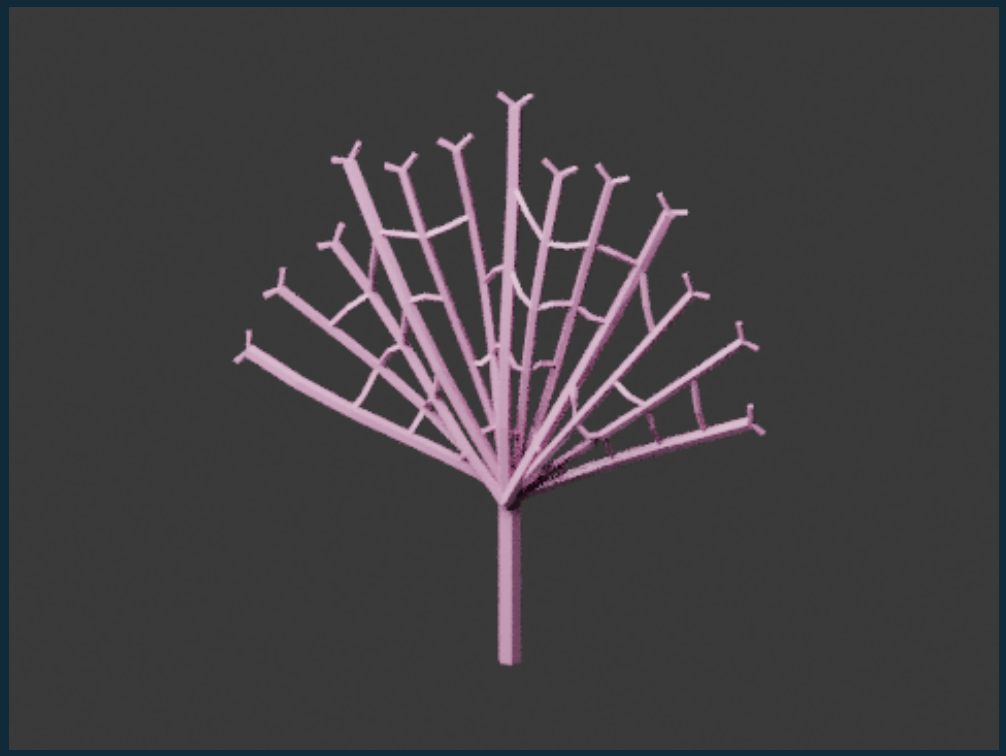
Replaced tiny subdivided eyes with surfaces that survive export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

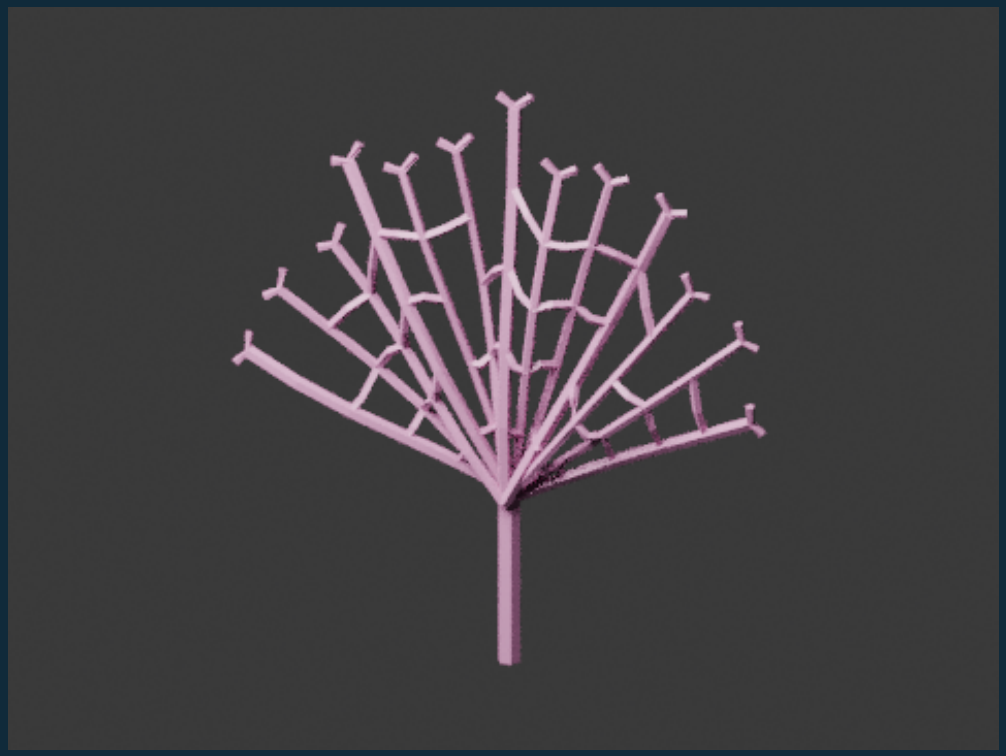
# Sea fan: export survival



**BEFORE**



**AFTER**



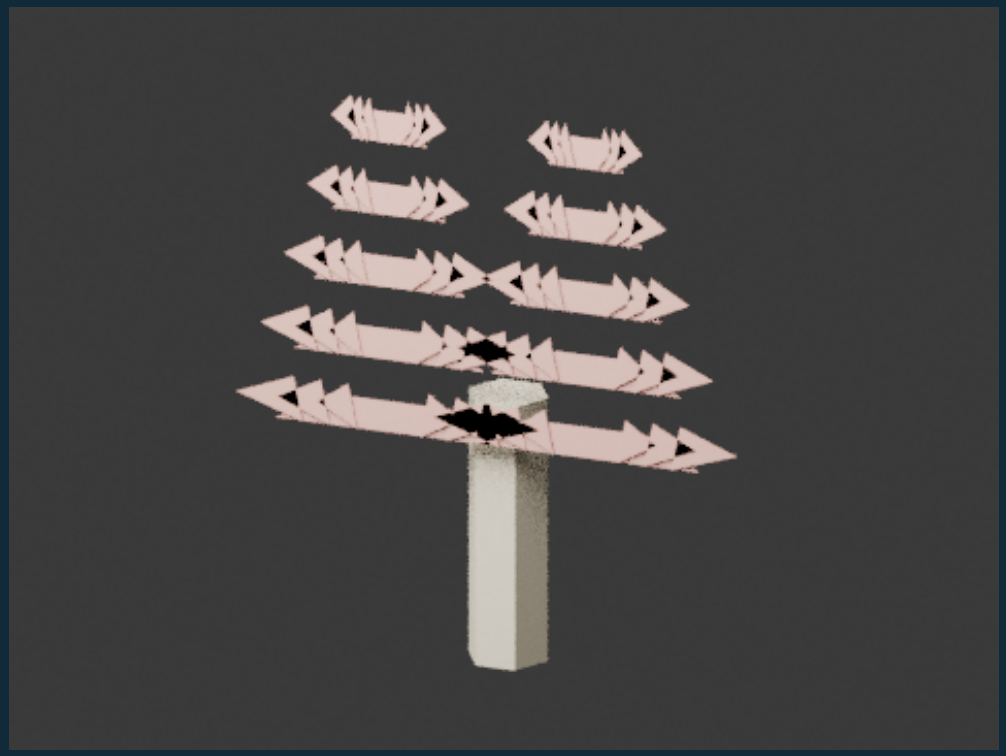
Thickened cross-branches to retain their end caps during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

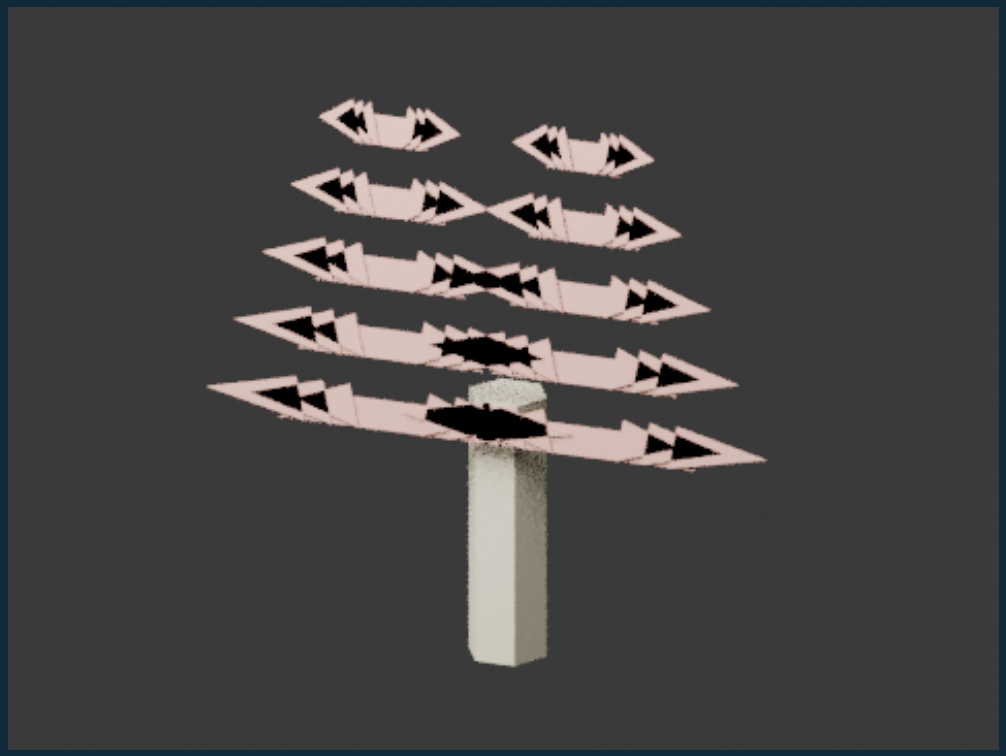
# Worm crown: export survival



## BEFORE



## AFTER



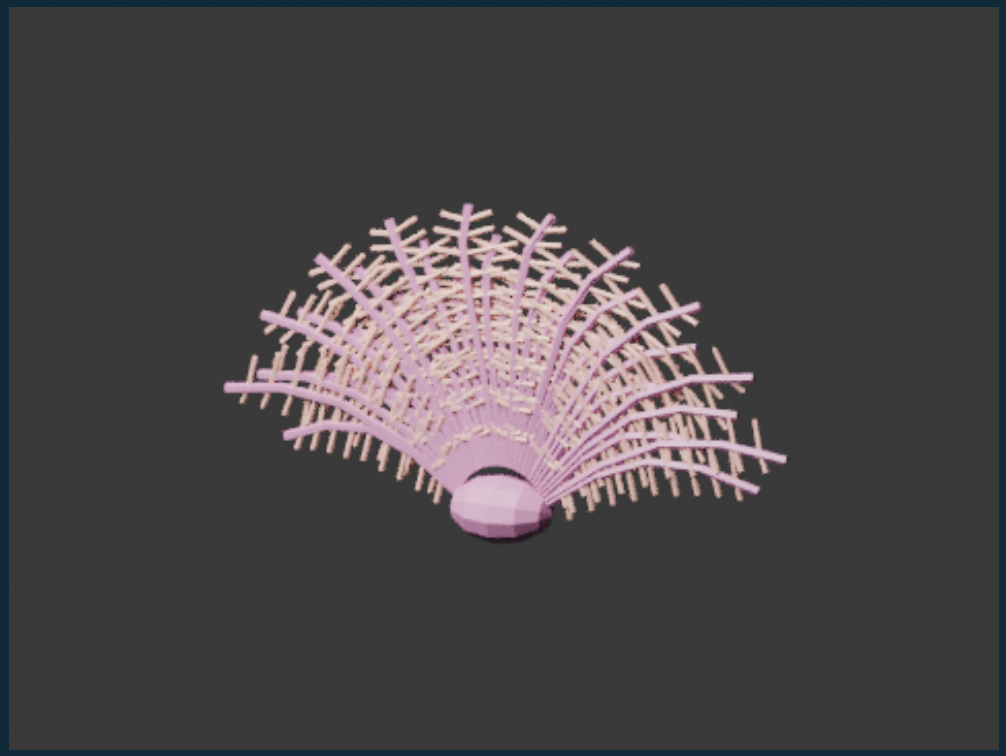
Extended the outer radioles to retain their triangles during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

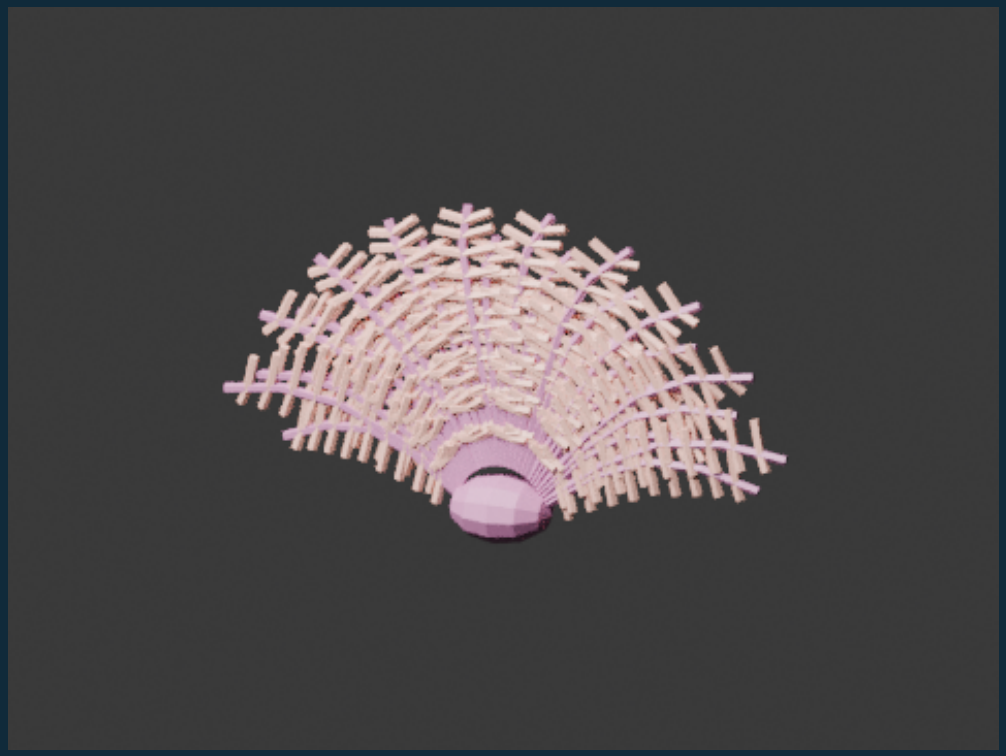
# Feather star: export survival



**BEFORE**



**AFTER**



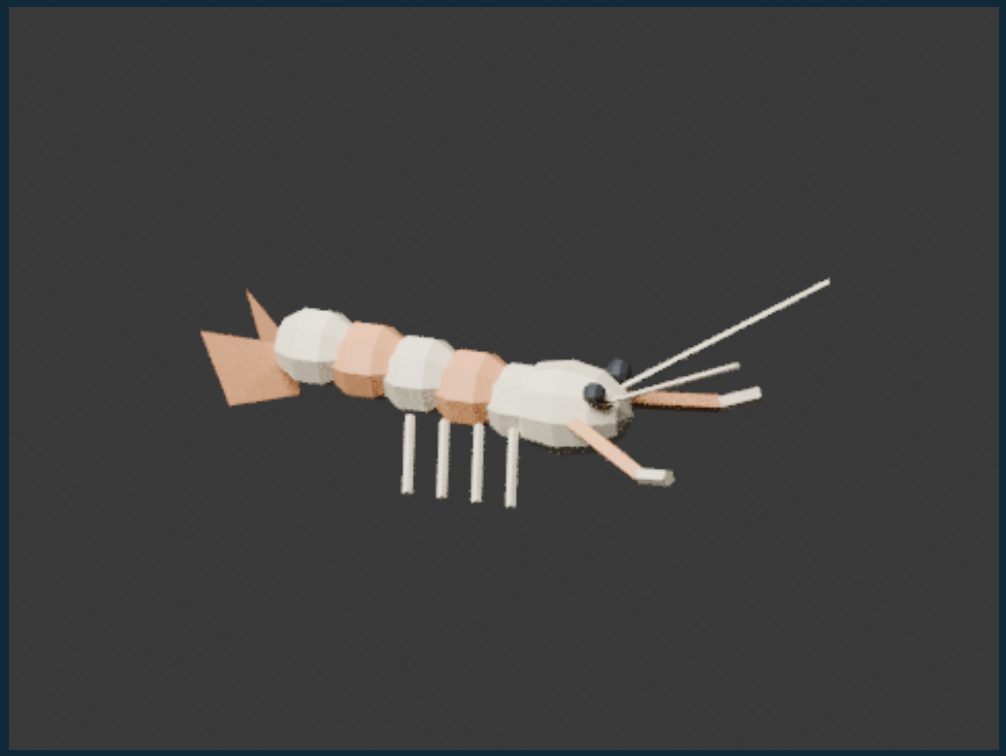
Thickened pinnules to preserve their end caps during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

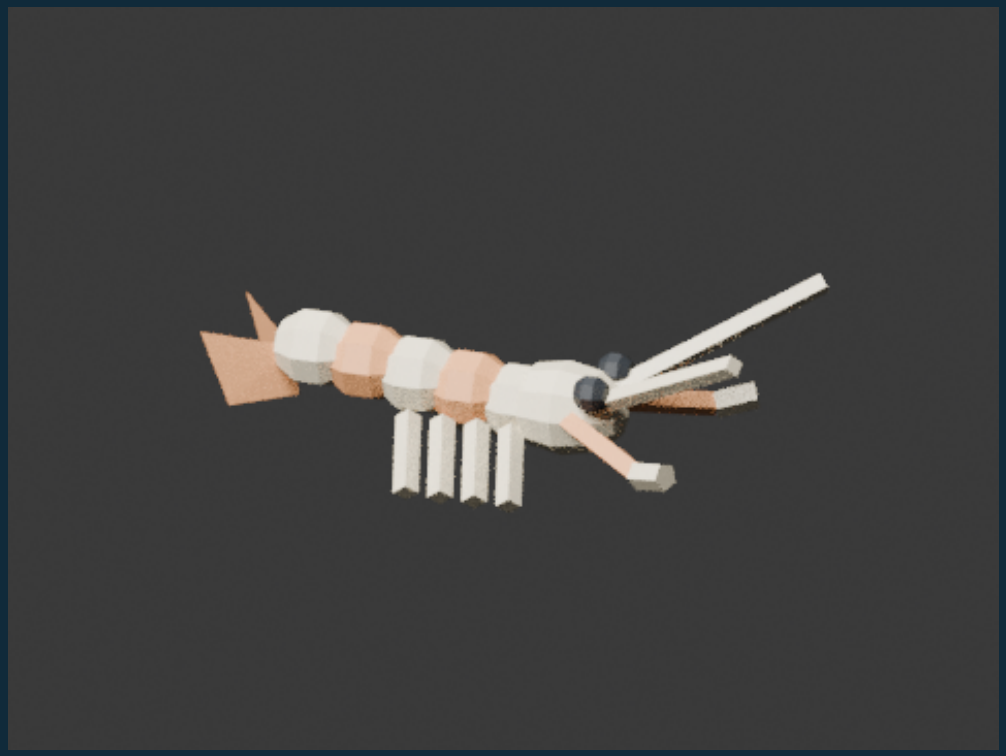
# Shrimp: export survival



## BEFORE



## AFTER



Thickened fine legs and antennae to preserve more faces during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Guard crab: export survival



## BEFORE

## AFTER

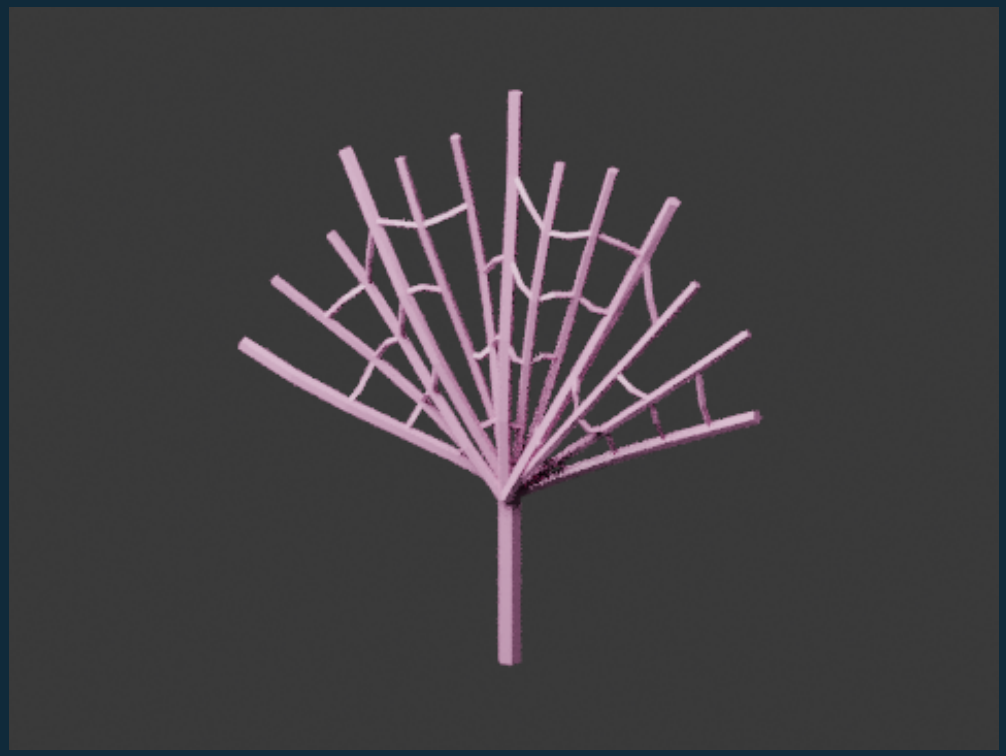
Thickened fine legs and pincers to preserve more faces during export.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

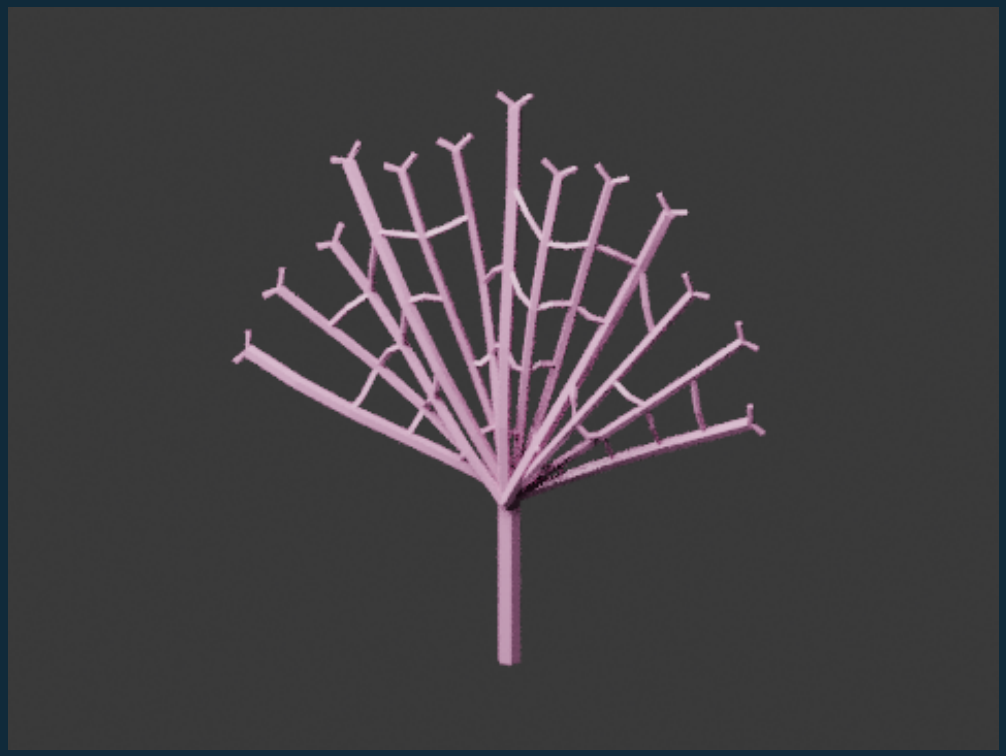
# Sea fan: branch tips



**BEFORE**



**AFTER**



Added split tips along the outer branches.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Humbug: face and fins



## BEFORE

## AFTER

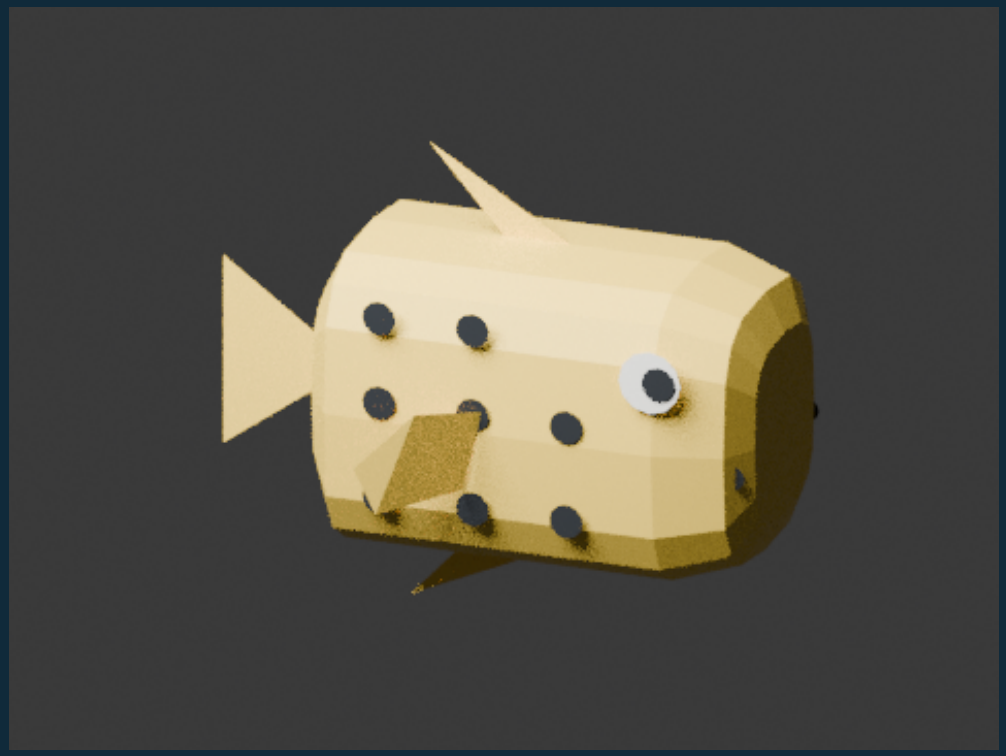
Added eye rings, pupils, mouth markings and layered pectoral fins.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Boxfish: tail and mouth



## BEFORE



## AFTER

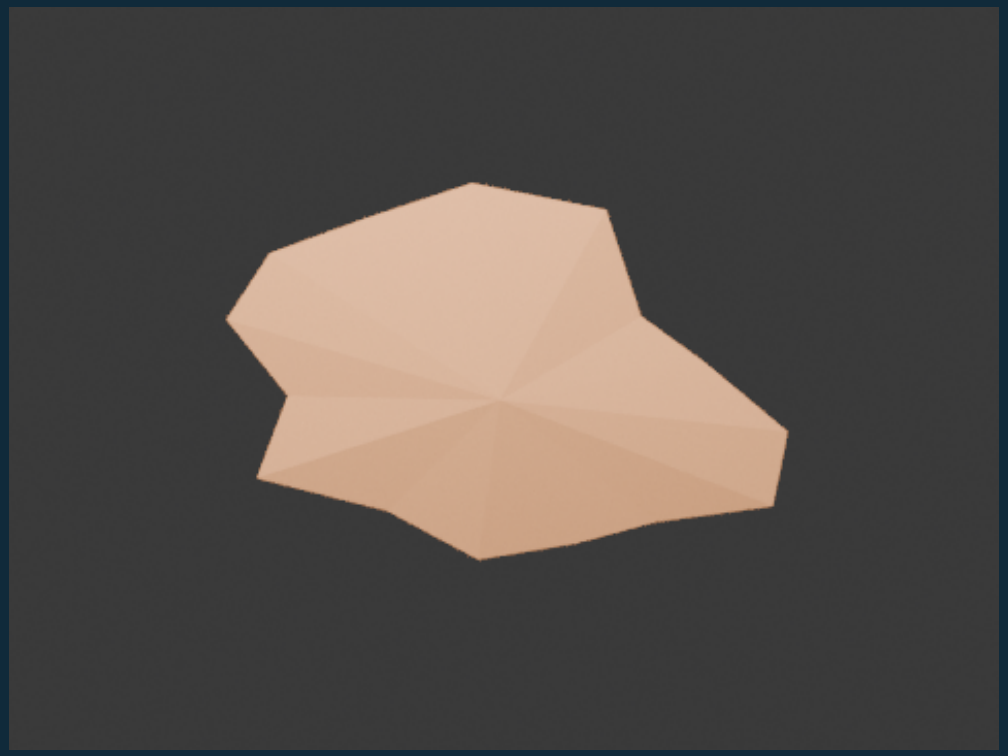
Rounded the tail fan and added a projecting snout and mouth.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Encrusting sponge



## BEFORE



## AFTER

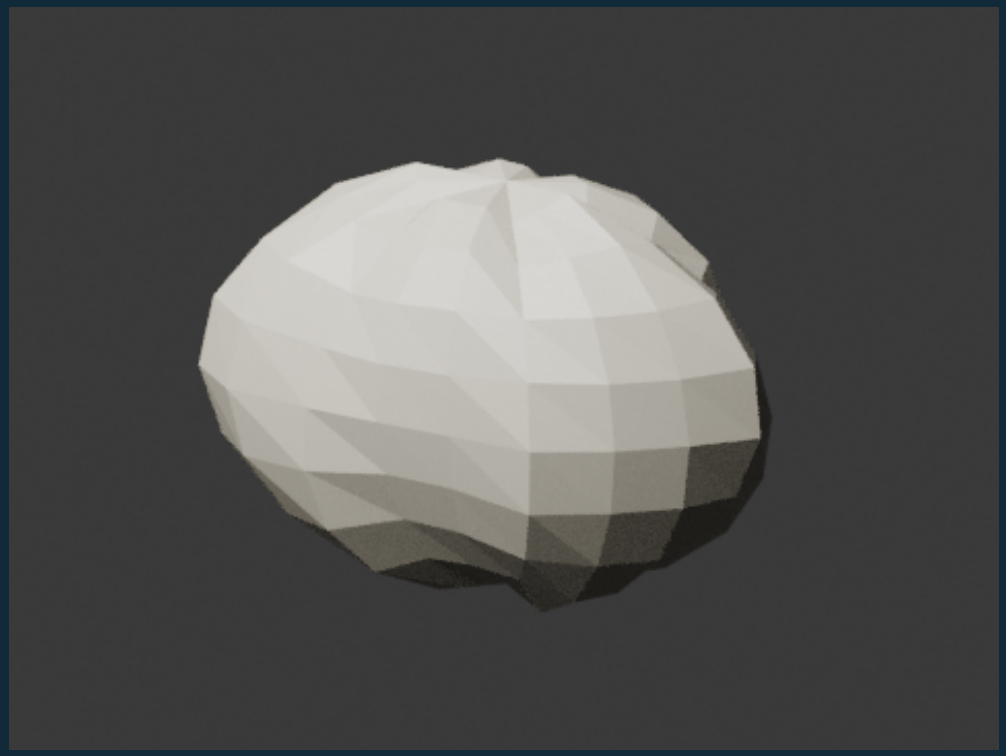
Added a gently uneven surface to the formerly flat crust.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

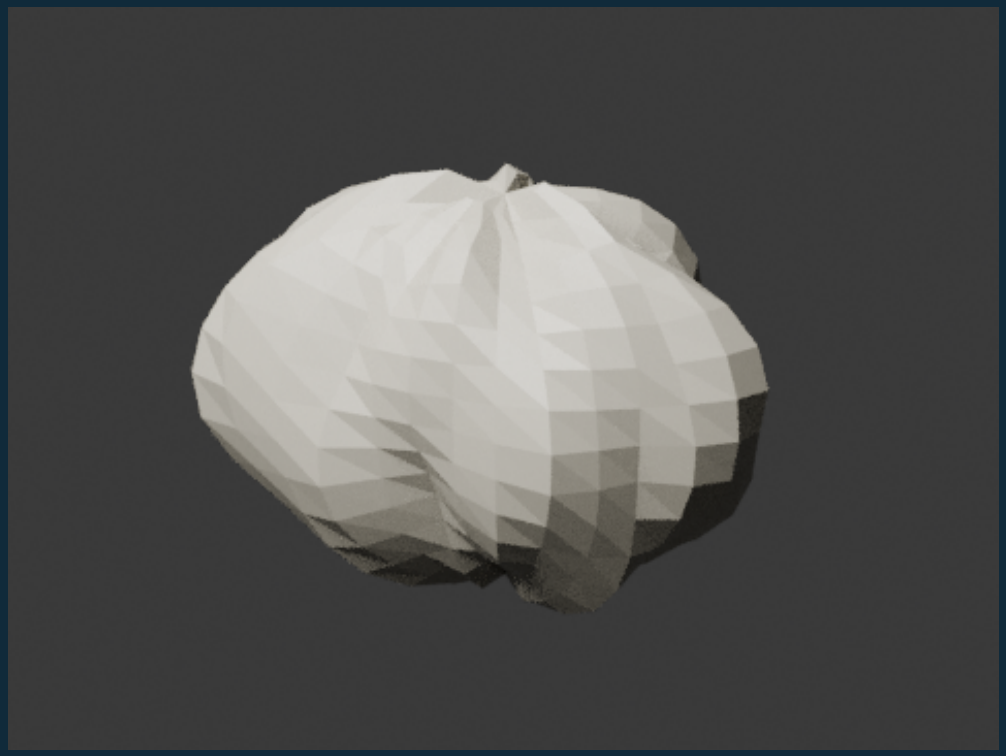
# Reef rock mass



**BEFORE**



**AFTER**



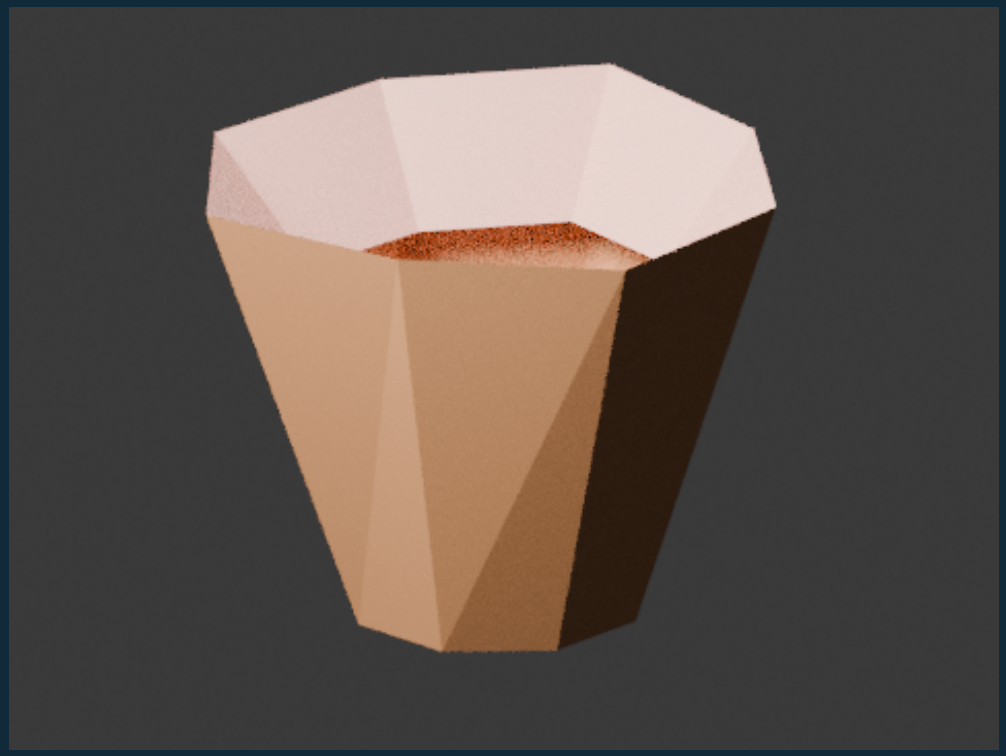
Added finer ledges and deeper irregular lobes; preview is before voxel remeshing.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Cup coral



## BEFORE



## AFTER

Added an oral disc and radiating tentacles.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Zoanthid colony



## BEFORE

## AFTER

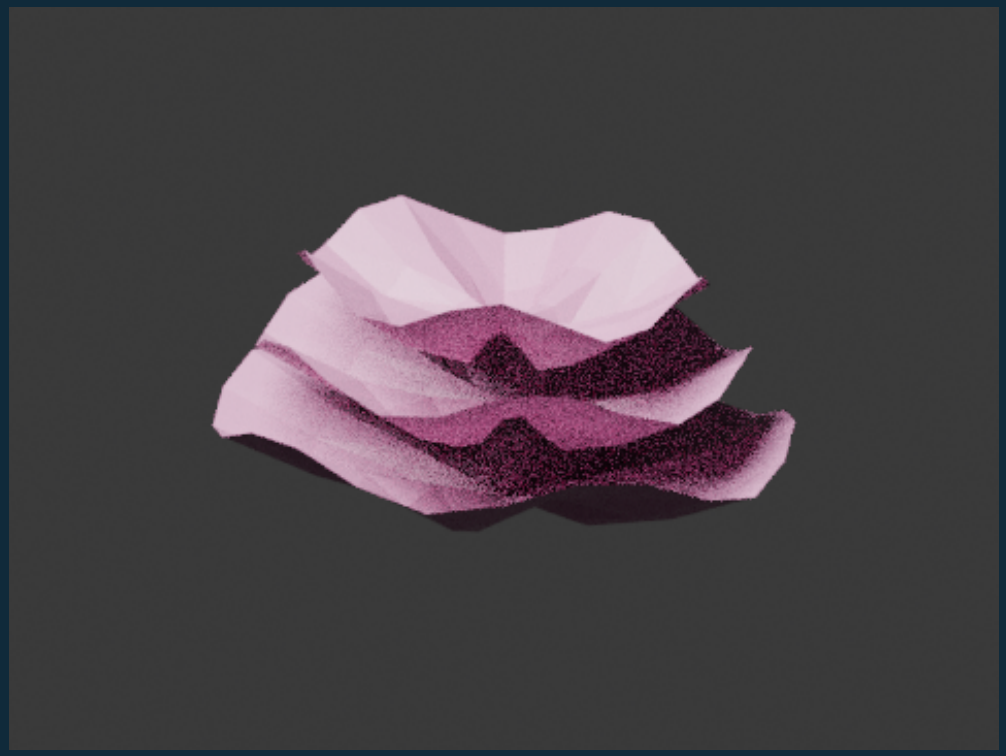
Added tentacles around each polyp disc.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Scroll coral



## BEFORE



## AFTER

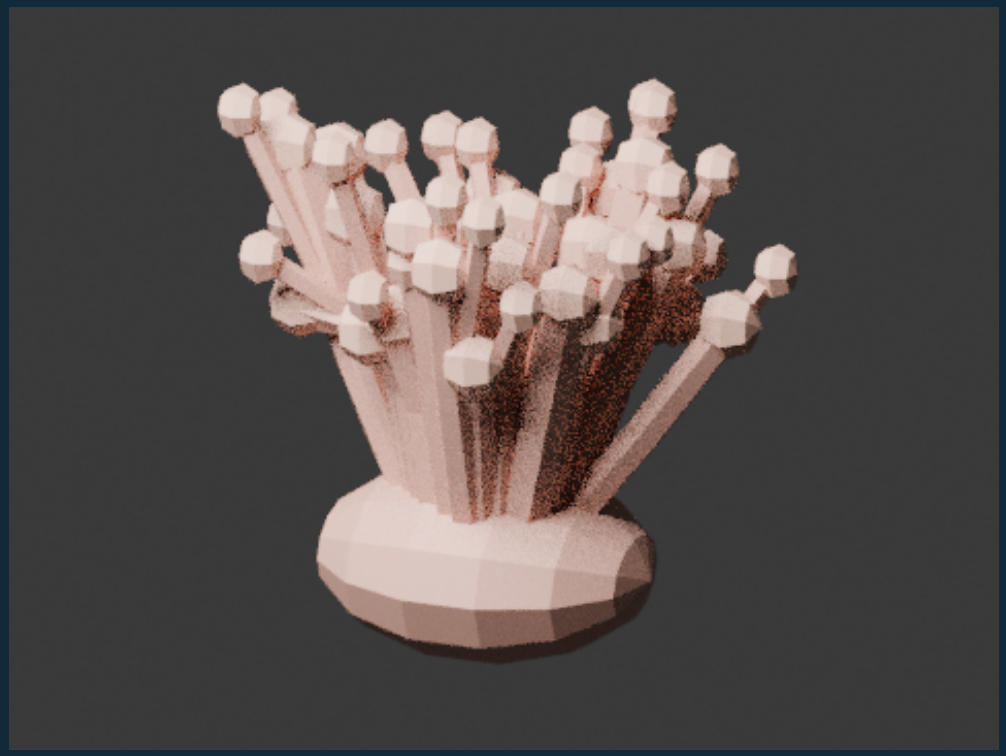
Added asymmetry, deeper curls and ruffled edges to the plates.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

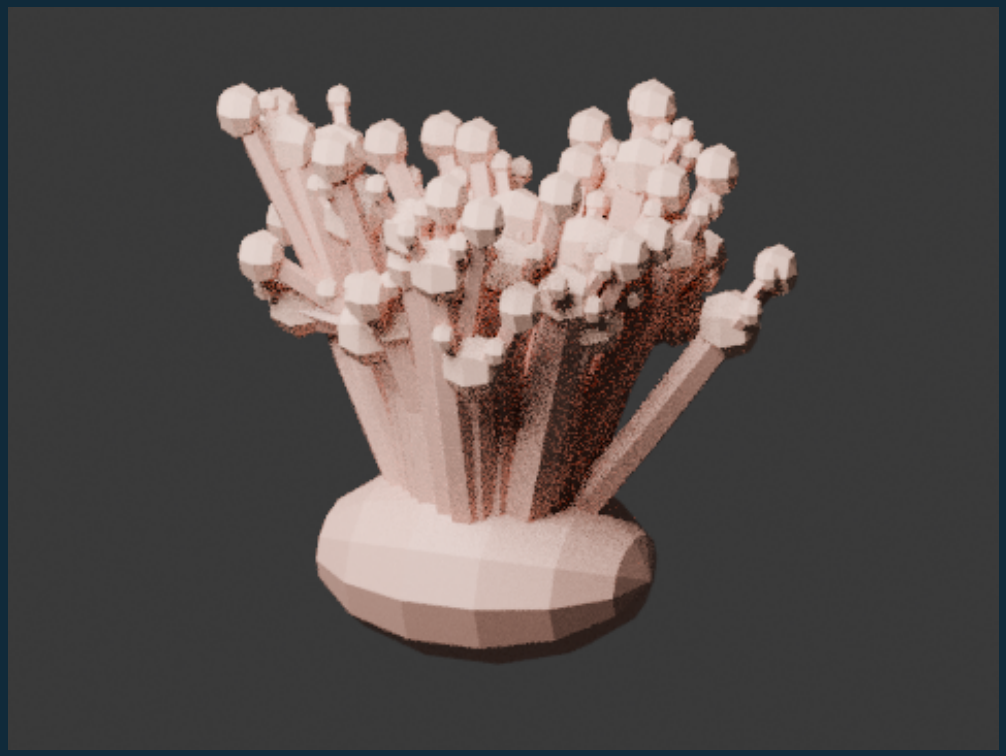
# Cauliflower coral



## BEFORE



## AFTER



Added irregular secondary branchlets to the colony tips.

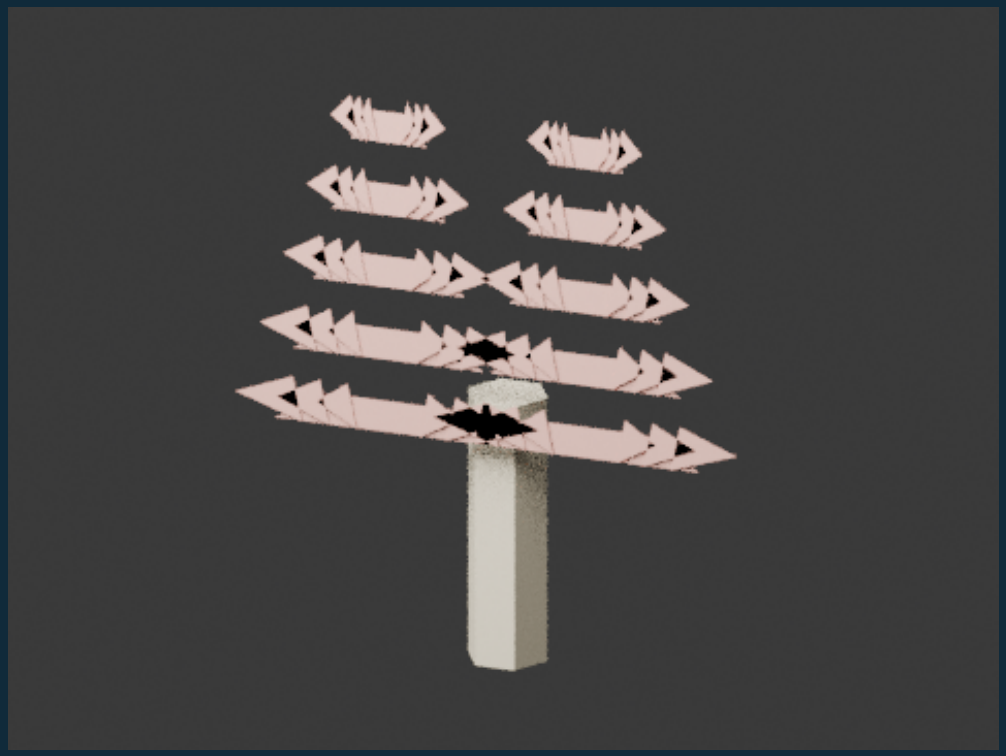
Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Christmas tree worm



## BEFORE

## AFTER



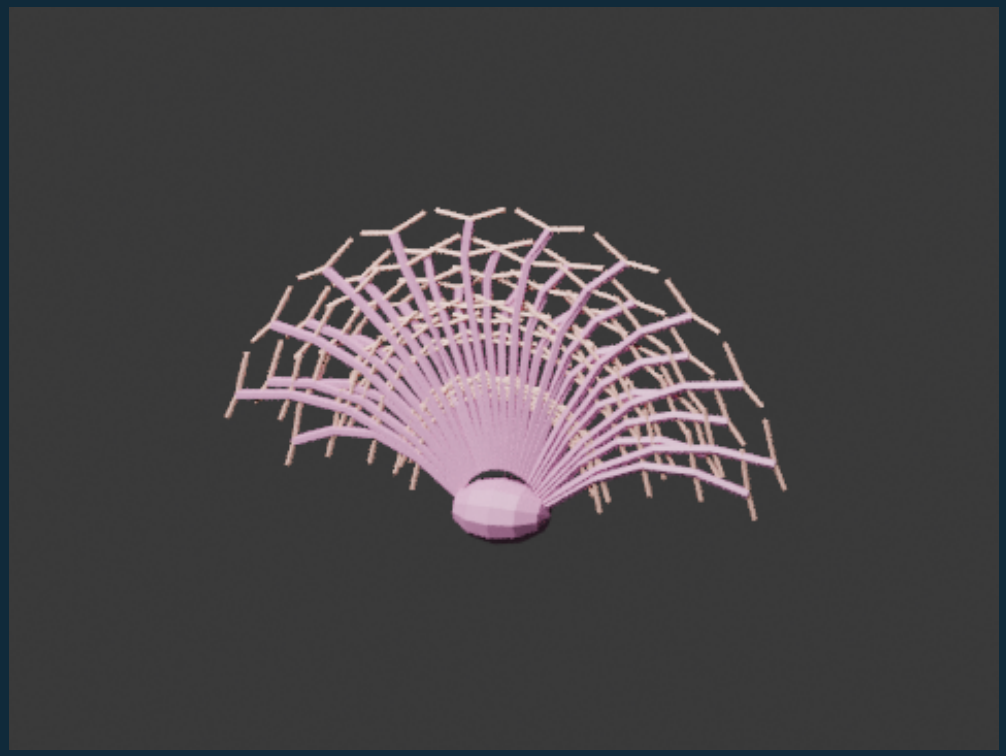
Added individual radioles to each crown tier.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

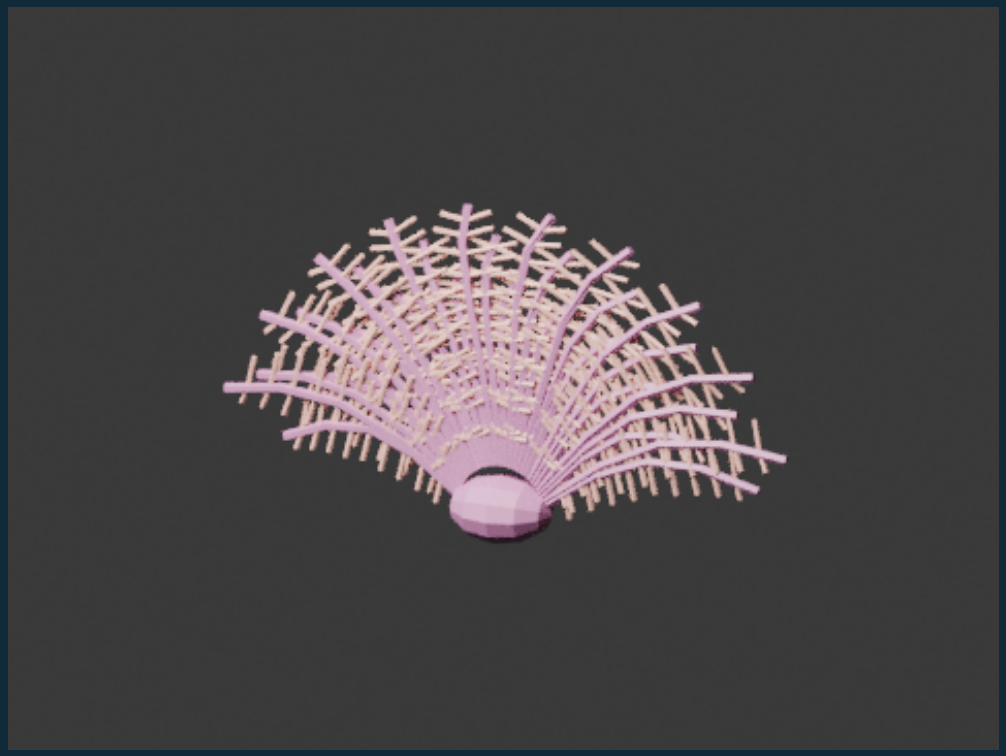
# Feather star



## BEFORE



## AFTER



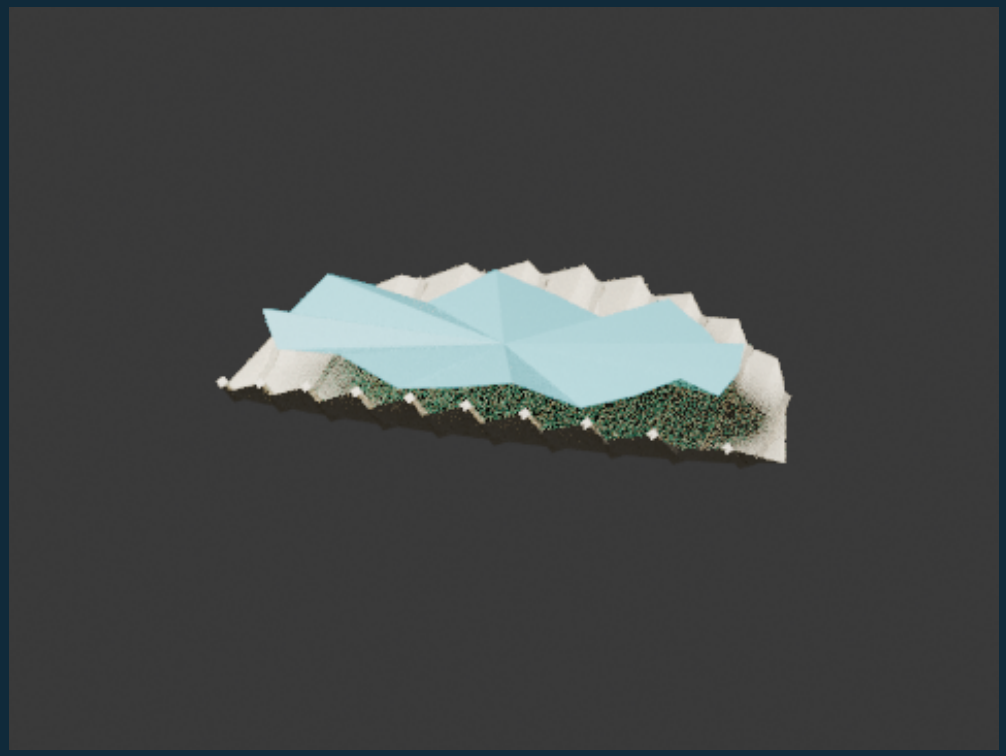
Added closely spaced pinnules along the arms.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

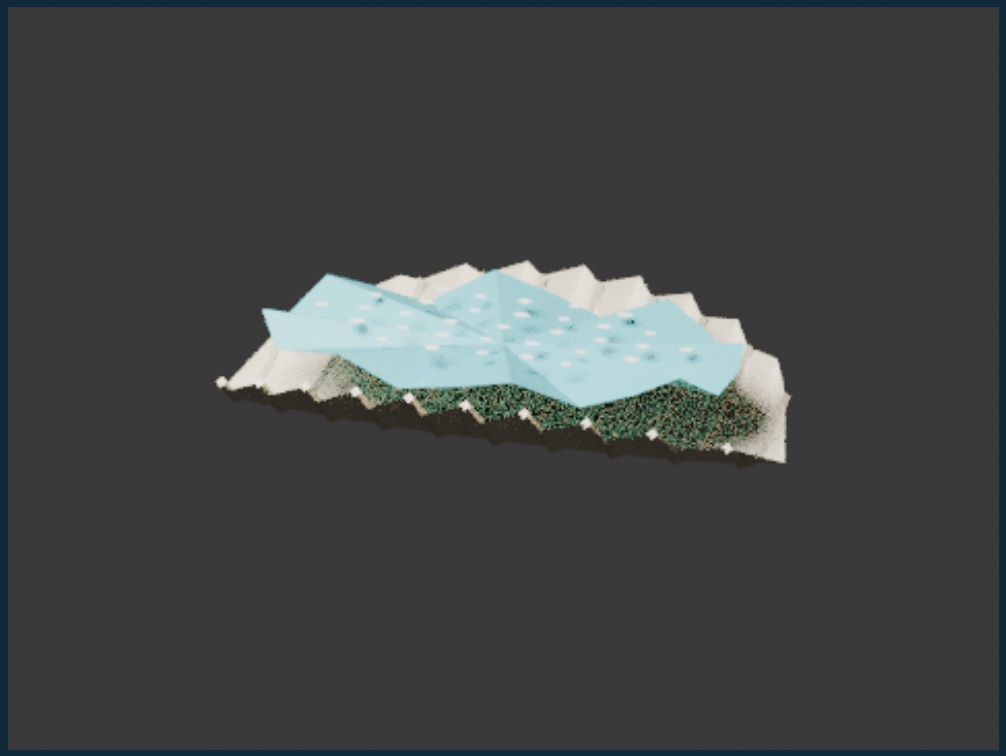
# Giant clam



## BEFORE



## AFTER



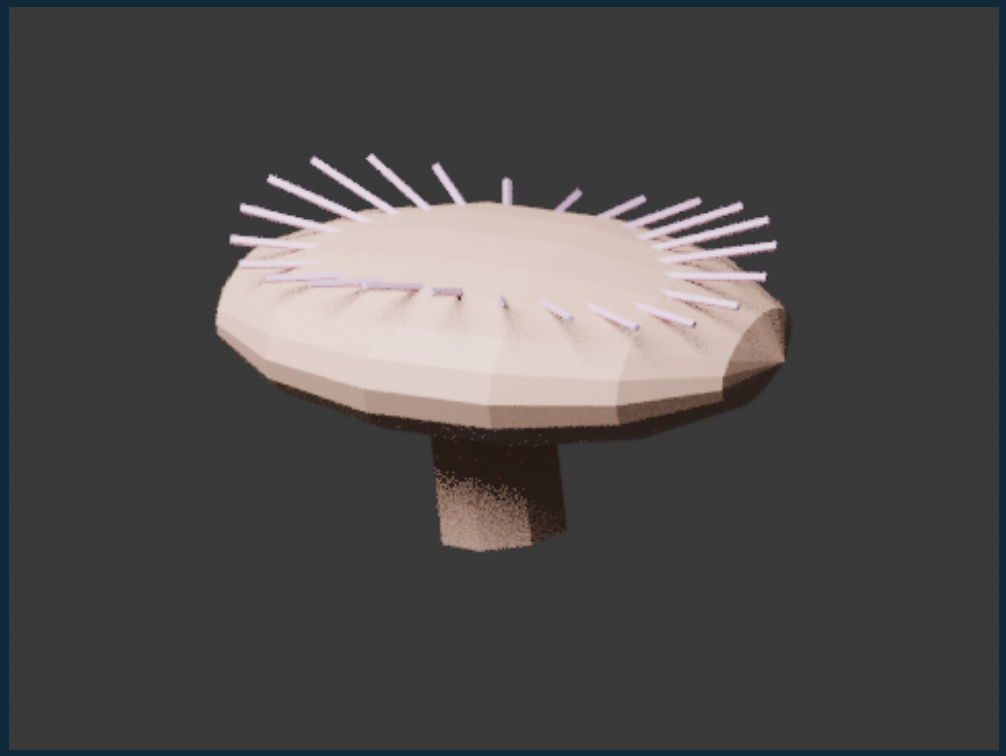
Added pale markings to the blue mantle.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

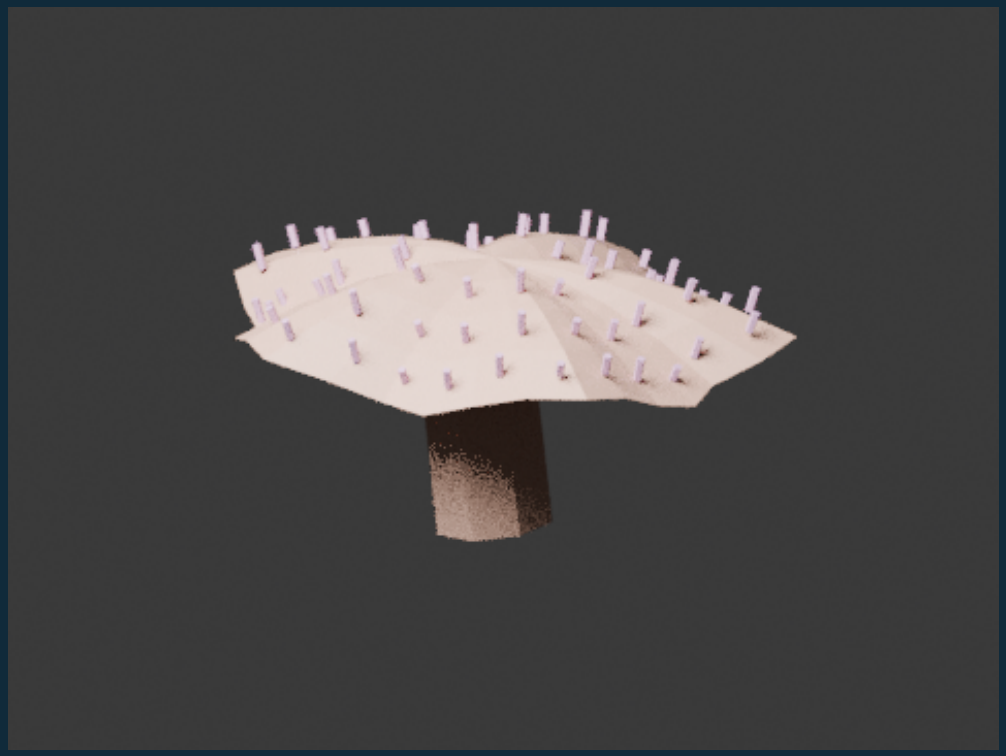
# Leather coral



## BEFORE



## AFTER



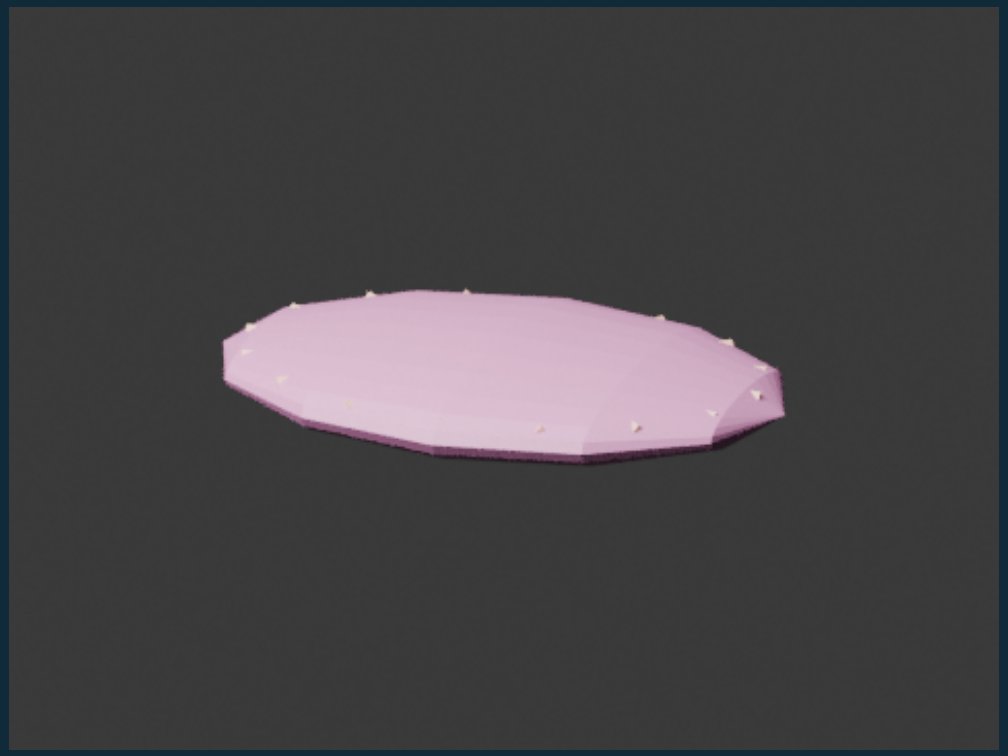
Replaced the smooth cap with a scalloped crown and projecting polyps.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

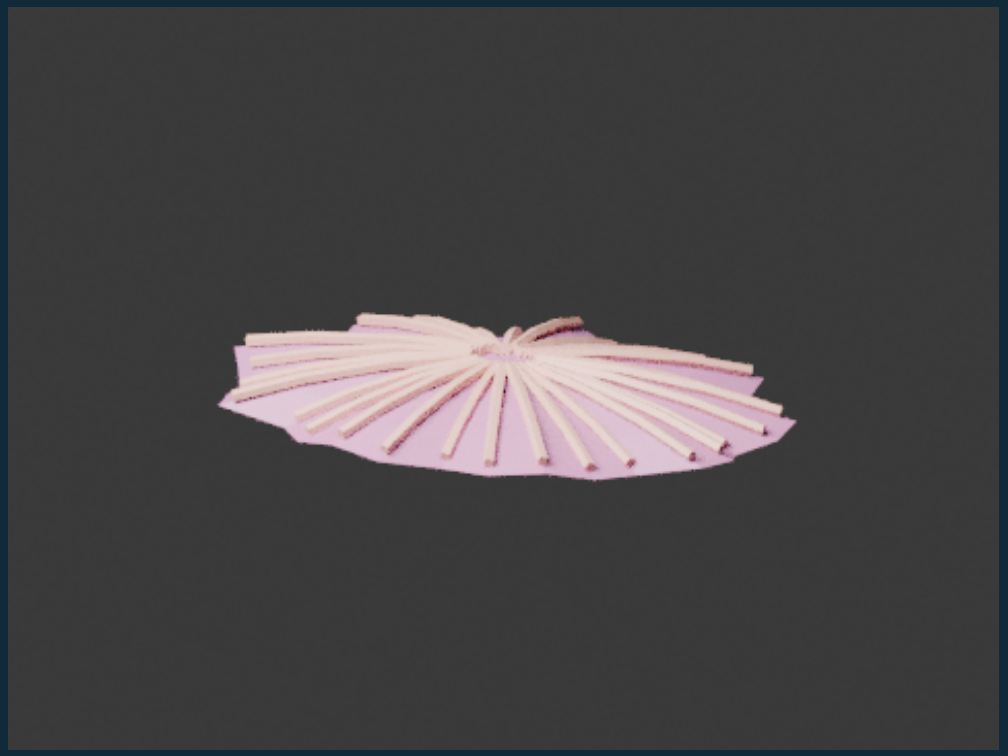
# Mushroom coral



## BEFORE



## AFTER



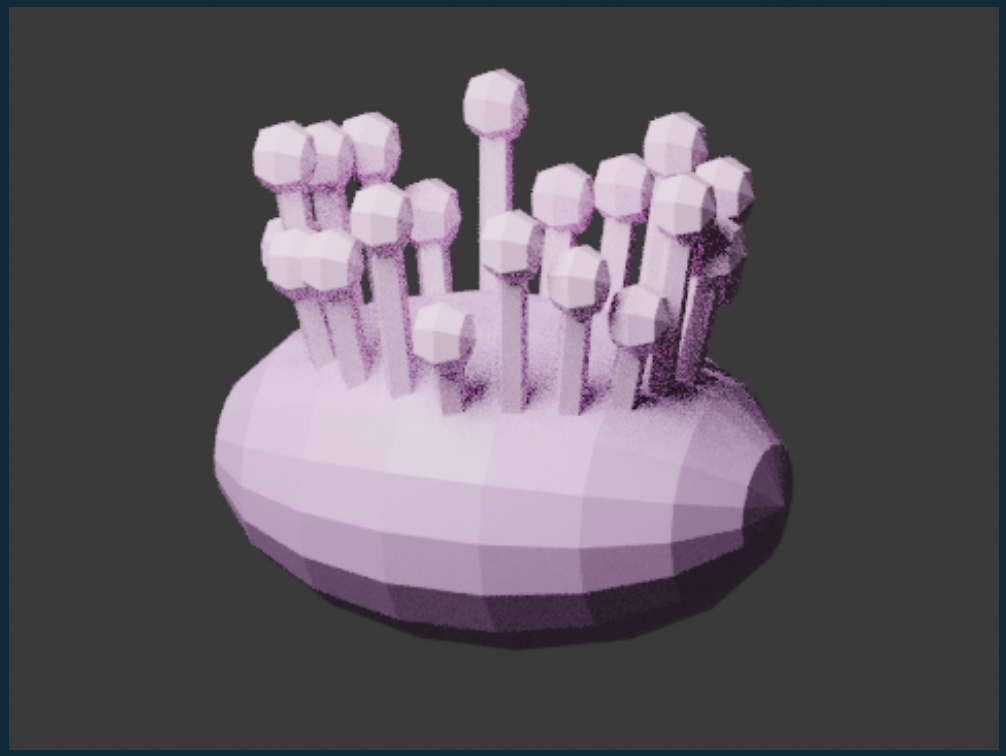
Replaced the smooth disc with a scalloped surface and radial ridges.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

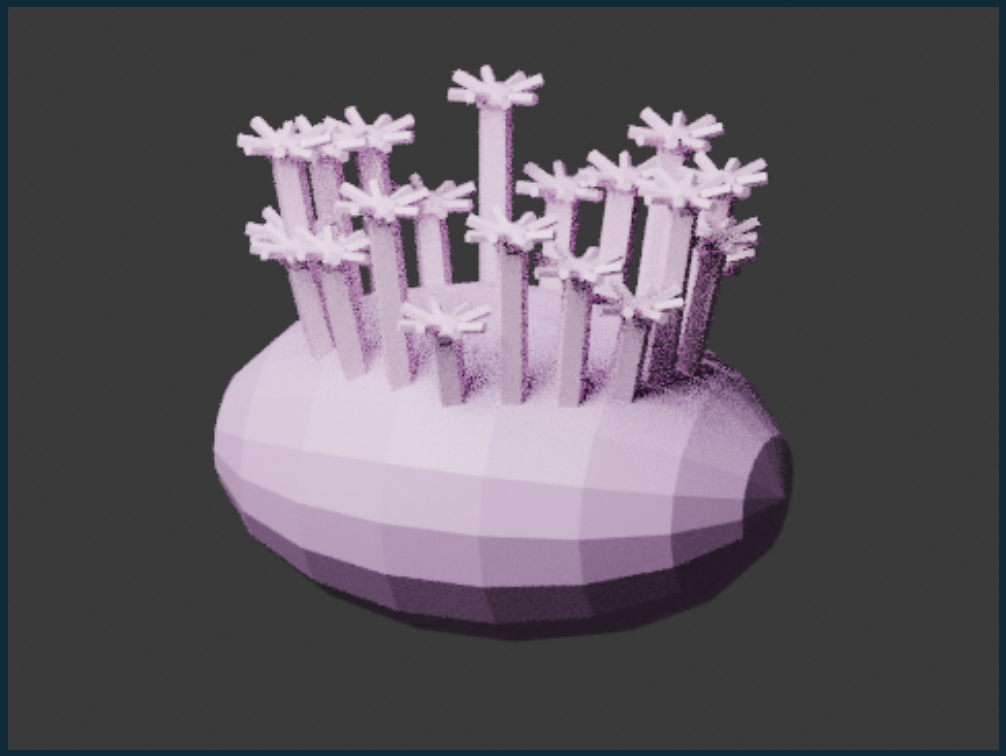
# Flowerpot coral



## BEFORE



## AFTER



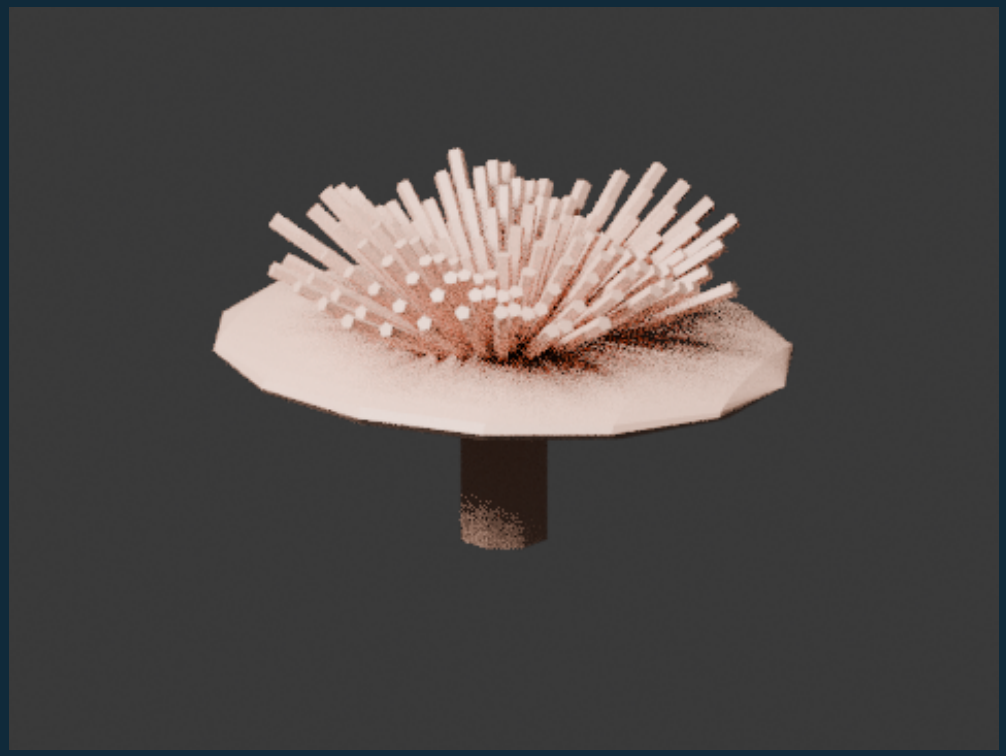
Replaced oval polyp tips with radiating tentacle crowns.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

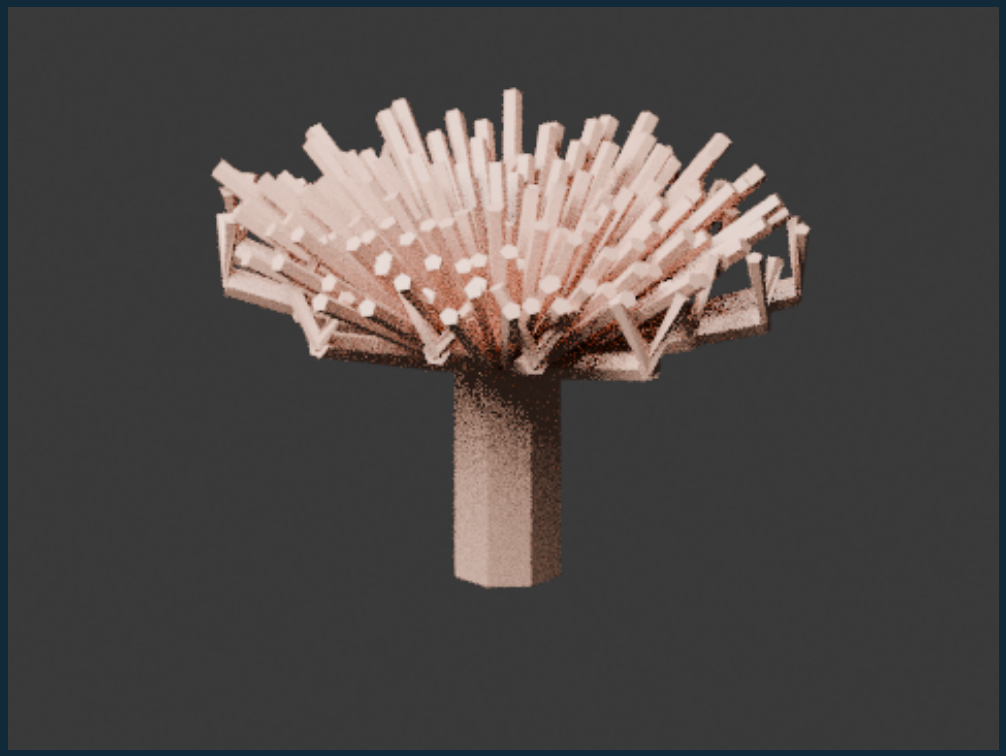
# Table coral



## BEFORE



## AFTER



Replaced the solid plate with branching supports and an irregular crown.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Boulder coral



**BEFORE**

**AFTER**

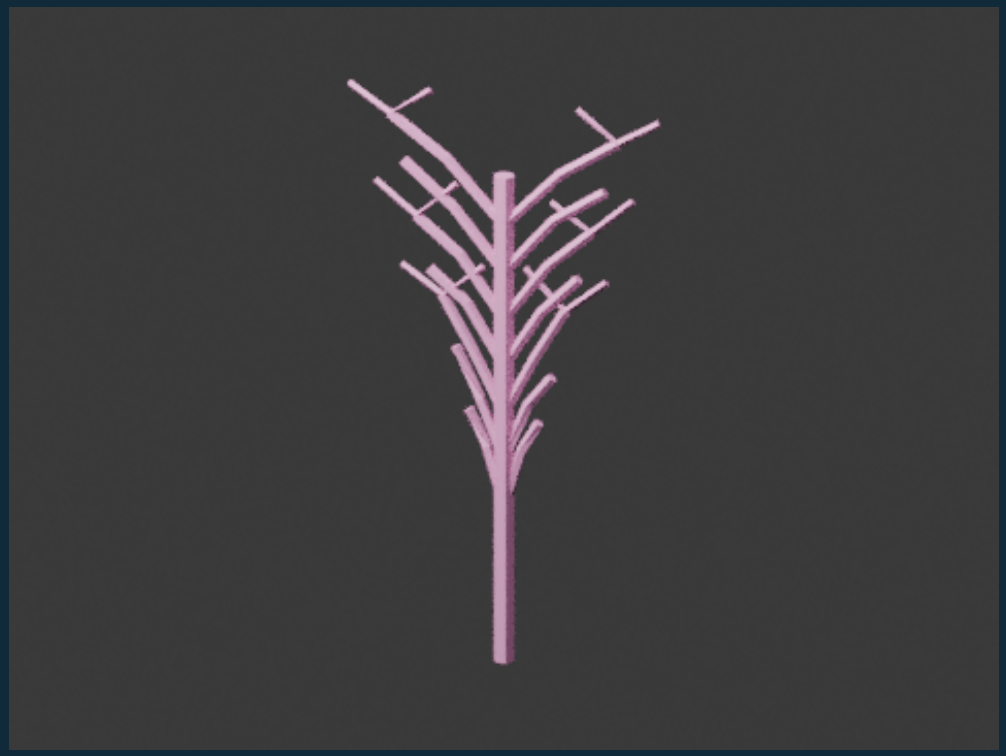
Added smooth, irregular growth contours and additional surface geometry.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

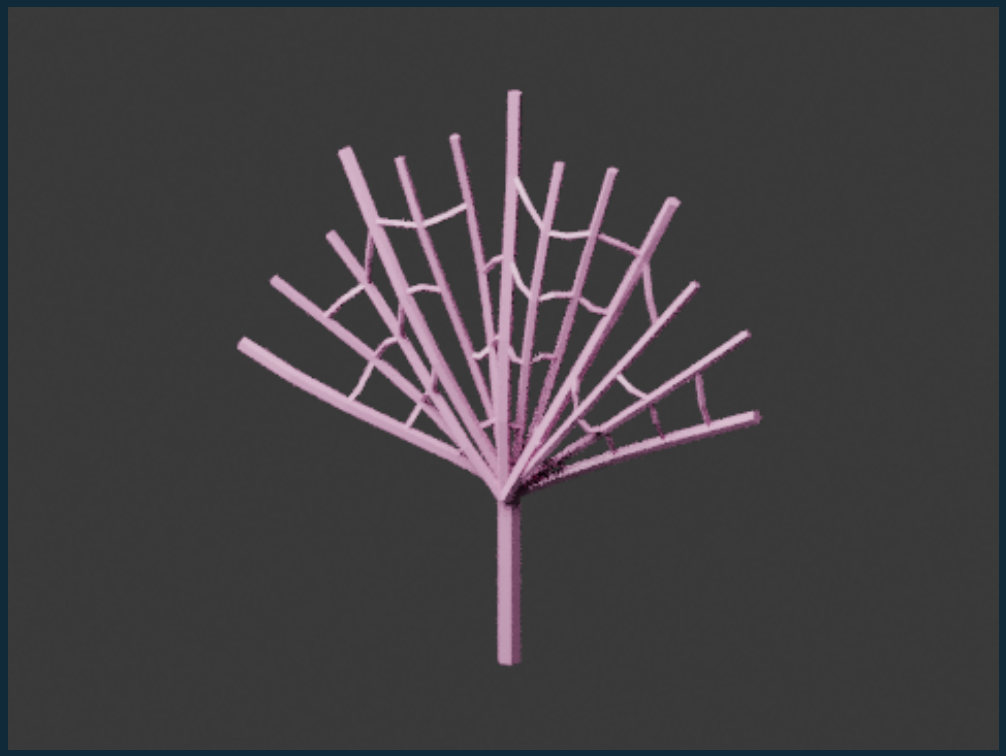
# Sea fan



## BEFORE



## AFTER



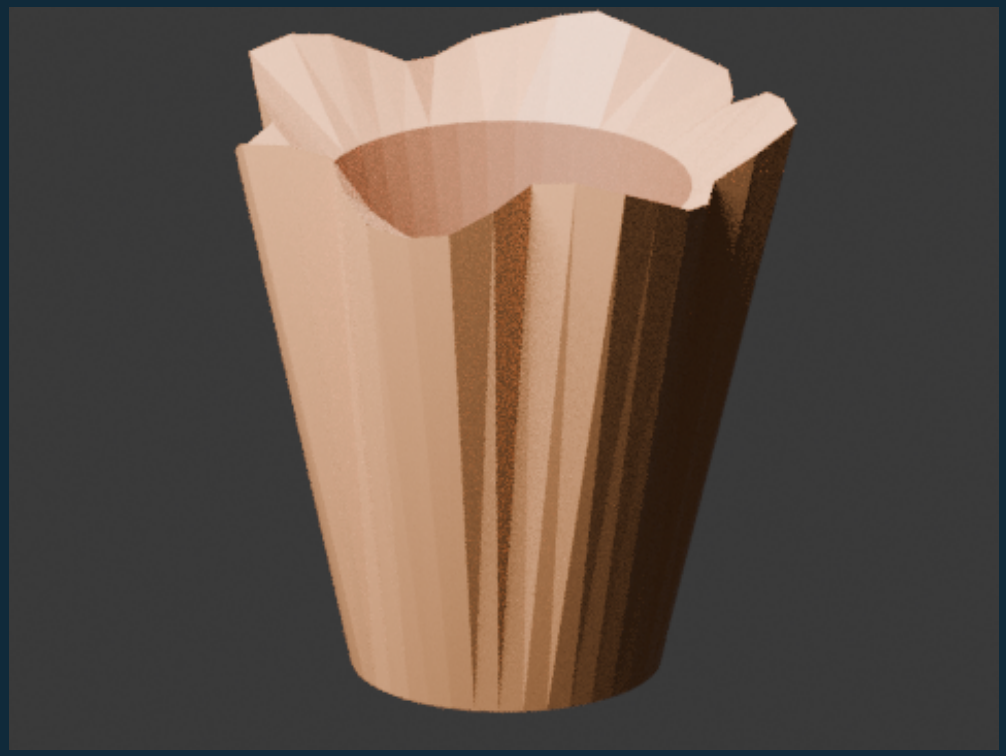
Replaced the sparse tree shape with radial branches and cross-branches.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

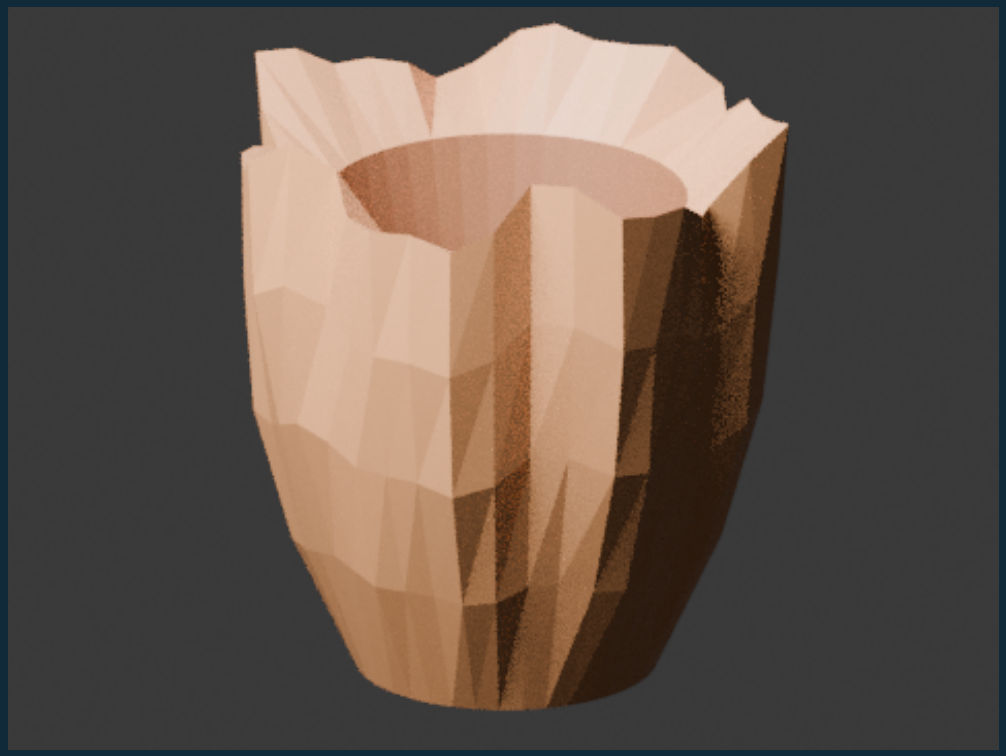
# Barrel sponge



## BEFORE



## AFTER



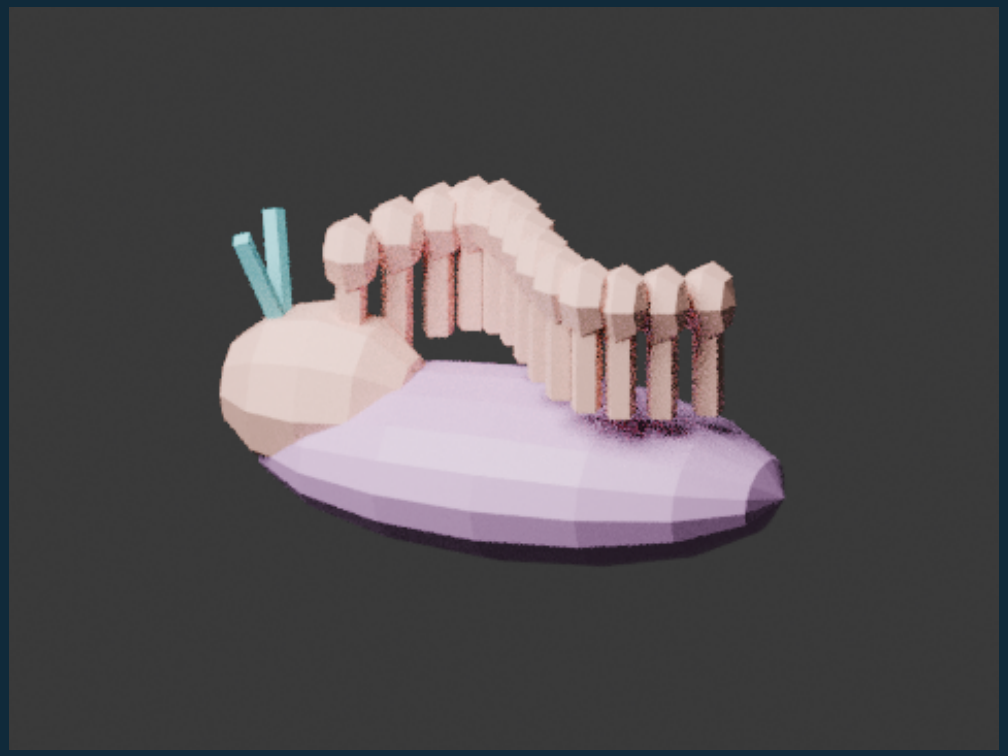
Added uneven, bulging wall rings and an irregular lip.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

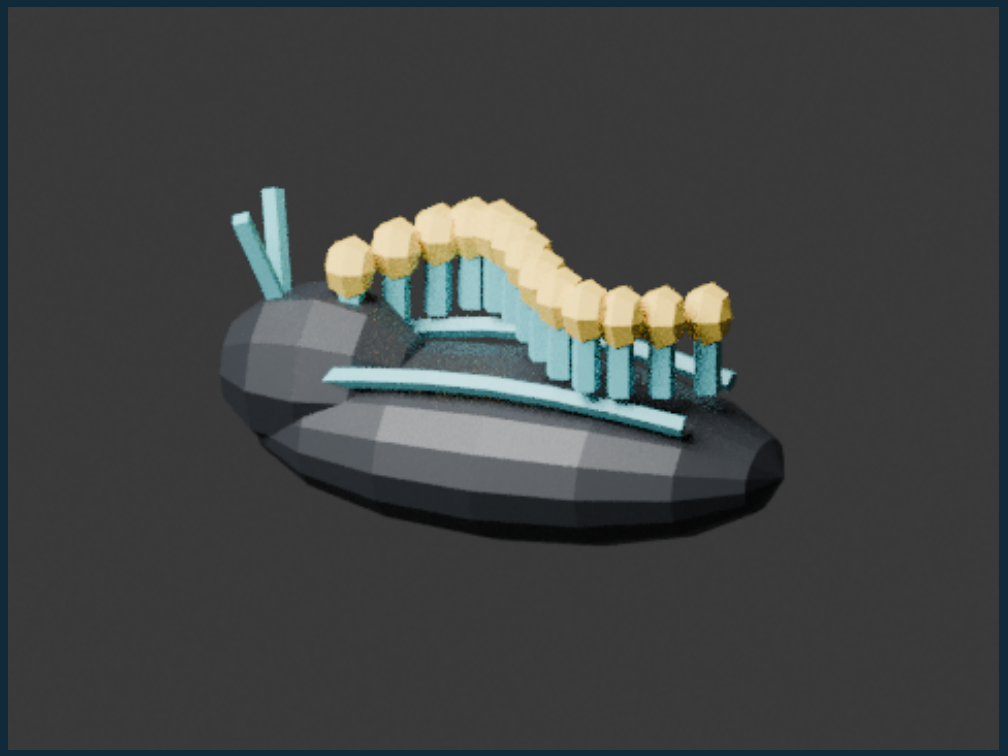
# Nudibranch



## BEFORE



## AFTER



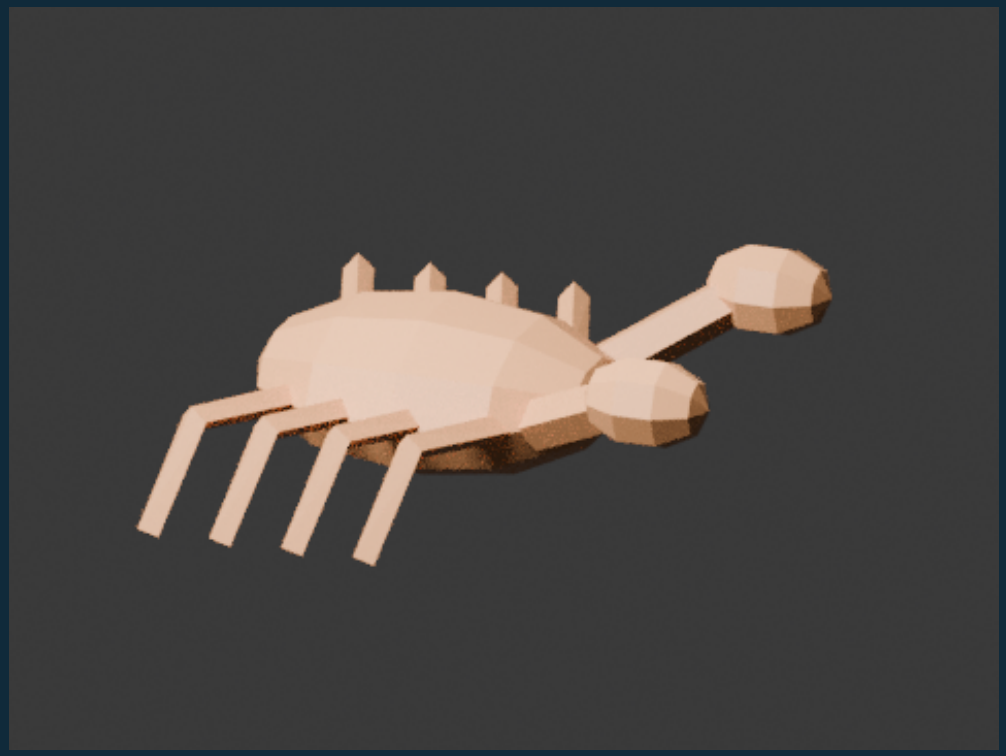
Added a dark mantle, pale ridges and yellow-tipped tubercles.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Coral guard crab



## BEFORE



## AFTER

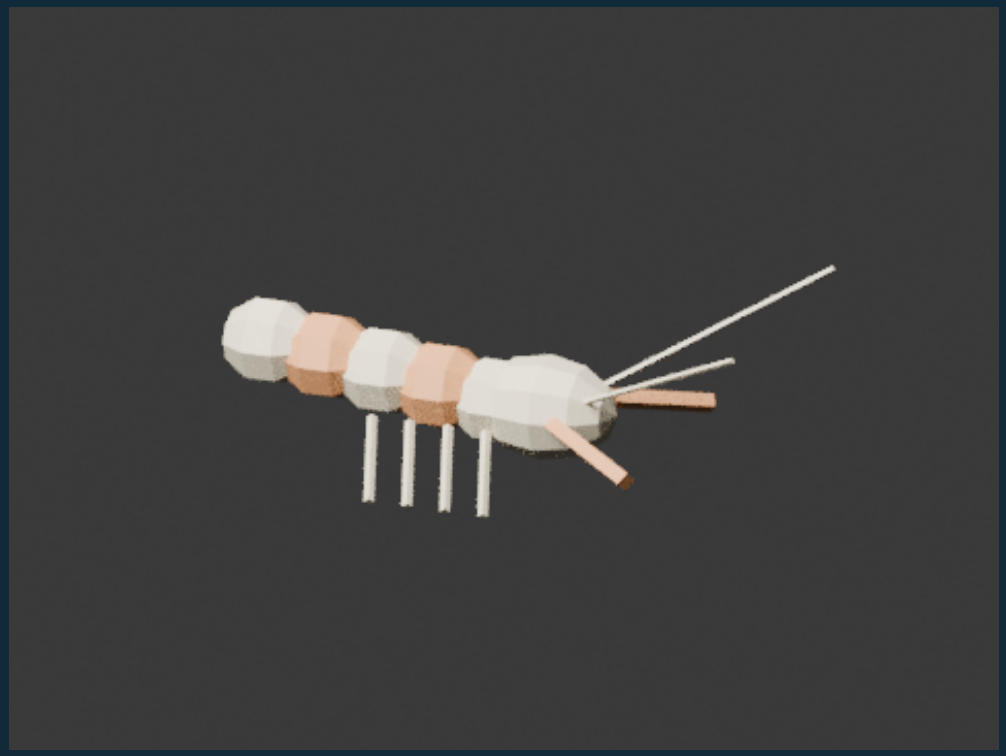
Added eyes and separate pincer shapes.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

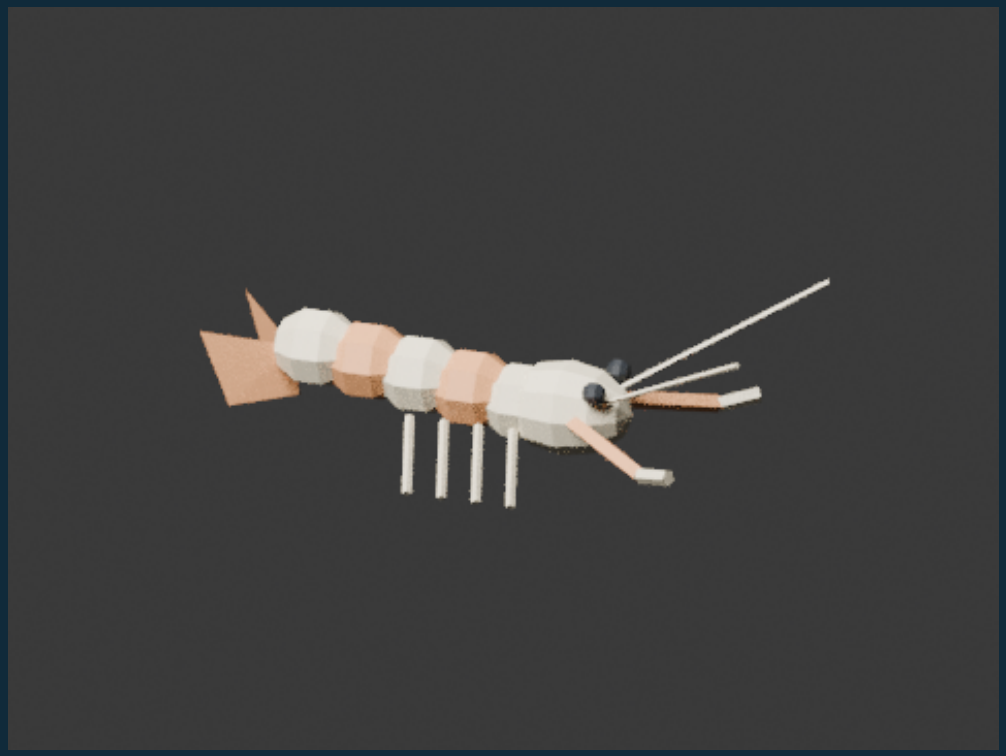
# Banded coral shrimp



## BEFORE



## AFTER



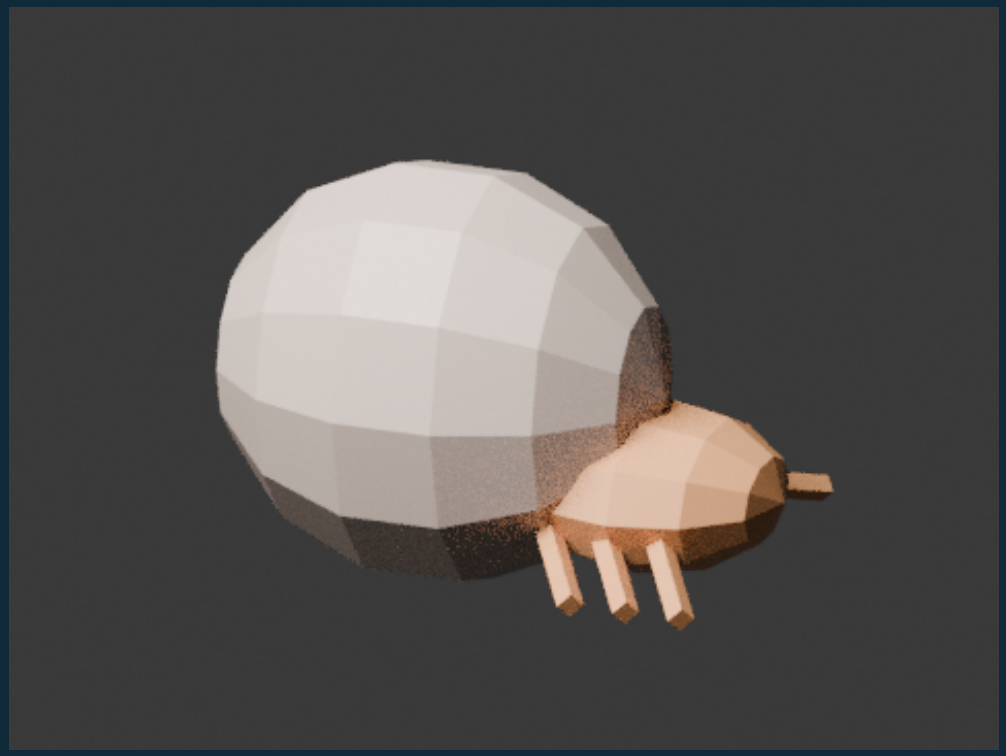
Added eyes, pincer details and a tail fan.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

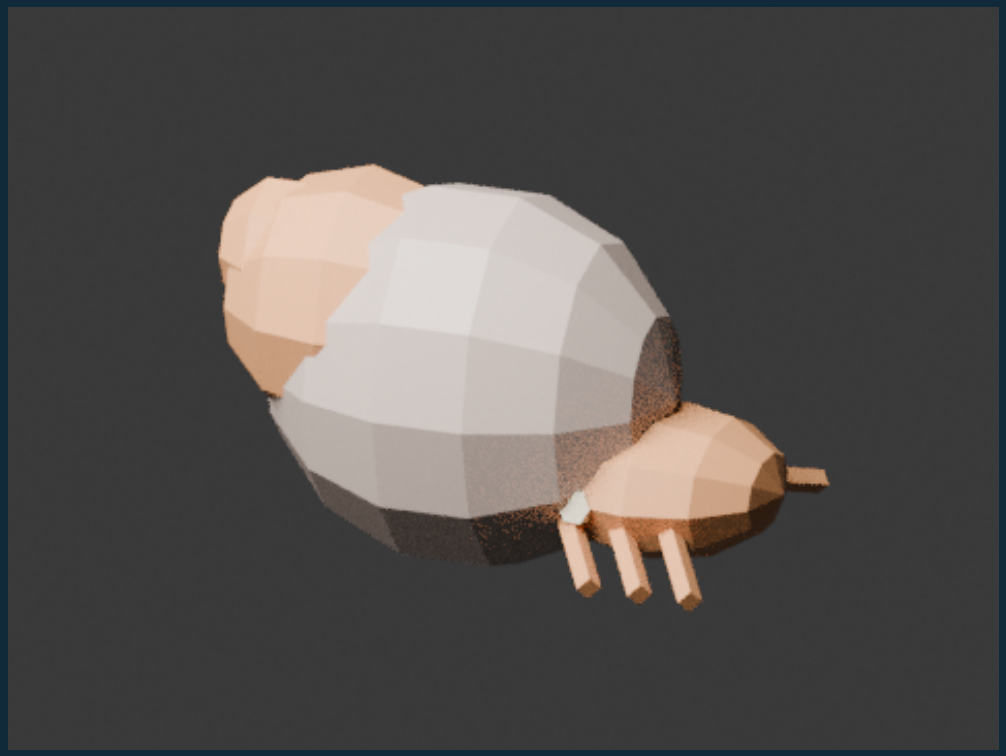
# Hermit crab



## BEFORE



## AFTER



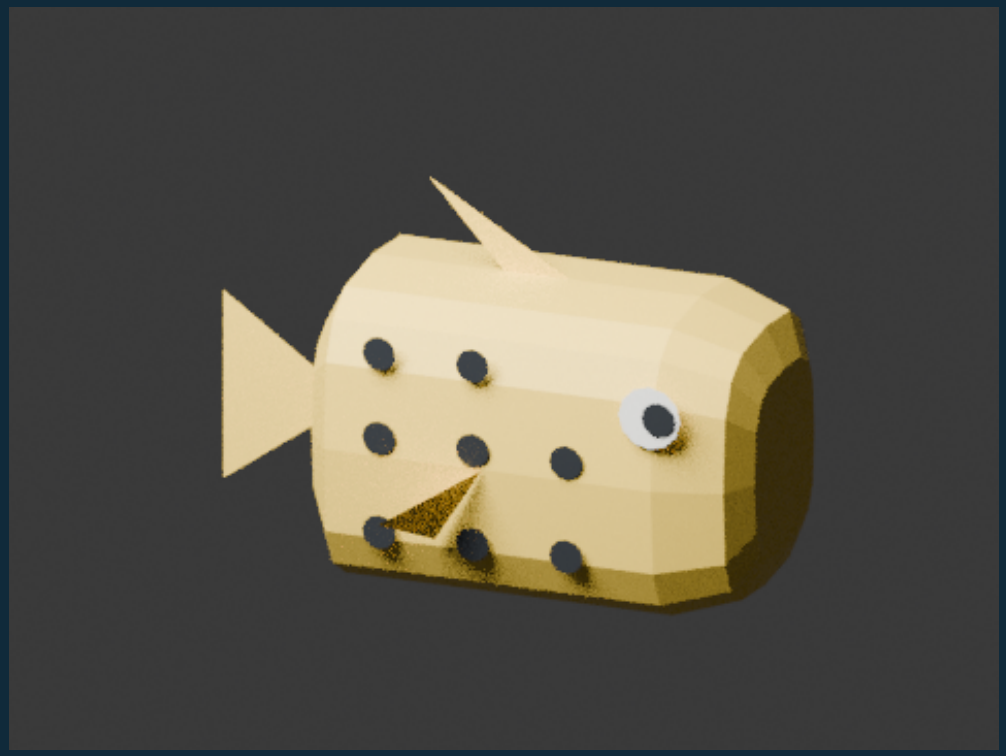
Added shell whorls and a pale lip.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

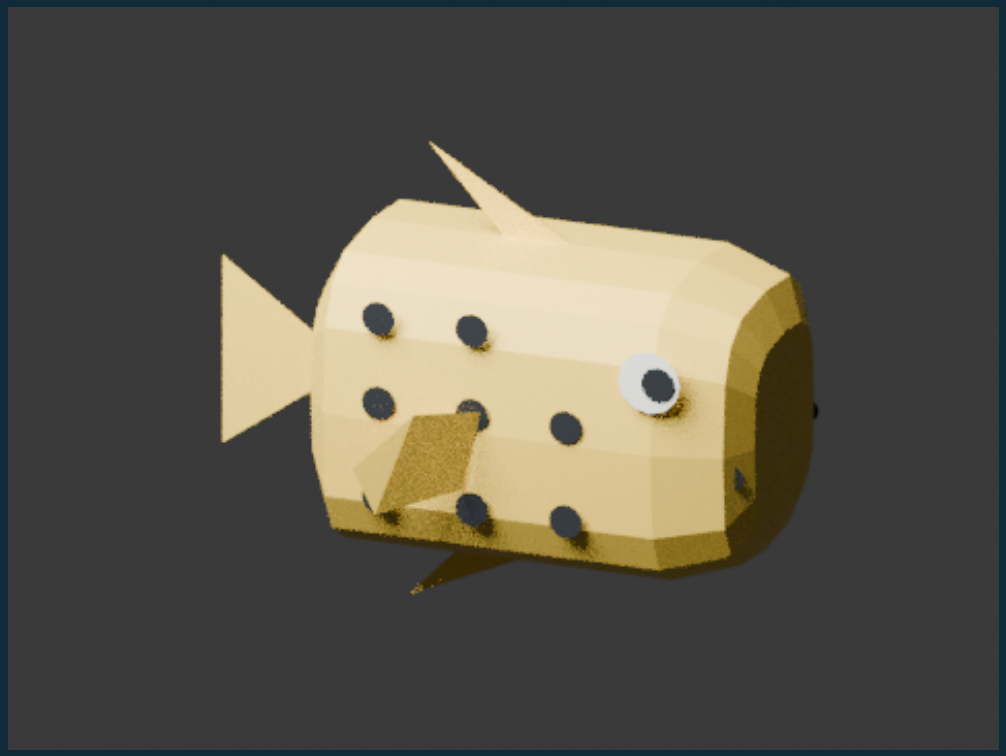
# Juvenile boxfish



## BEFORE



## AFTER



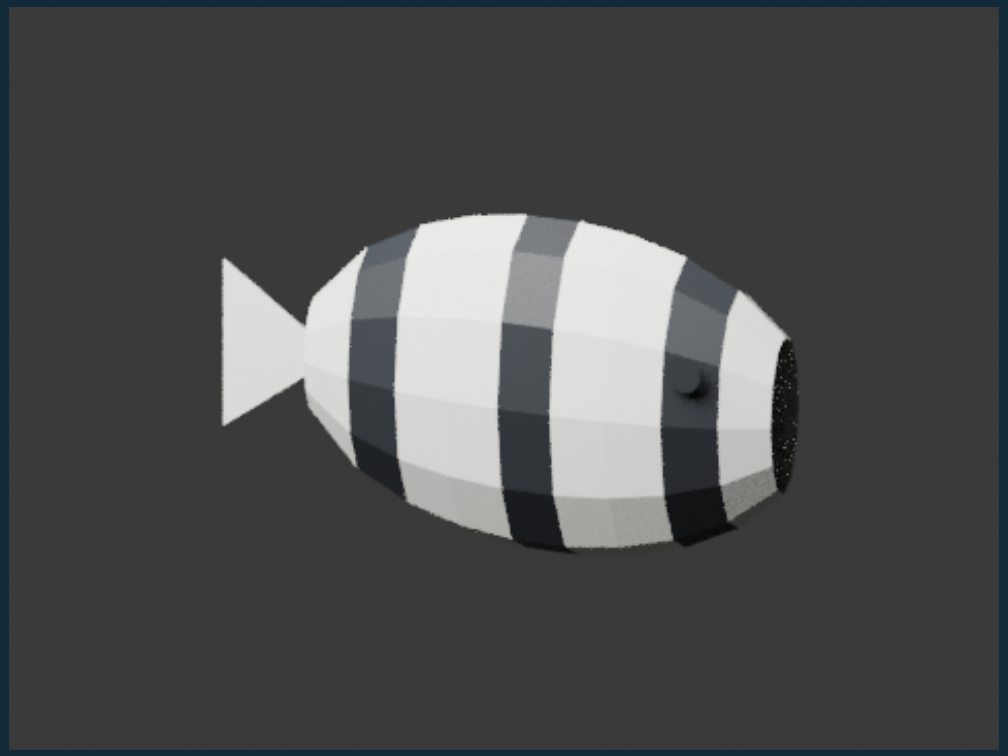
Added fuller pectoral and anal fins and a mouth detail.

Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Humbug damselfish



## BEFORE



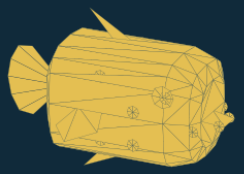
## AFTER

Added a forked tail and dorsal, anal and pectoral fins.

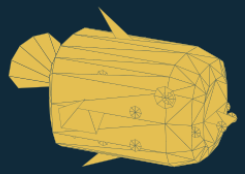
Before: source before the recorded refinement. After: source immediately after it. Blender model previews use simple color swatches, not the game atlas or in-tank lighting; framing is adjusted per model. No game rebuild was performed.

# Boxfish: independent fins

**BEFORE / NEUTRAL**



**AFTER / POSED**

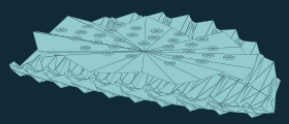


Neutral bind pose and an example fin pose.

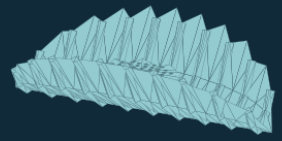
Exported mesh and reference deformation; plain swatches, not in-game textures or lighting.

# Giant clam: valves and mantle

**BEFORE / NEUTRAL**



**AFTER / POSED**



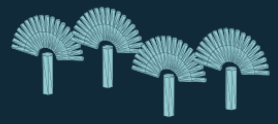
Neutral open pose and an example closed pose.

Exported mesh and reference deformation; plain swatches, not in-game textures or lighting.

# Feather dusters: new crowns

**BEFORE / NEUTRAL**

**AFTER / POSED**

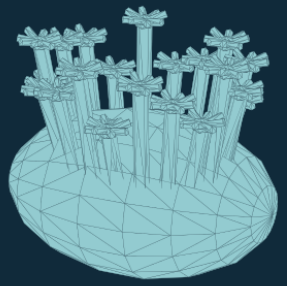


New crowns at rest and retracted; these animals were absent before this refinement.

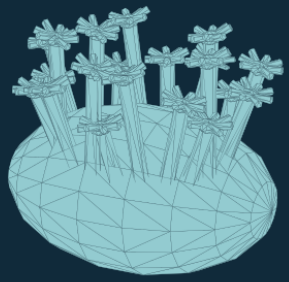
Exported mesh and reference deformation; plain swatches, not in-game textures or lighting.

# Flowerpot coral: polyp sway

**BEFORE / NEUTRAL**



**AFTER / POSED**

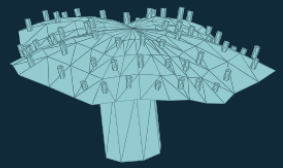


Neutral coral and an example of the polyps swaying.

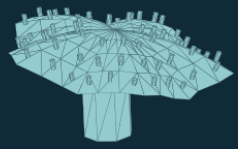
Exported mesh and reference deformation; plain swatches, not in-game textures or lighting.

# Leather coral: deflation

**BEFORE / NEUTRAL**



**AFTER / POSED**

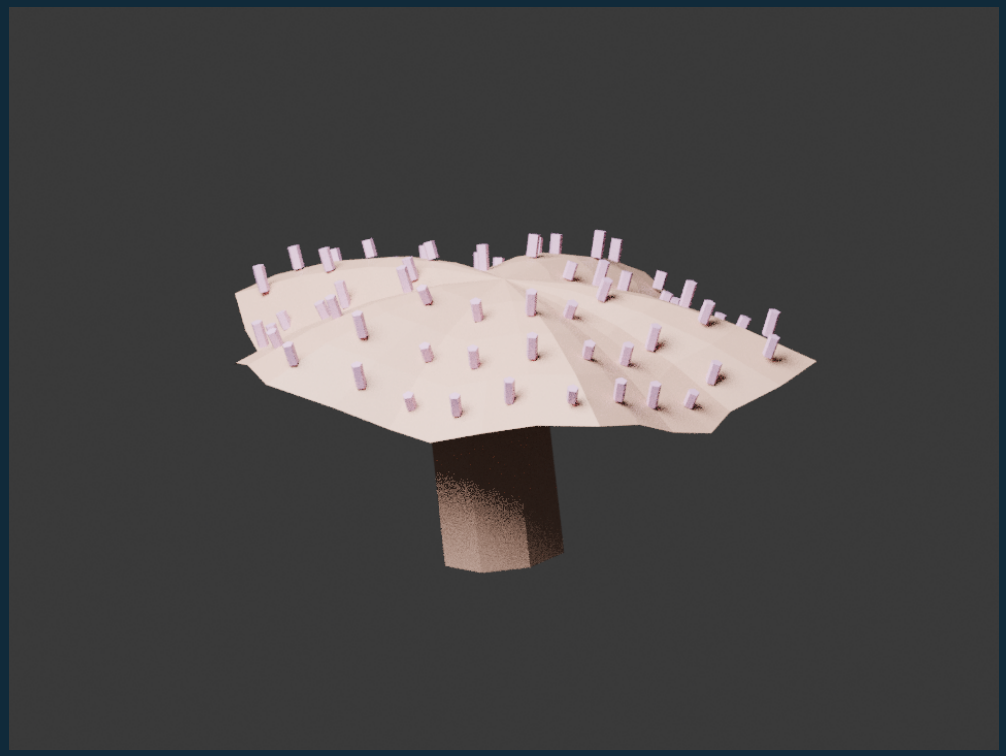


Neutral coral and an example deflated pose.

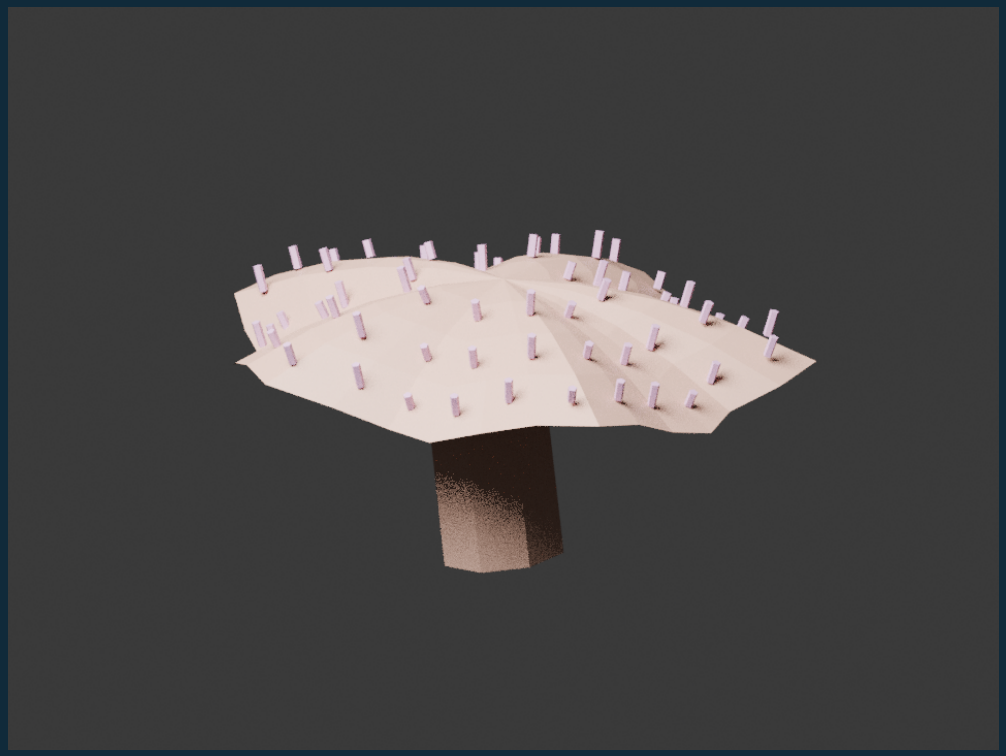
Exported mesh and reference deformation; plain swatches, not in-game textures or lighting.

# Leather coral: restore original polyp detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

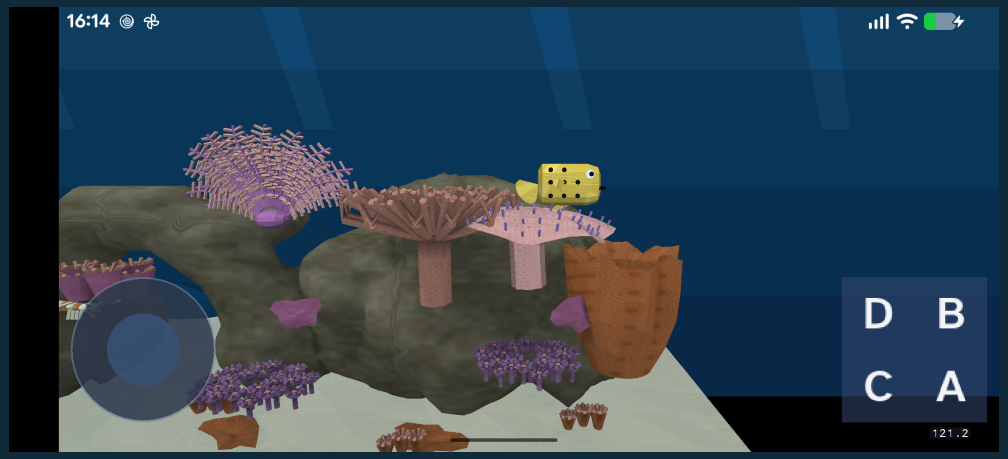


Restored the recorded original 0.010 polyp radius from the export workaround (0.013).

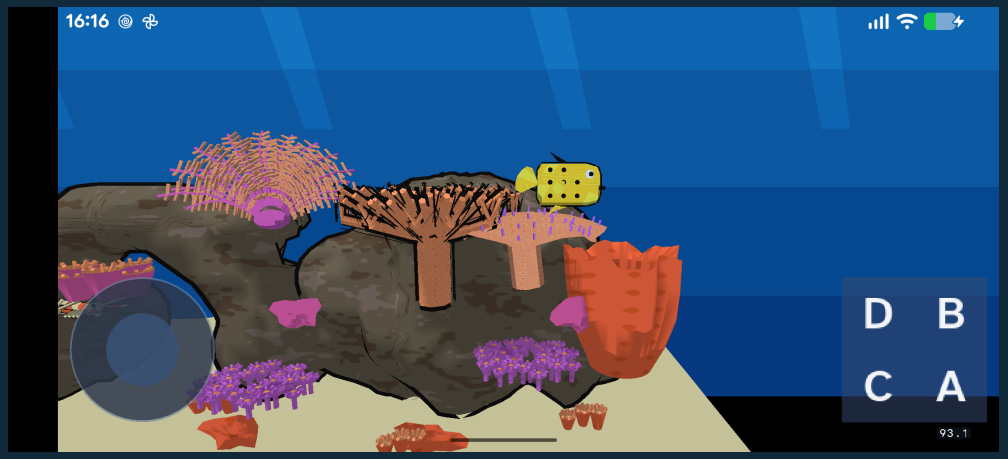
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Leather coral: in-game close-up review

## REALISTIC




## ANIME



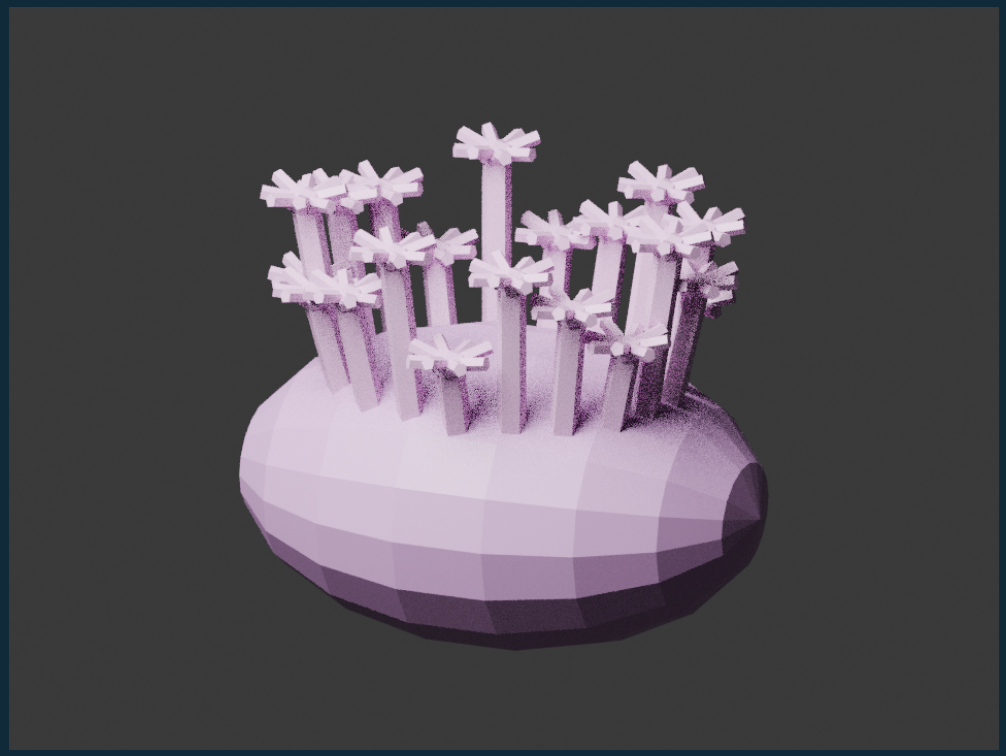

Pixel captures use temporary camera/player placement with the rebuilt production mesh and textures.

The slim purple polyps are visible on the pink leather coral; its neighbors and the boxfish overlap the view.

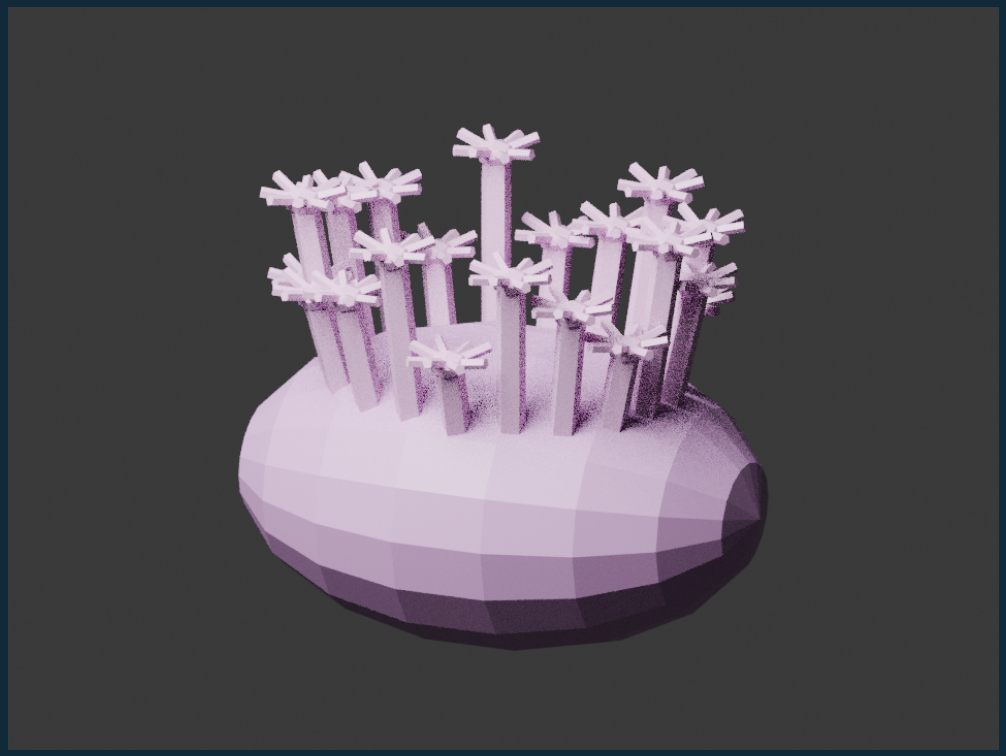
These are after captures in the two styles; the previous page shows the source-mesh before/after comparison.

# Flowerpot coral: restore original tentacle detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

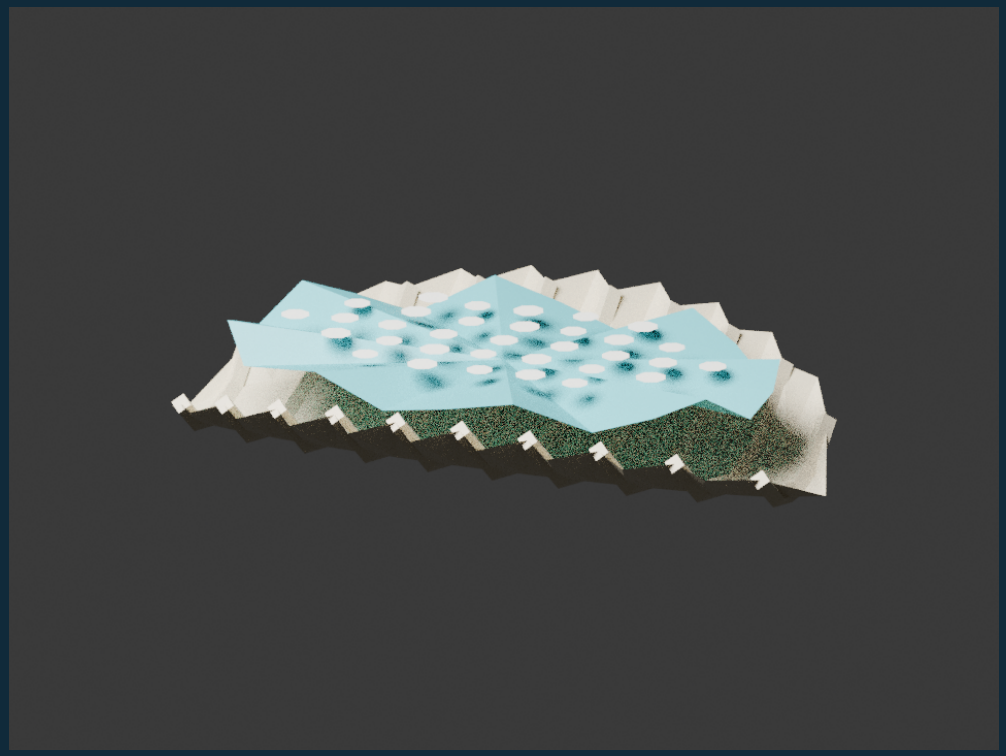


Restored the recorded original 0.010 tentacle radius from the export workaround (0.013).

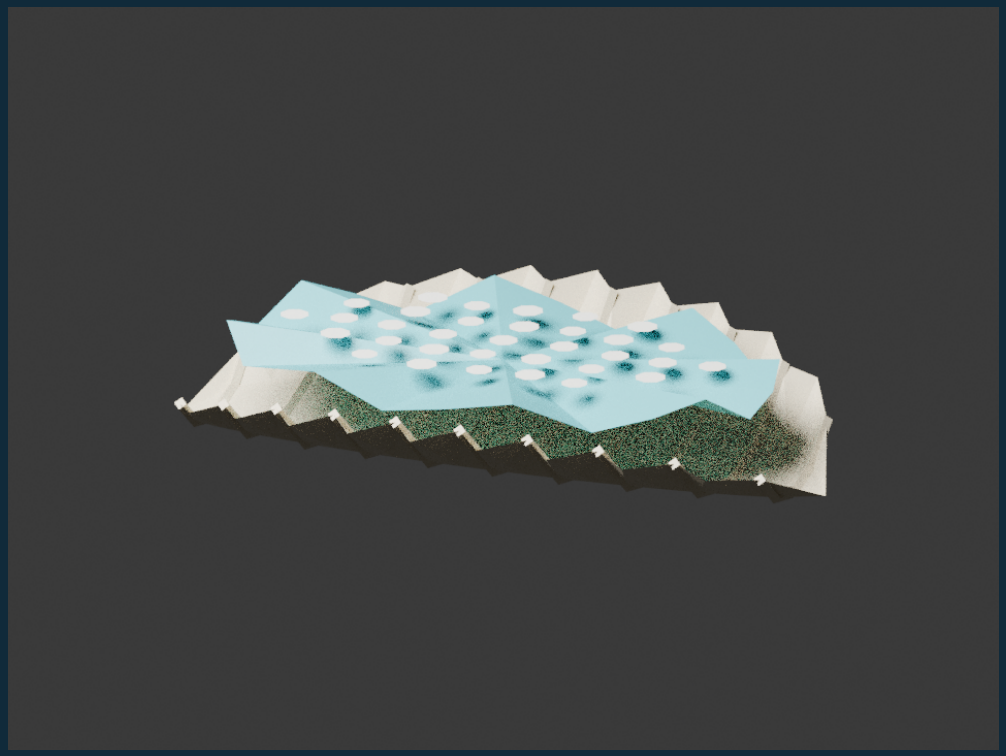
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Giant clam: restore original shell ribs

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL



Restored the recorded original 0.008 rib radius from the export workaround (0.013).

Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Mushroom coral: restore original detail

## BEFORE: THICKENED

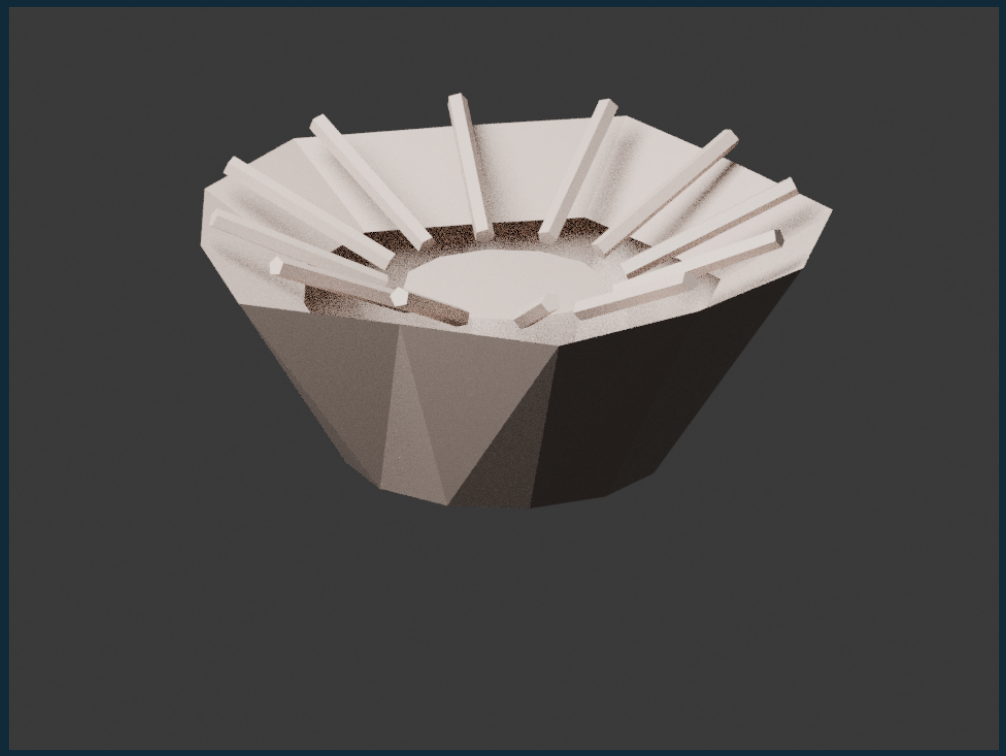
## AFTER: ORIGINAL DETAIL

Restored the recorded original .010 detail radius from the export workaround (0.013).

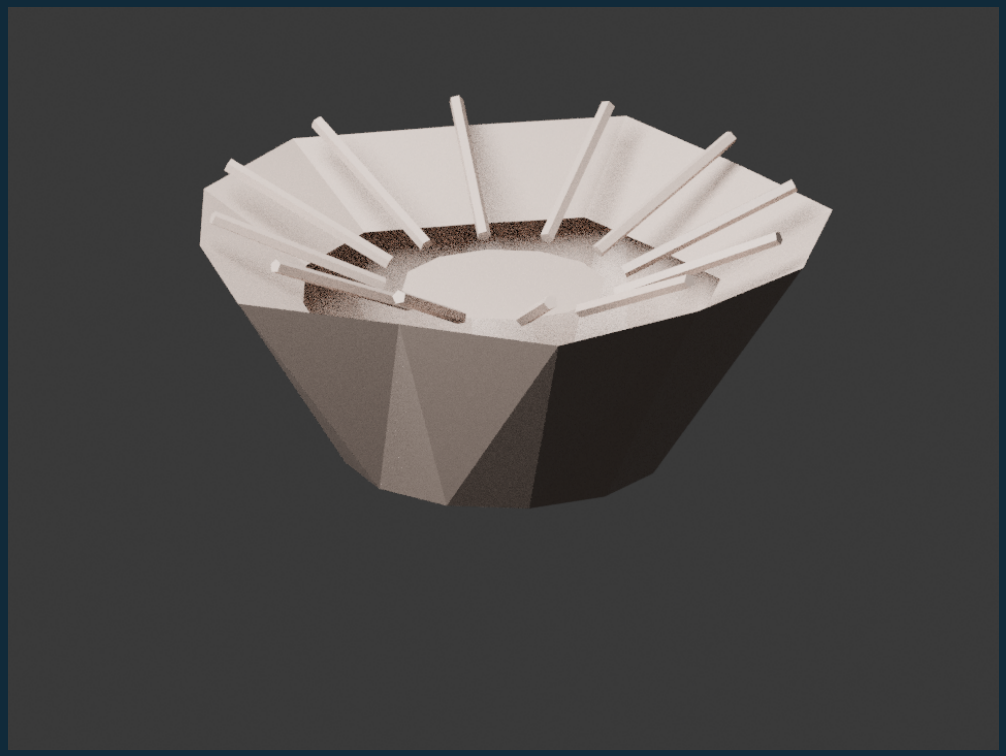
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Cup coral: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

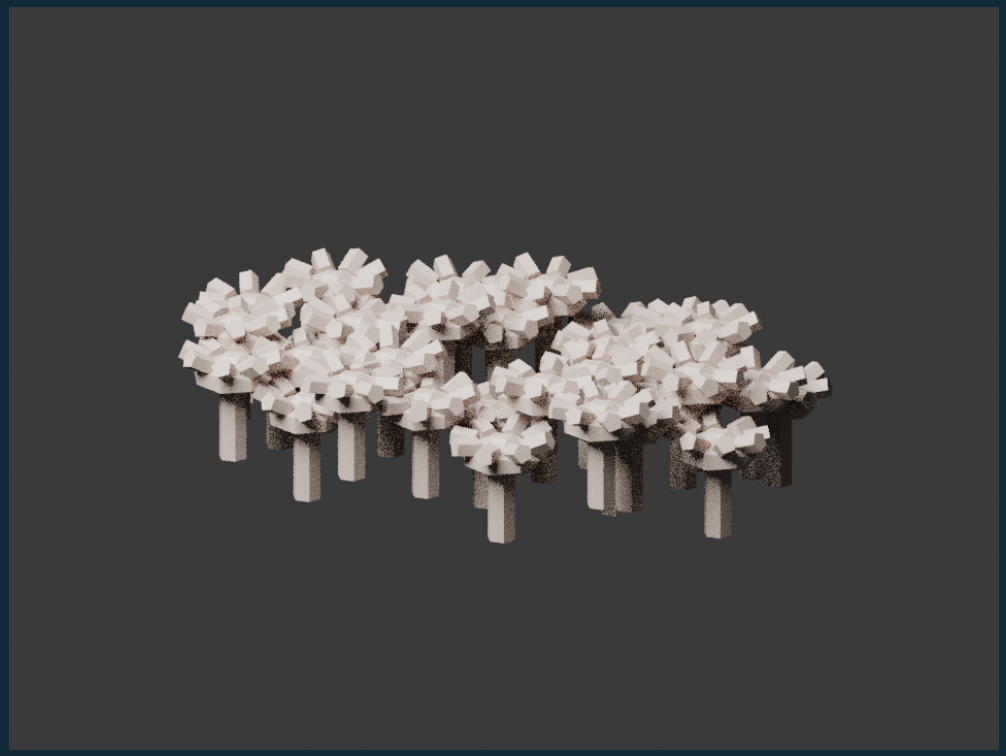


Restored the recorded original .009 detail radius from the export workaround (0.013).

Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Zoanthids: restore original detail

## BEFORE: THICKENED



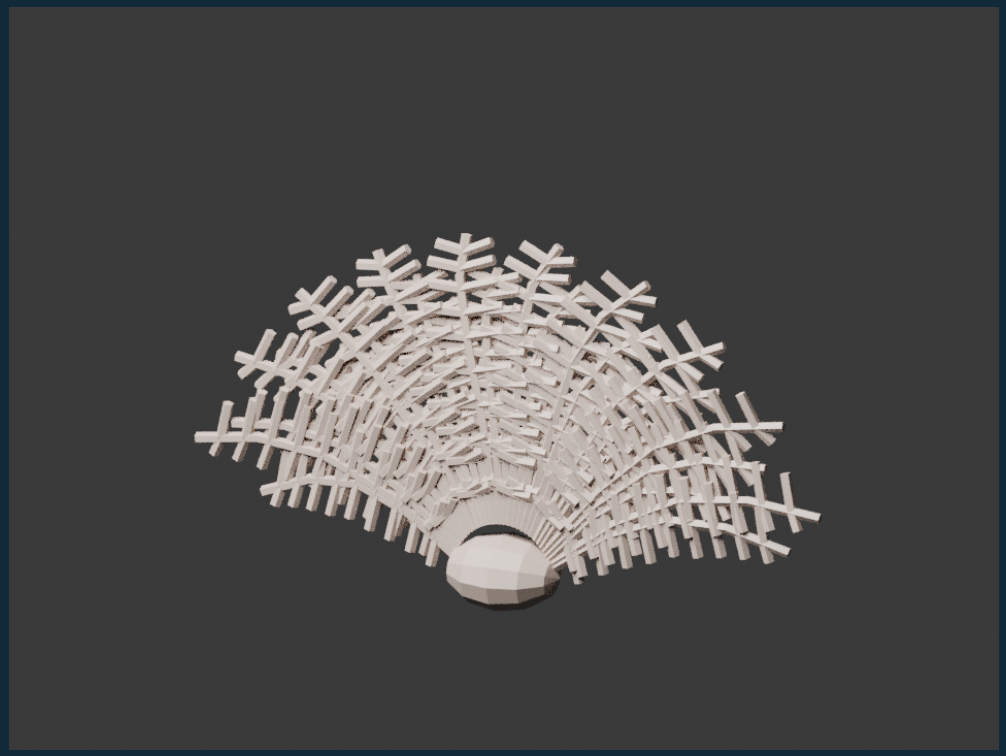
## AFTER: ORIGINAL DETAIL

Restored the recorded original .009 detail radius from the export workaround (0.013).

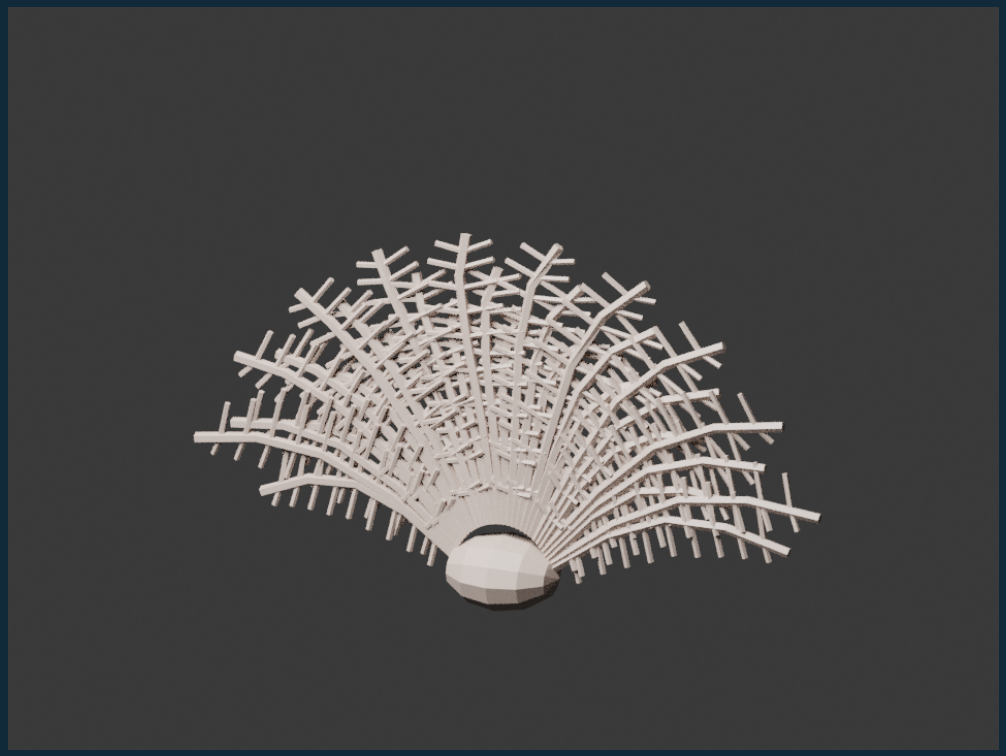
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Feather star: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

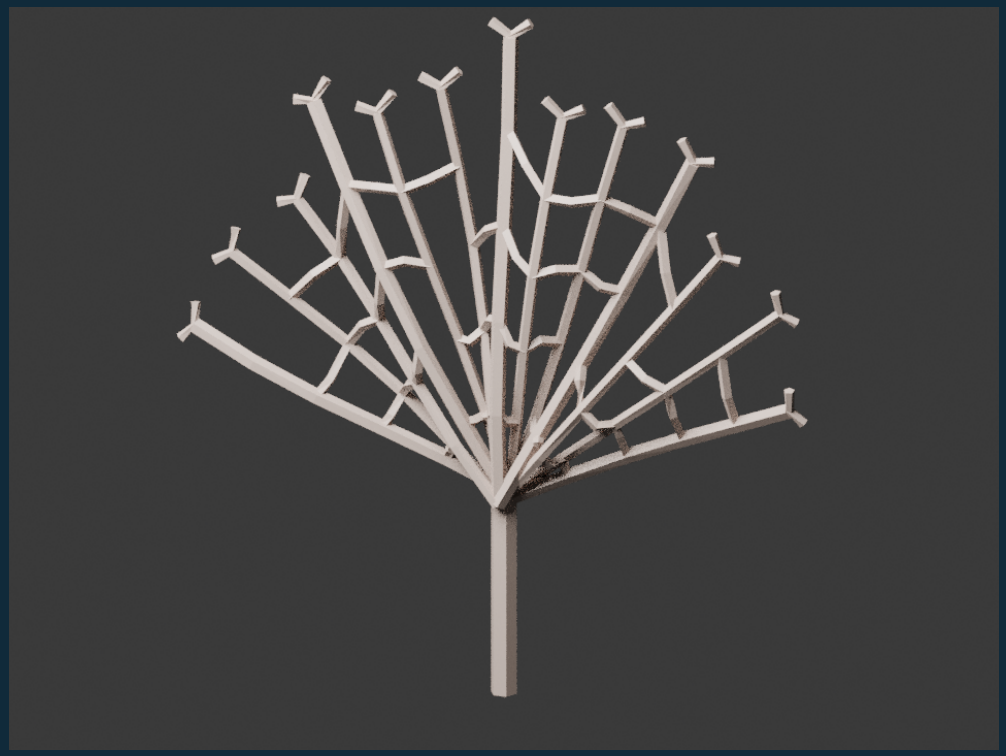


Restored the recorded original .007 detail radius from the export workaround (0.013).

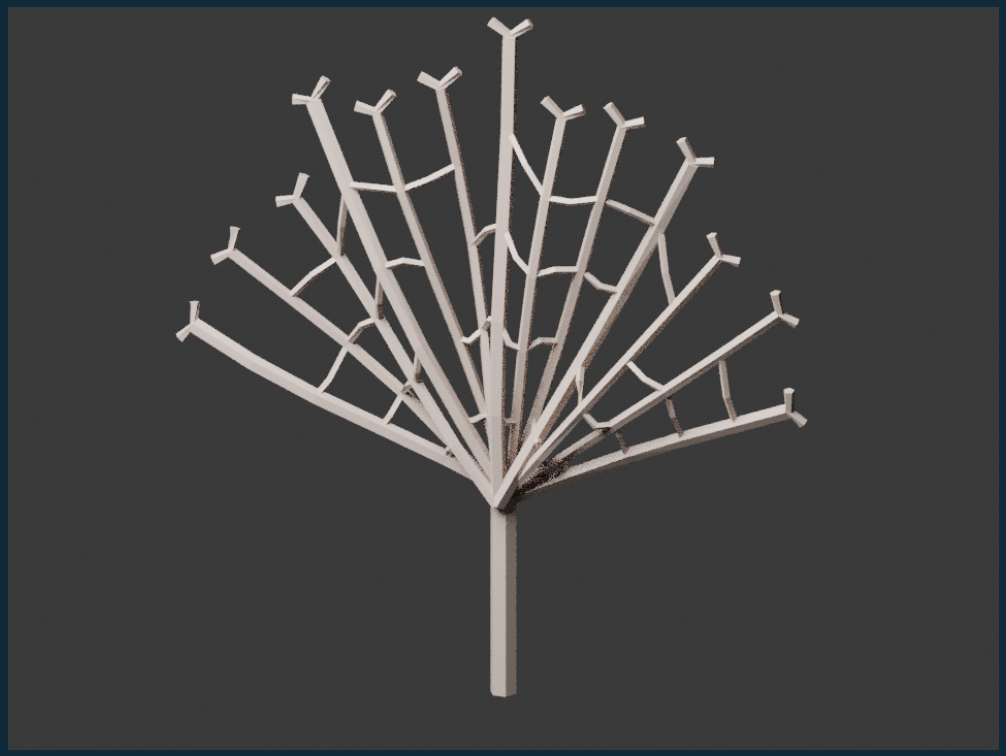
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Sea fan cross branches: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

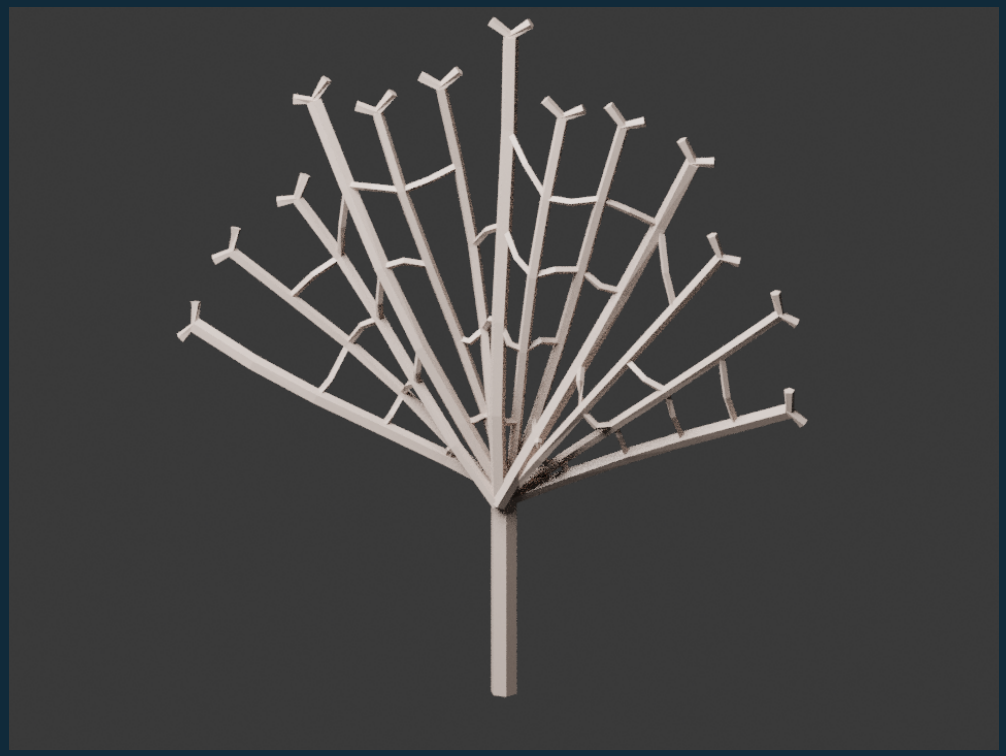


Restored the recorded original .009 recorded dimensions.

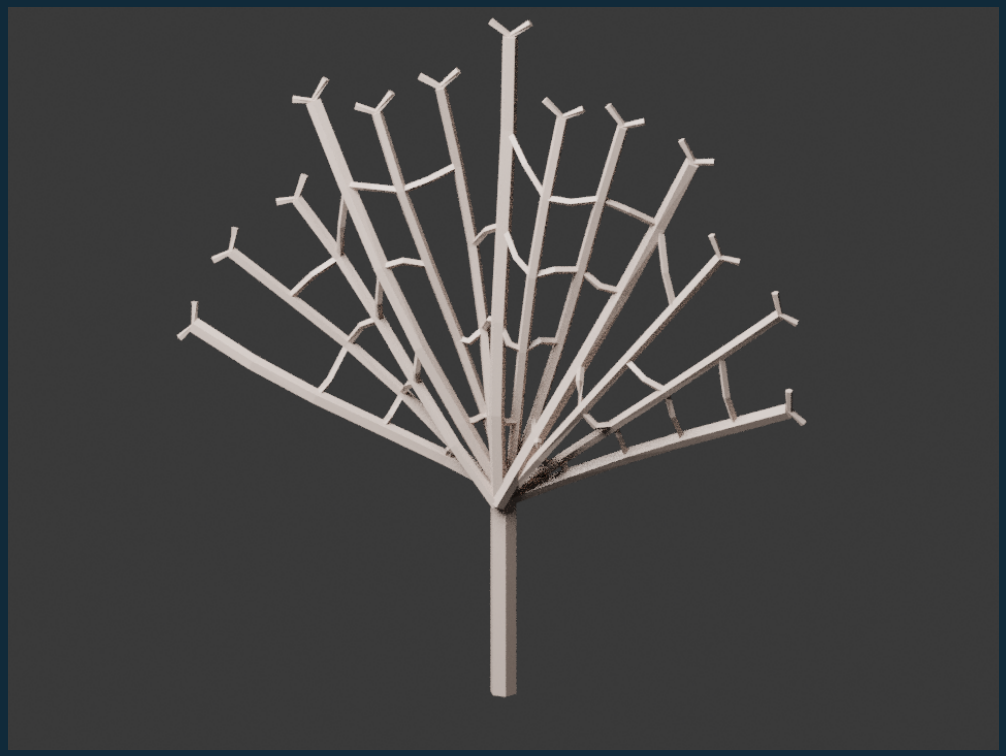
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Sea fan split branches: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

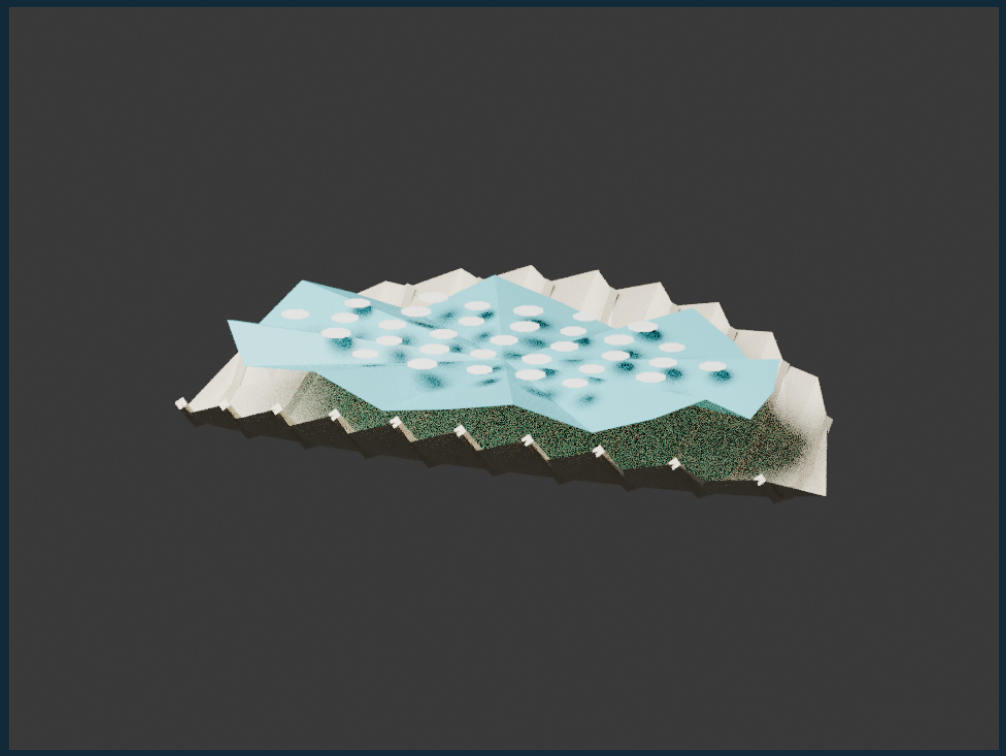


Restored the recorded original .009 recorded dimensions.

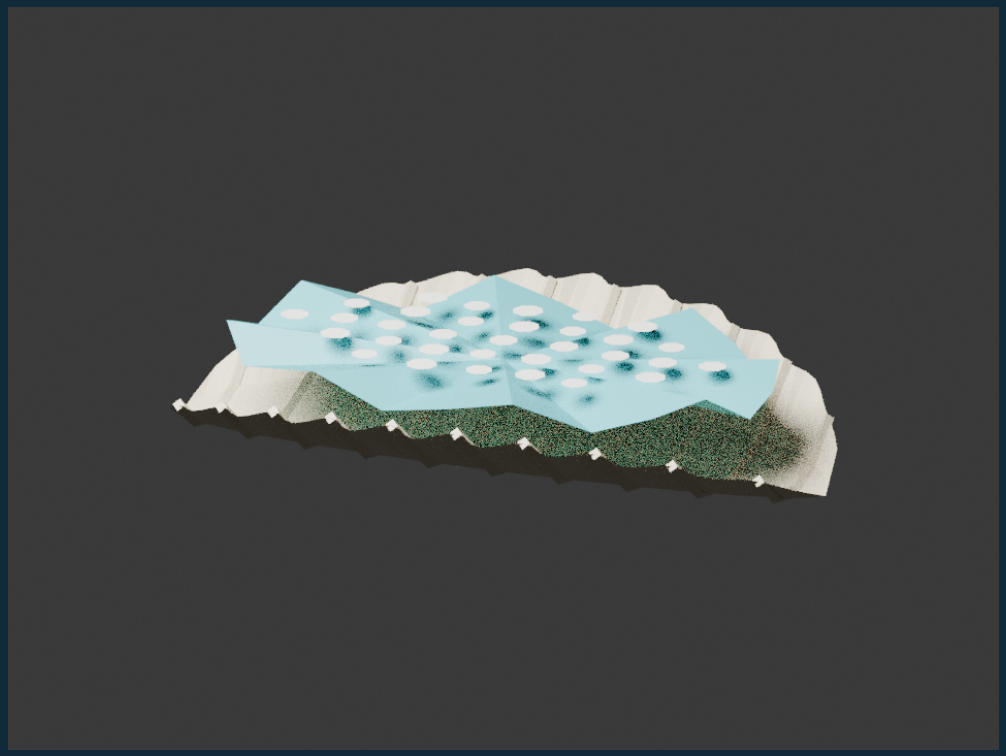
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Giant clam: soften shell silhouette

## BEFORE: SHARP SHELL



## AFTER: SOFTER SHELL

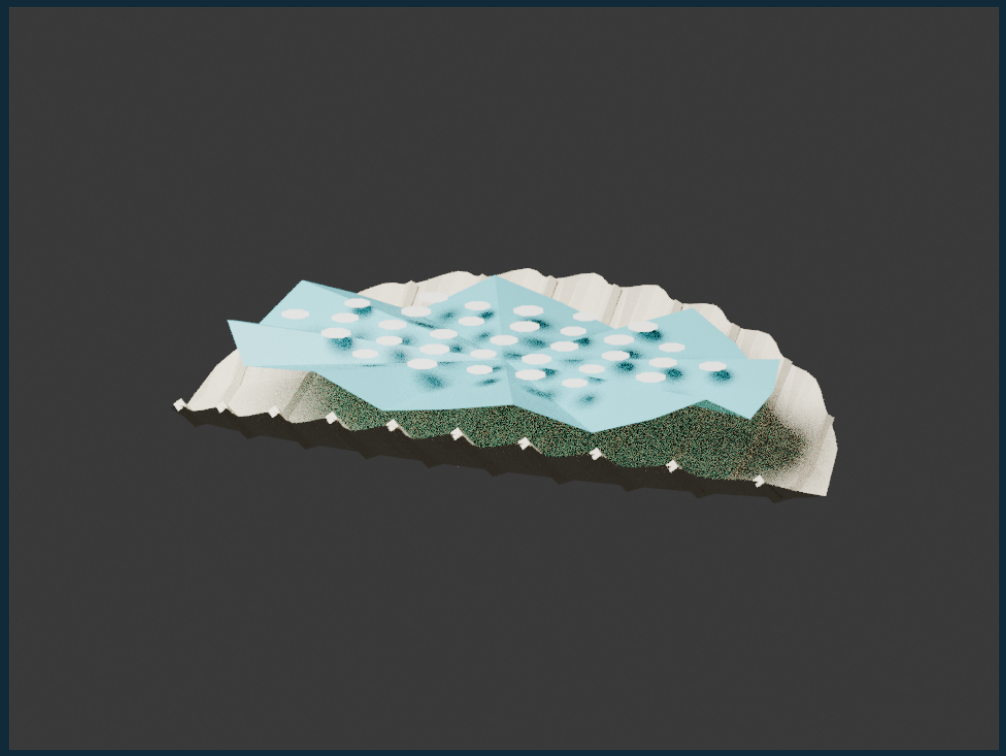


Added shell contour samples and softened alternating scallops; kept ten ribs per valve.

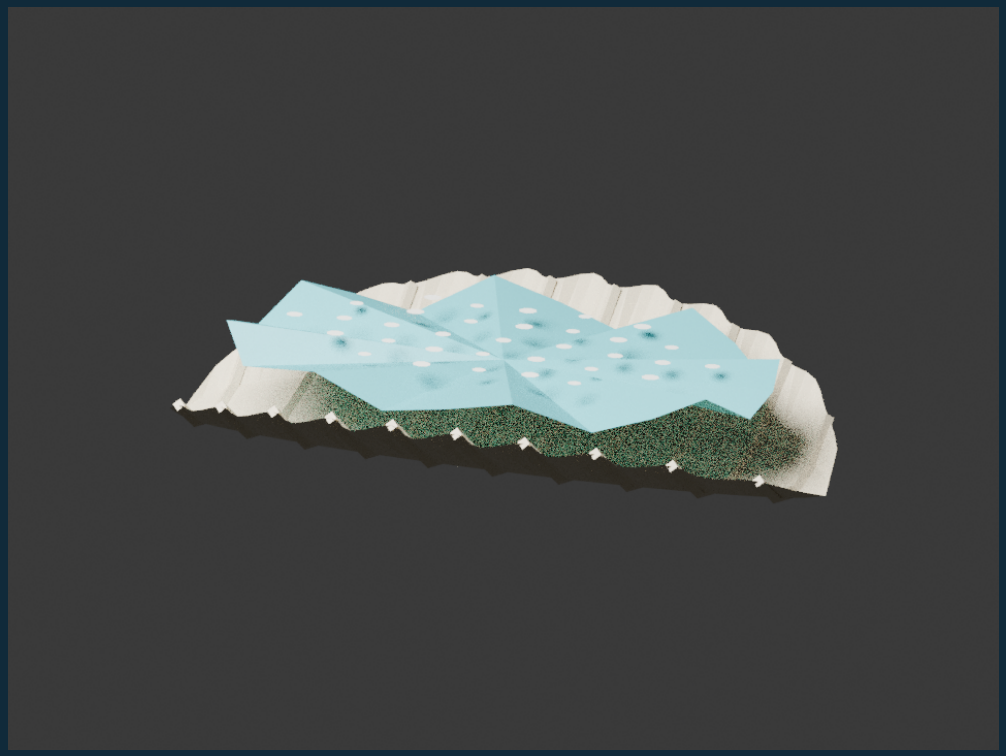
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Giant clam mantle: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

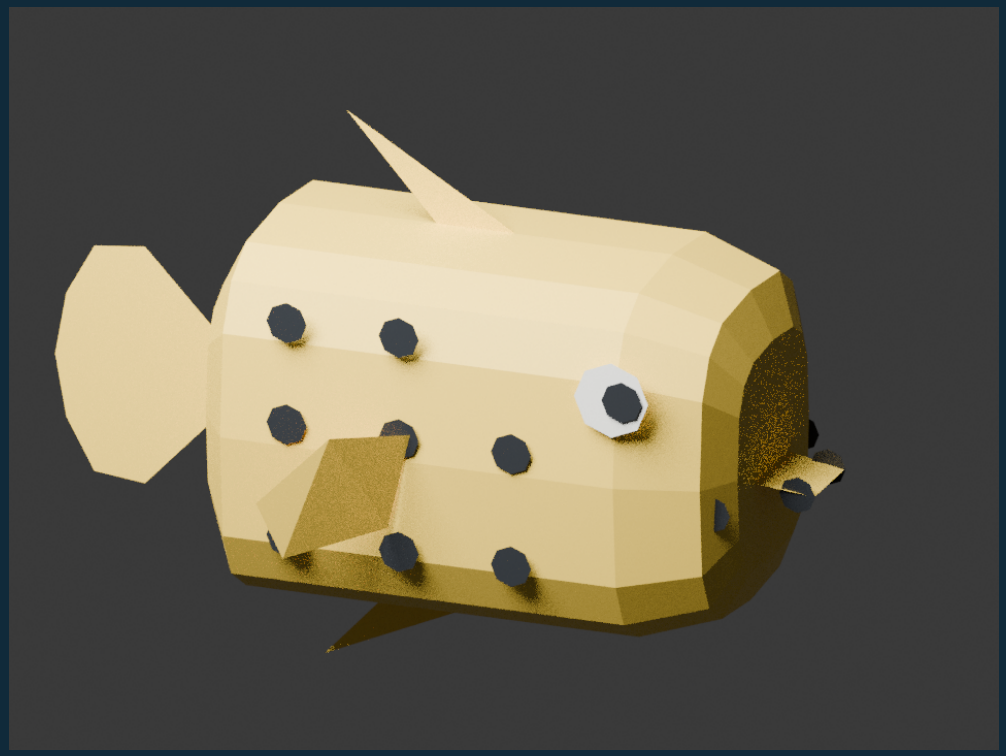


Restored the recorded dimensions and geometry from before the export workaround.

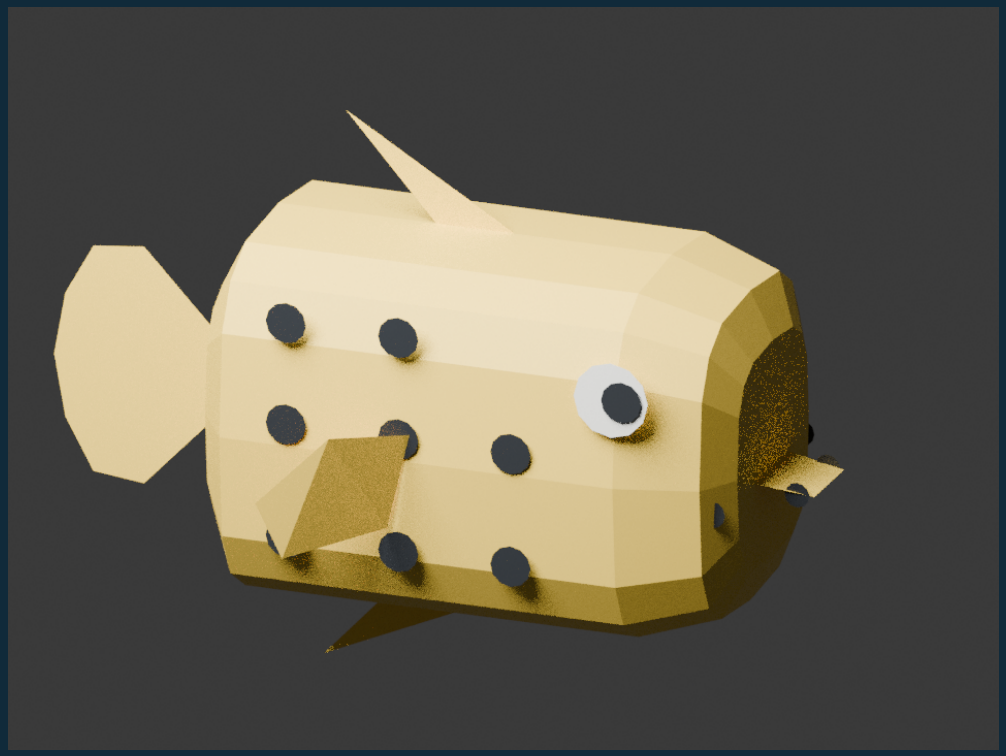
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Boxfish markings: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

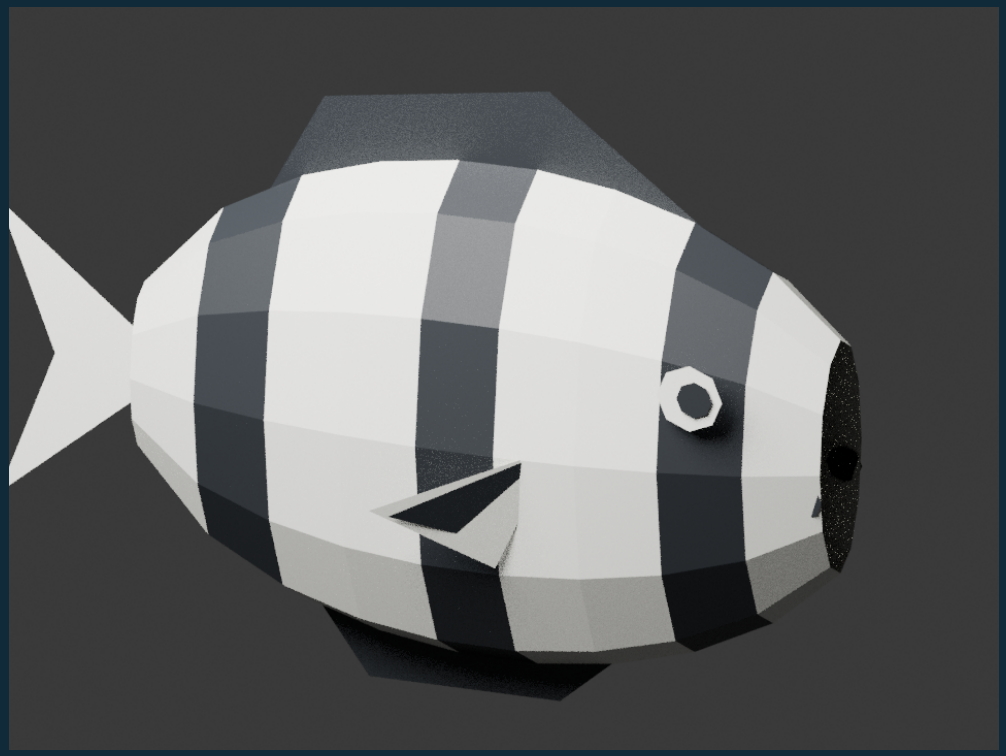


Restored the original marking sizes and polygon shapes before the export workaround.

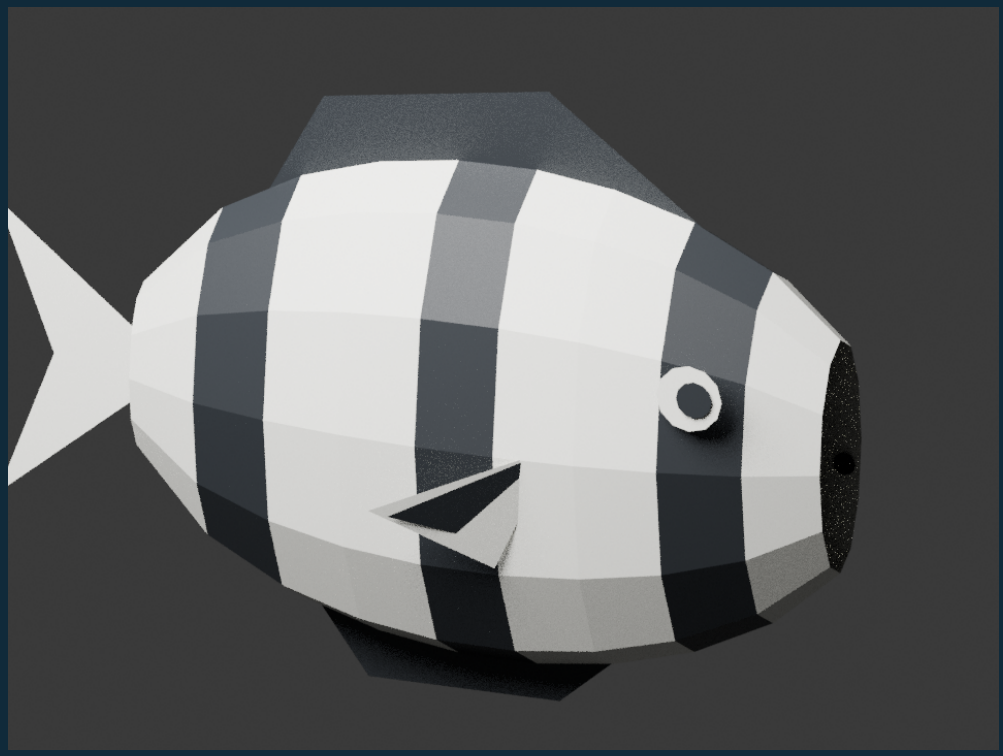
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Humbug markings: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL



Restored the original marking sizes and polygon shapes before the export workaround.

Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Guard crab eyes: restore original detail

## BEFORE: THICKENED

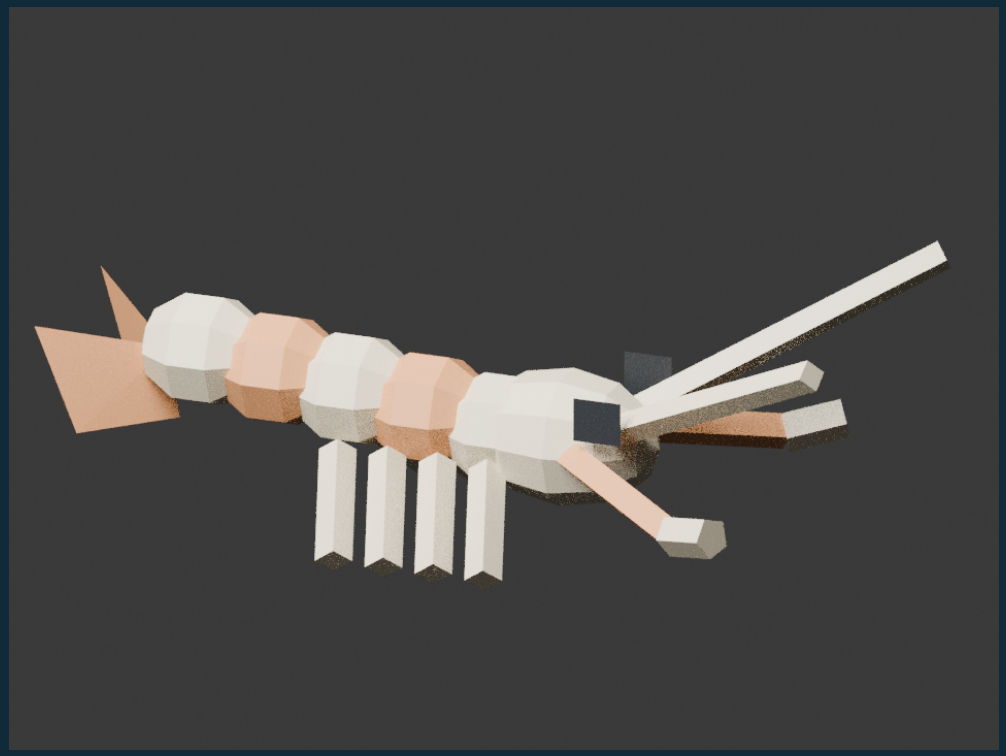
## AFTER: ORIGINAL DETAIL

Restored the recorded dimensions and geometry from before the export workaround.

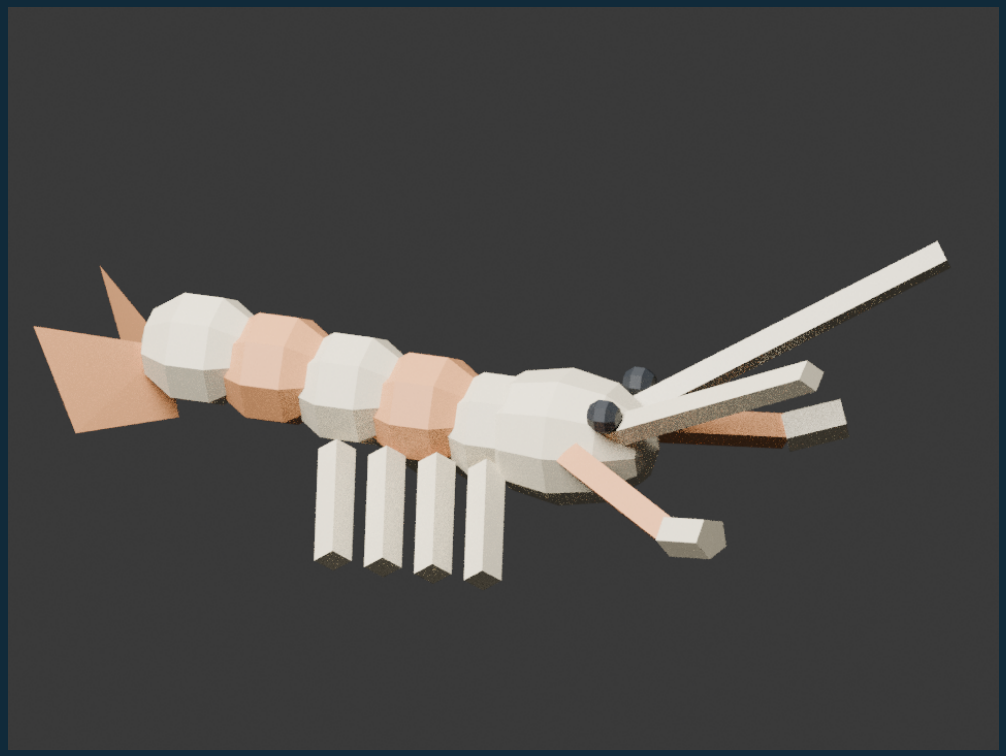
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Shrimp eyes: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL



Restored the recorded dimensions and geometry from before the export workaround.

Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Guard crab claws: restore original detail

## BEFORE: THICKENED

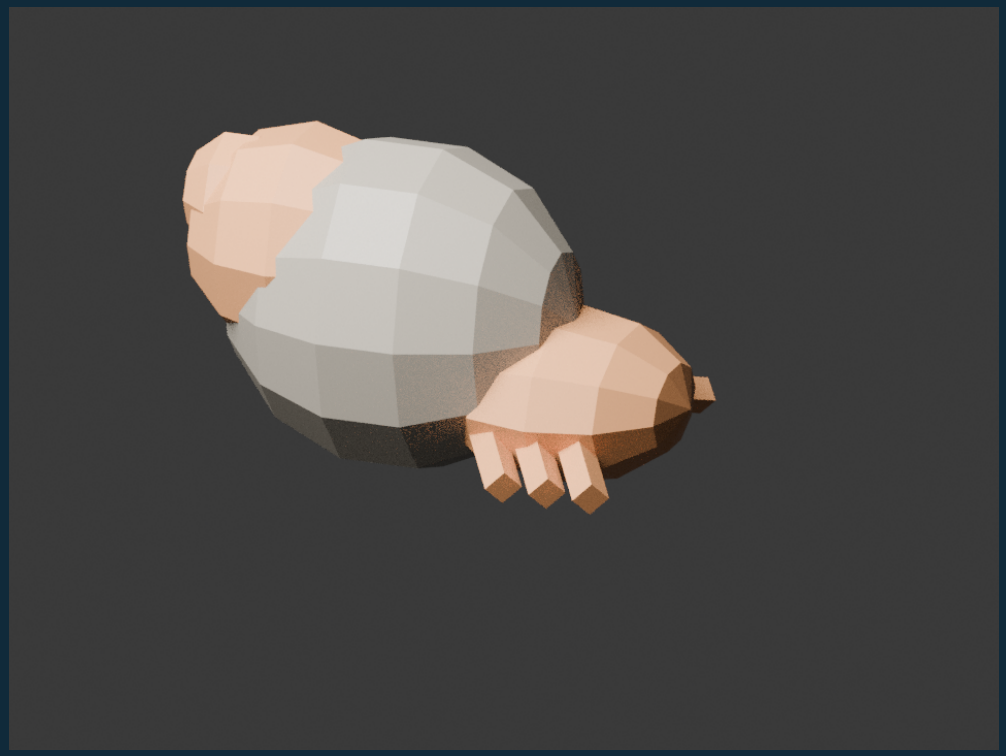
## AFTER: ORIGINAL DETAIL

Restored the recorded dimensions and geometry from before the export workaround.

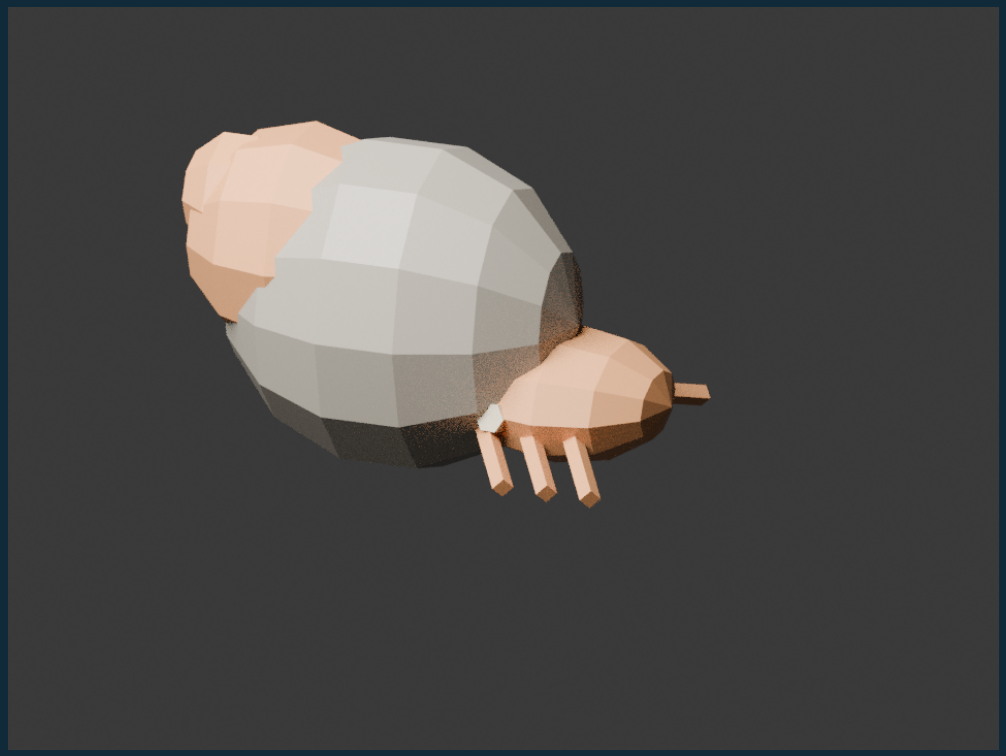
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Hermit crab: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL

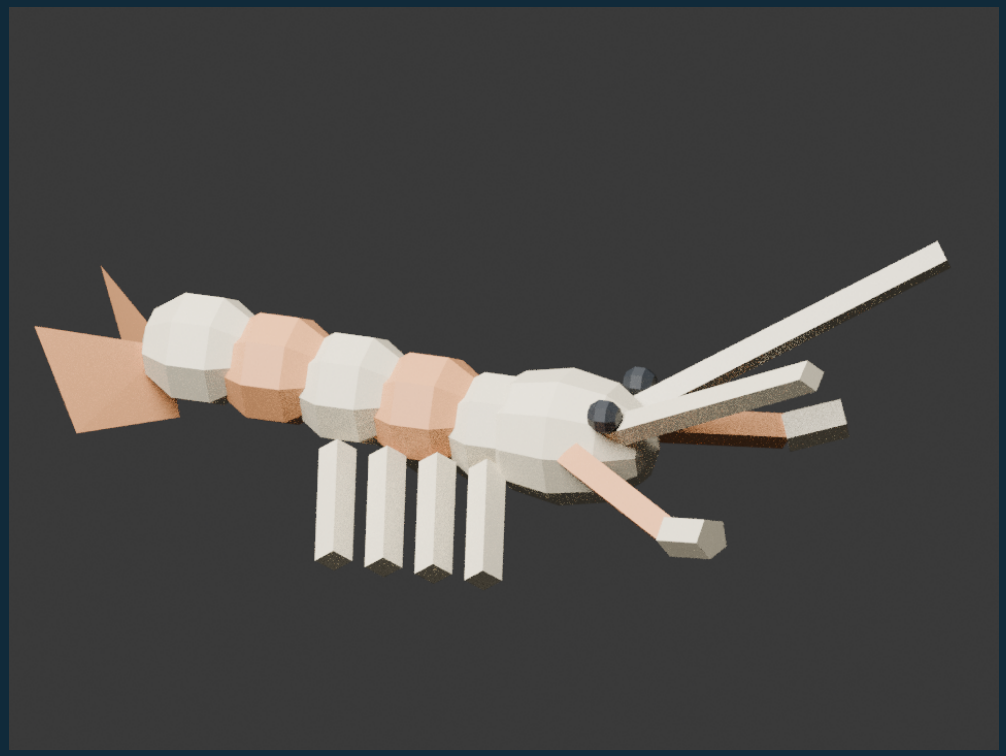


Restored the recorded dimensions and geometry from before the export workaround.

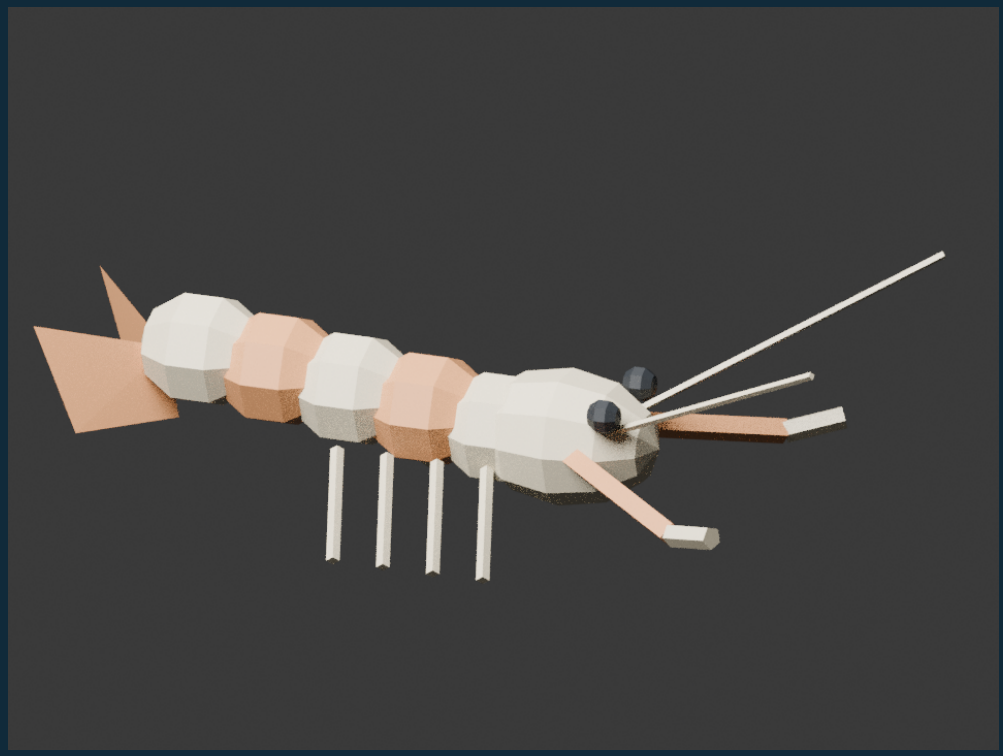
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Shrimp legs and antennae: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL



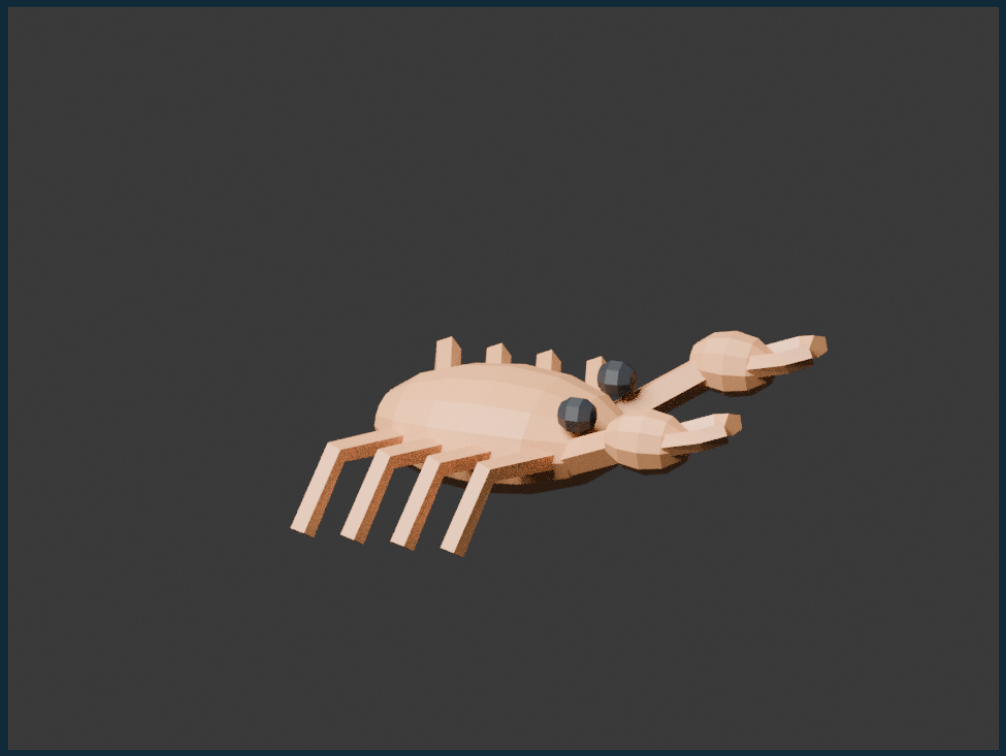
Restored the recorded dimensions and geometry from before the export workaround.

Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Guard crab legs and pincers: restore original detail

## BEFORE: THICKENED

## AFTER: ORIGINAL DETAIL

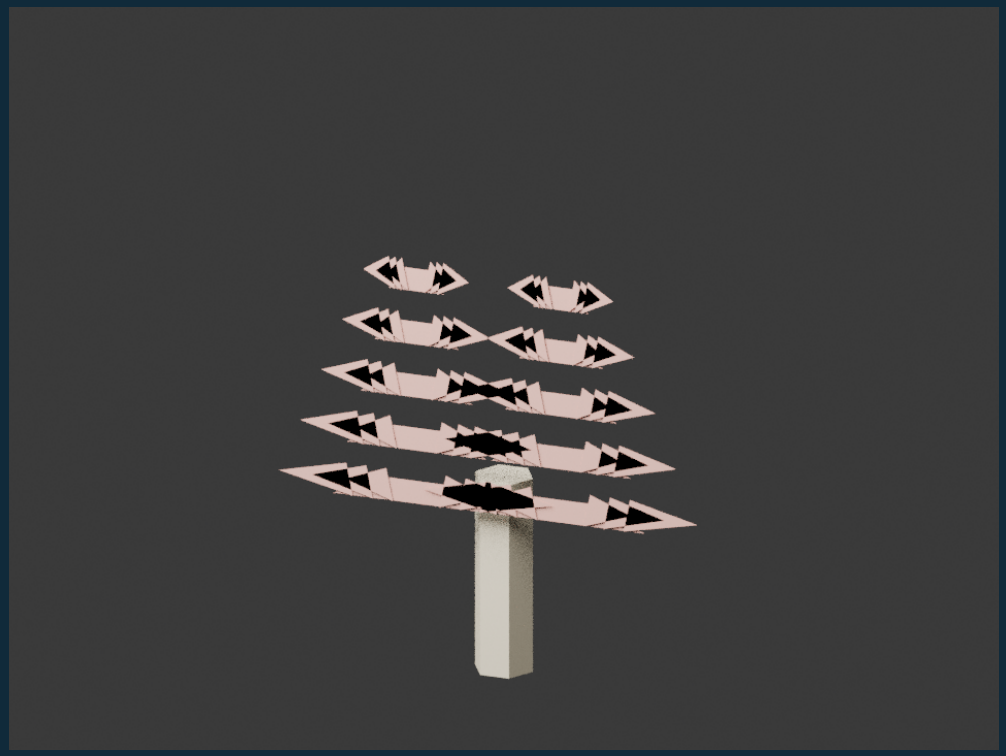


Restored the recorded dimensions and geometry from before the export workaround.

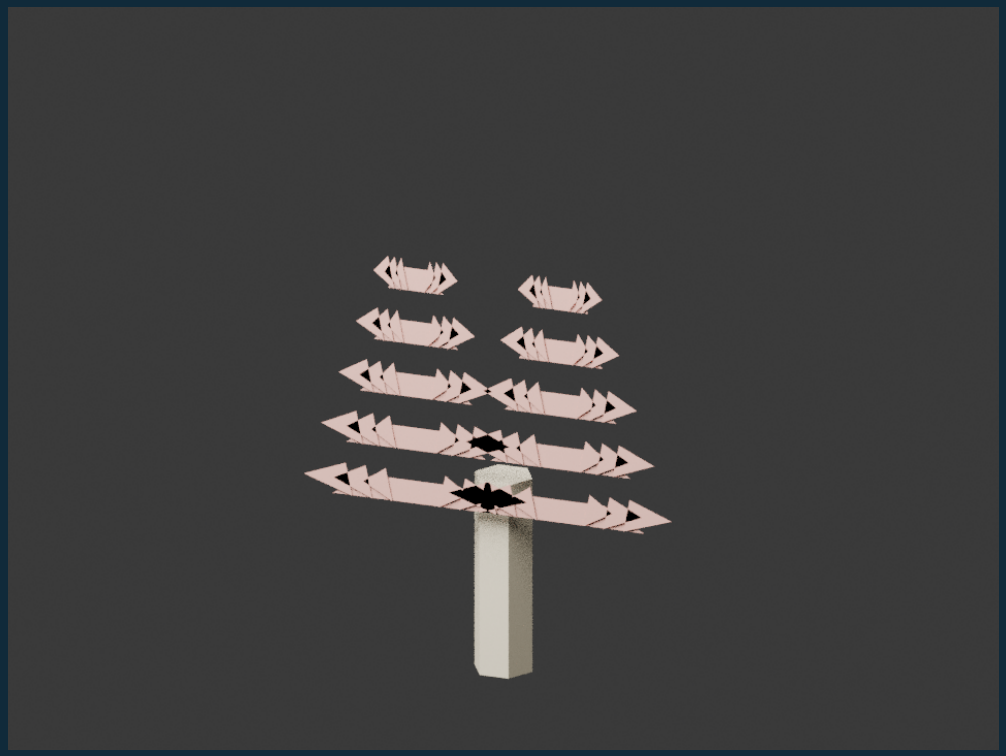
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Worm radioles: restore original detail

## BEFORE: THICKENED



## AFTER: ORIGINAL DETAIL



Restored the recorded original original recorded dimensions.

Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Final restoration: Pixel textured close-up review — flowerpot

**REALISTIC**

**ANIME**

Diagnostic camera; restored crowns are visible. Faceting and texture stretching remain.

# Final restoration: Pixel textured close-up review — closed clam

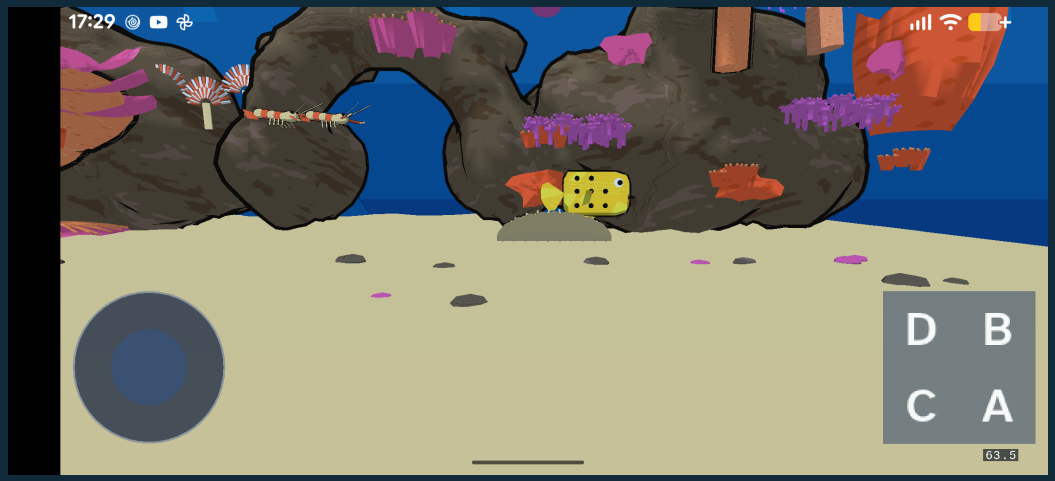
**REALISTIC: BEFORE**

**REALISTIC: AFTER**

The revised shell edge is smoother; shell shading remains flat and gray.

# Final restoration: Pixel textured close-up review — closed clam, anime

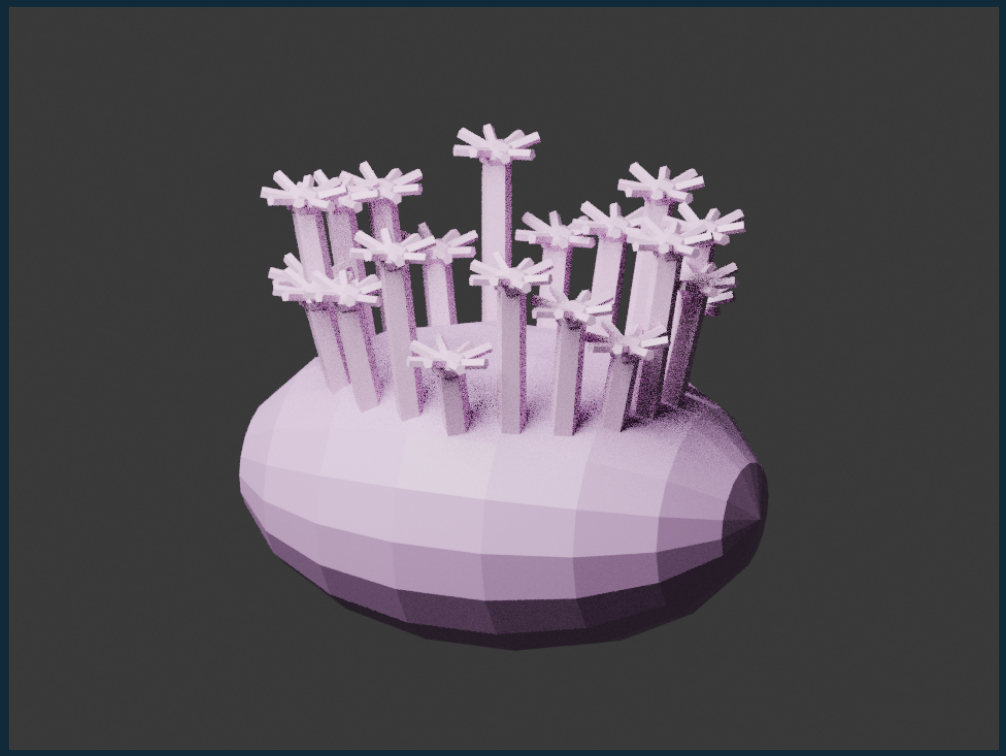
**ANIME: AFTER**



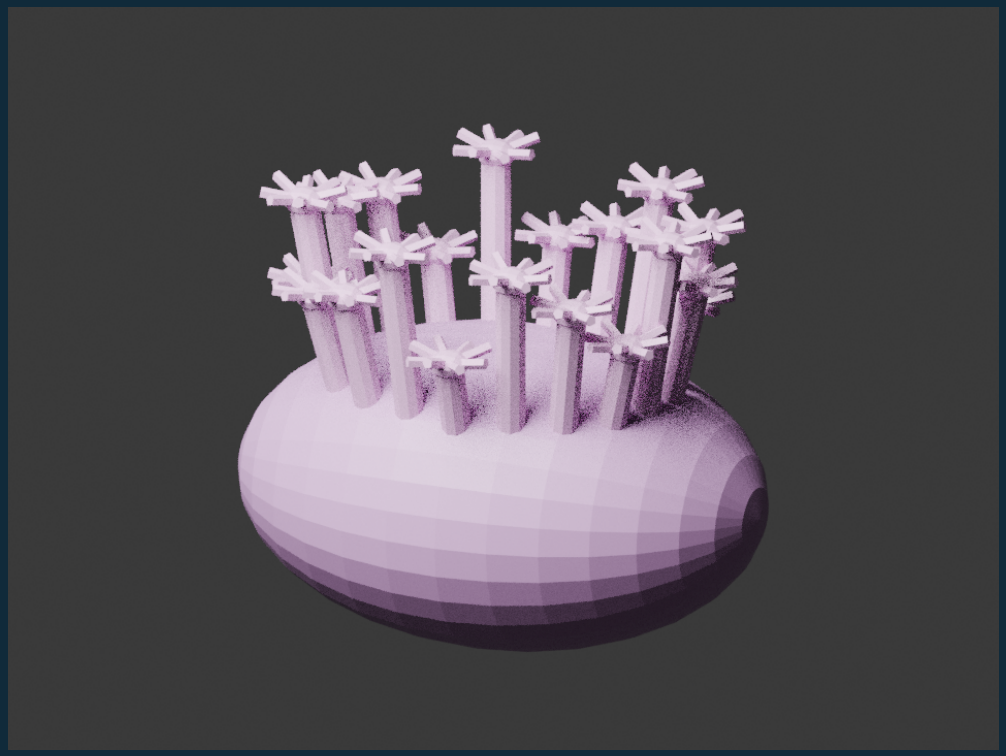

Diagnostic fish placement triggers closure; no matching earlier anime close-up is available.

# Flowerpot coral: refine base and stalk surfaces

## BEFORE: COARSE SURFACE



## AFTER: FINER SURFACE

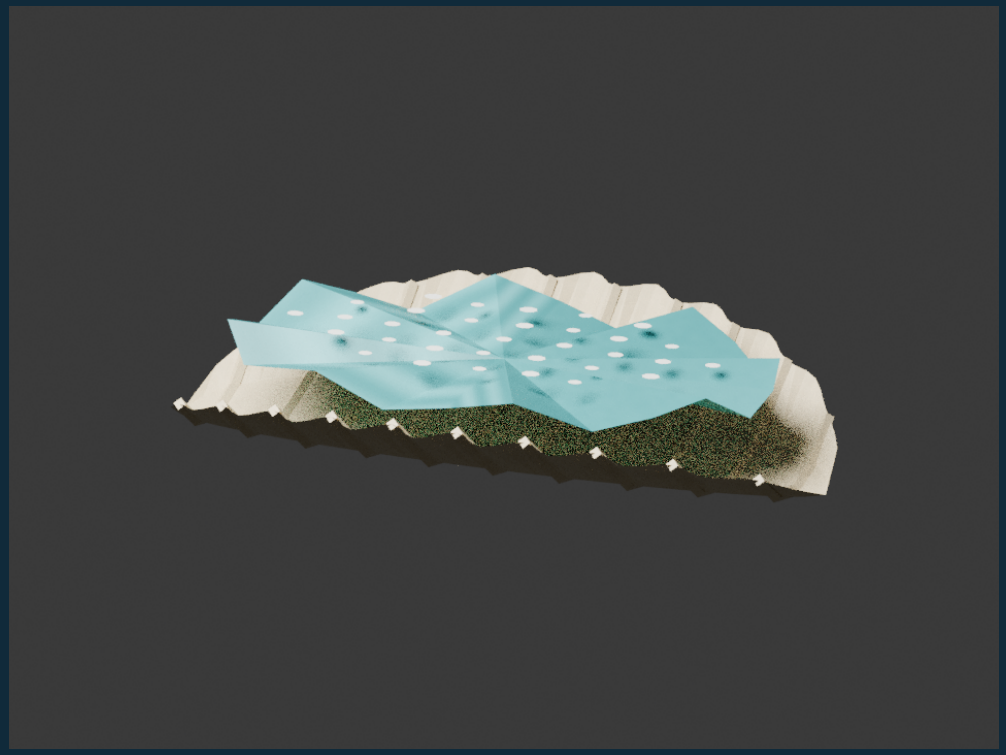


Increased base and stalk segments; preserved the original tentacle dimensions and rig.

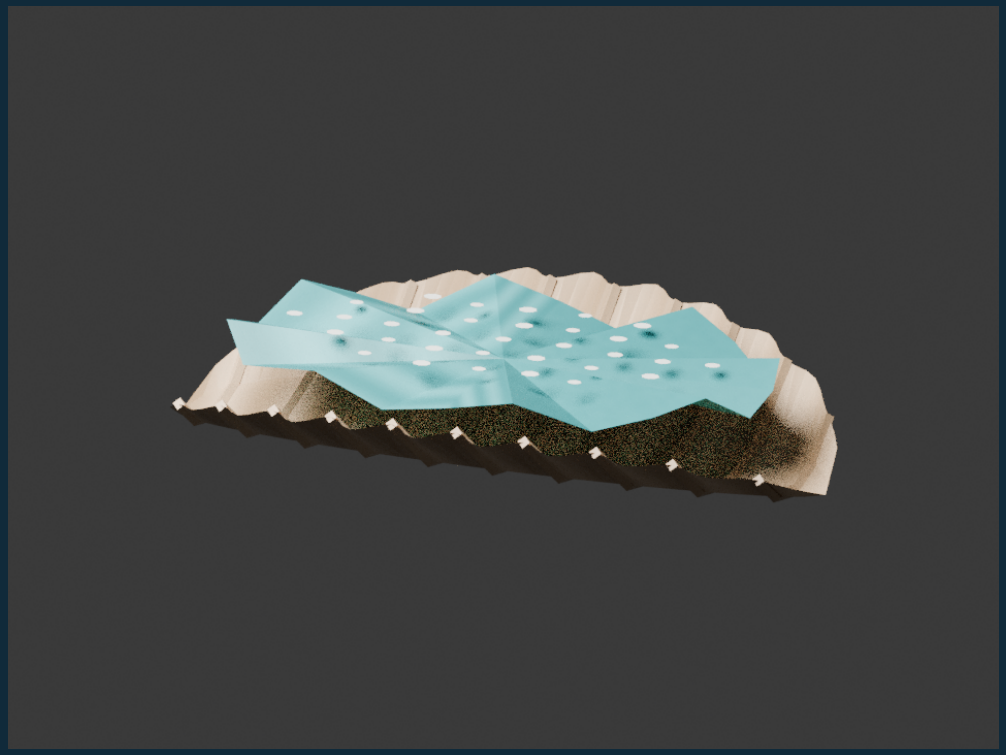
Source-mesh comparison with simple swatches; textured, lit in-game review remains pending.

# Giant clam: dedicated shell texture

## BEFORE: SAND TEXTURE



## AFTER: SHELL TEXTURE

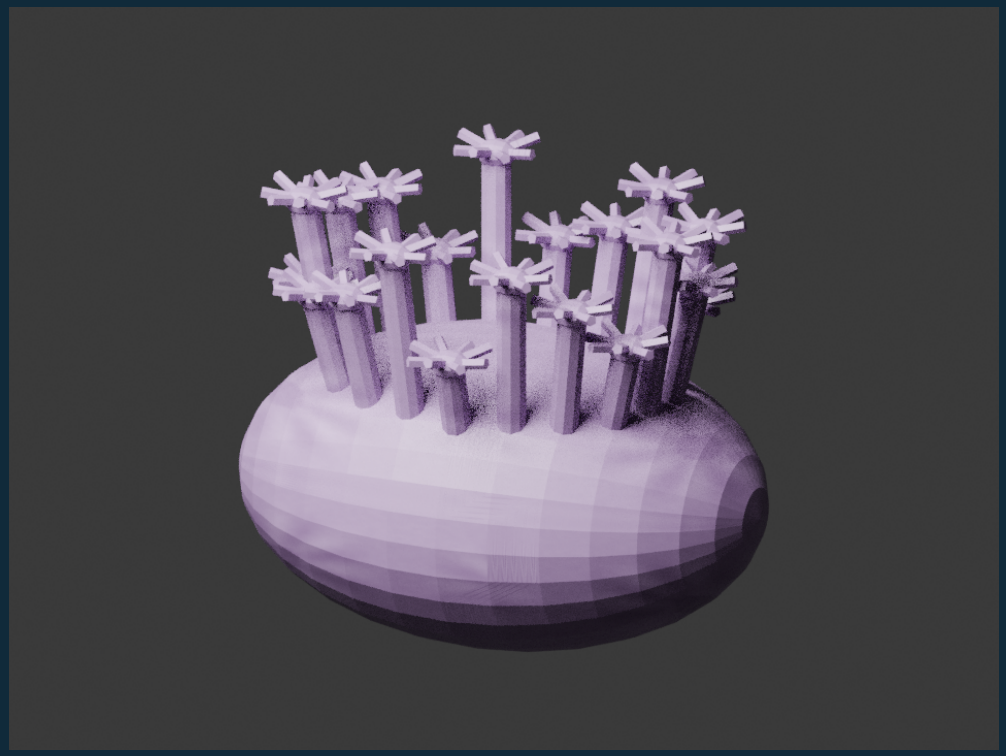


Replaced the generic sand mapping with a dedicated, ribbed shell texture.

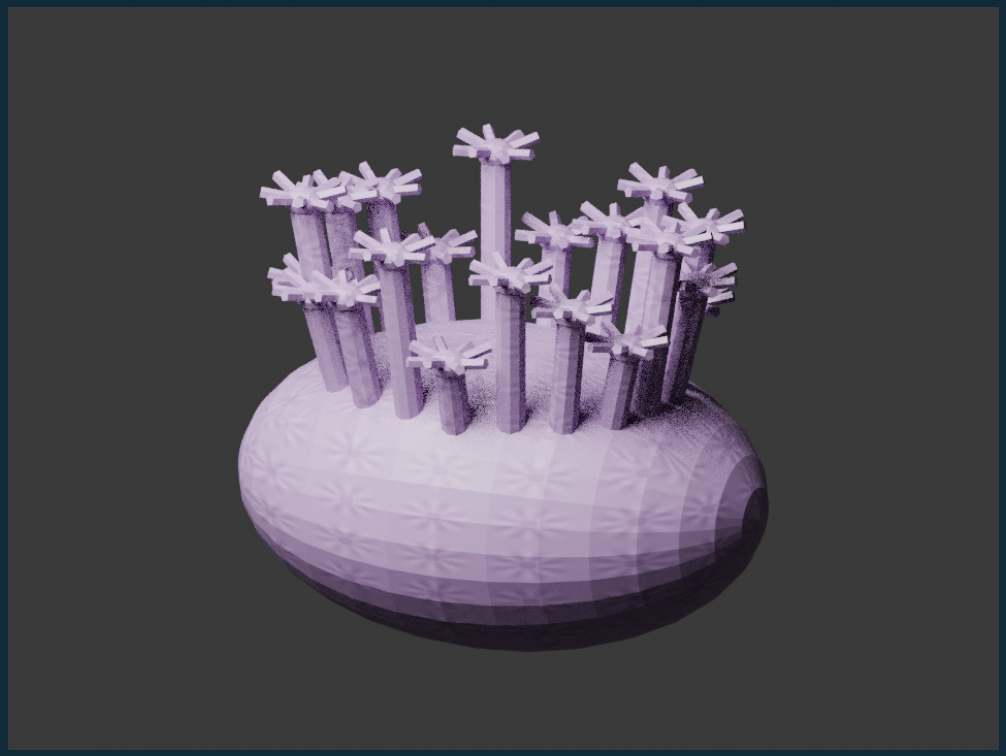
Textured source-mesh comparison; textured, lit in-game review remains pending.

# Flowerpot coral: texture projection

## BEFORE: SINGLE PROJECTION



## AFTER: AXIS PROJECTION

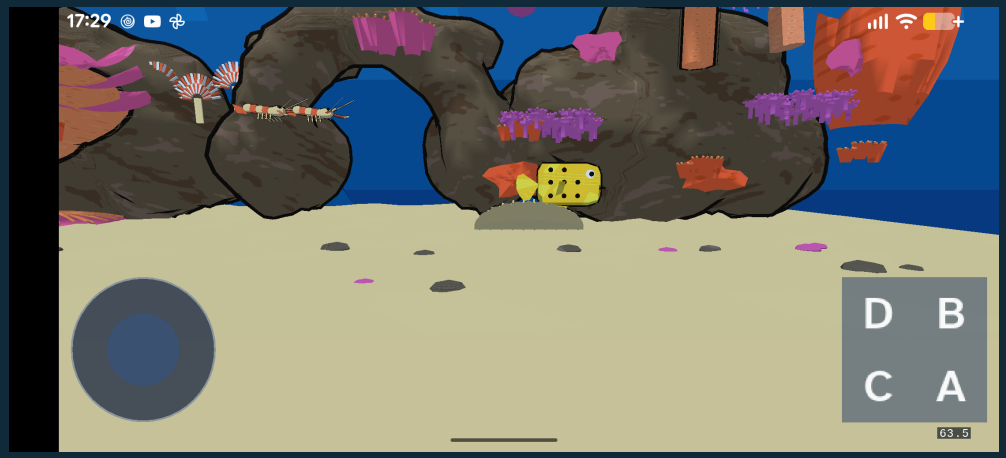


Compare the original world projection with the new face-oriented atlas mapping.

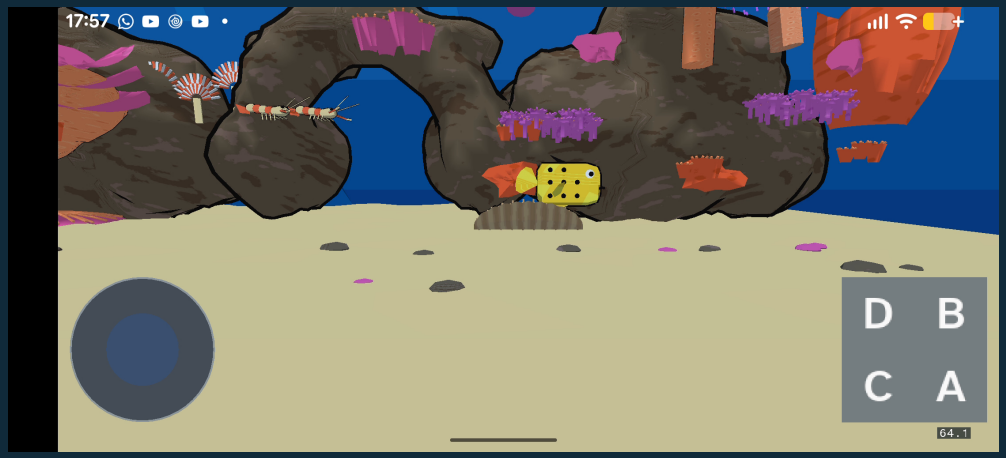
Textured source comparison; textured, lit in-game review remains pending.

# Giant clam: anime game texture comparison

## BEFORE: SAND TEXTURE




## AFTER: SHELL TEXTURE



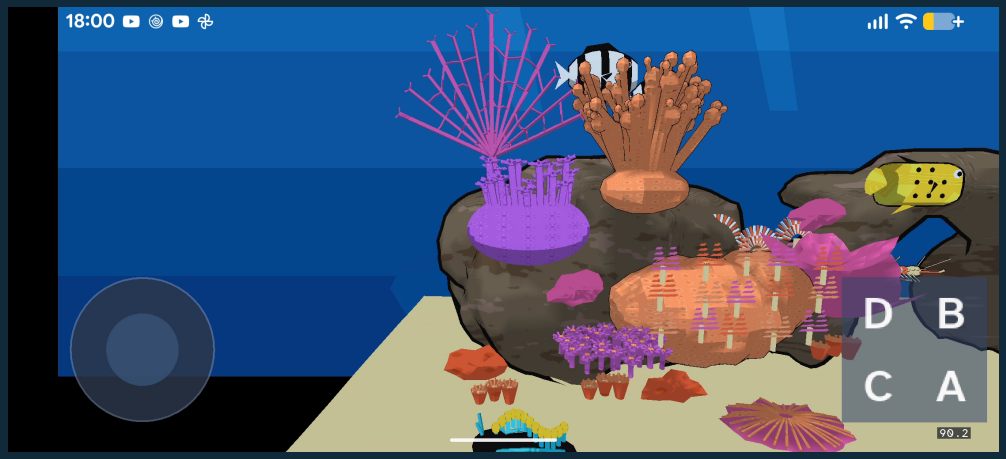

Actual Pixel captures: the shell now has visible striped color variation.

Diagnostic camera and fish placement; both views use the game atlas and lighting.

# Flowerpot: anime game surface comparison

## BEFORE: OLD SURFACE

## AFTER: NEW SURFACE



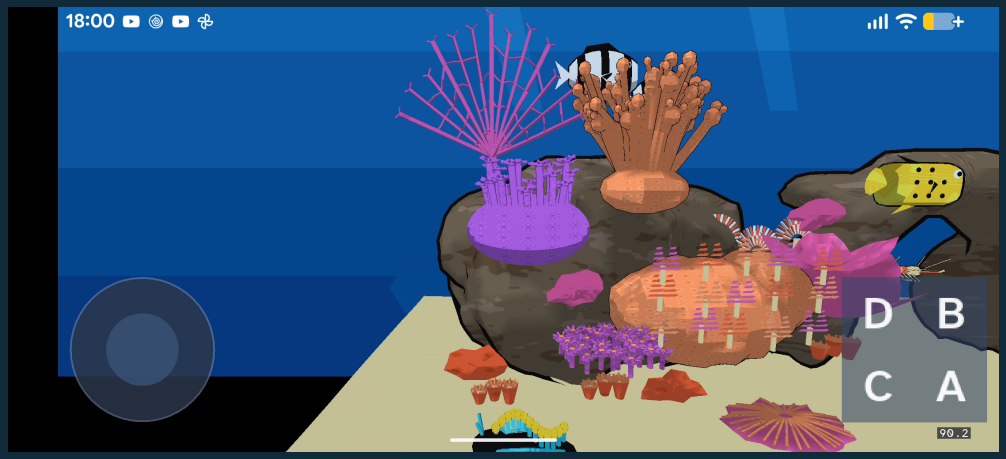

Actual Pixel captures: smoother base and more coherent flowerpot pattern.

Diagnostic camera and fish placement; both views use the game atlas and lighting.

# Branching coral: anime outline comparison

## BEFORE: WIDE HULL

## AFTER: NARROW HULL




Actual Pixel captures: reduced black hull artifacts on the branching coral.

Diagnostic camera and fish placement; both views use the game atlas and lighting.

# Flowerpot: realistic game surface comparison

## BEFORE: OLD SURFACE

## AFTER: NEW SURFACE

Actual Pixel captures: smoother base and more coherent flowerpot pattern.

Diagnostic camera and fish placement; both views use the game atlas and lighting.

# Giant clam: realistic game texture comparison

## BEFORE: SAND TEXTURE

## AFTER: SHELL TEXTURE

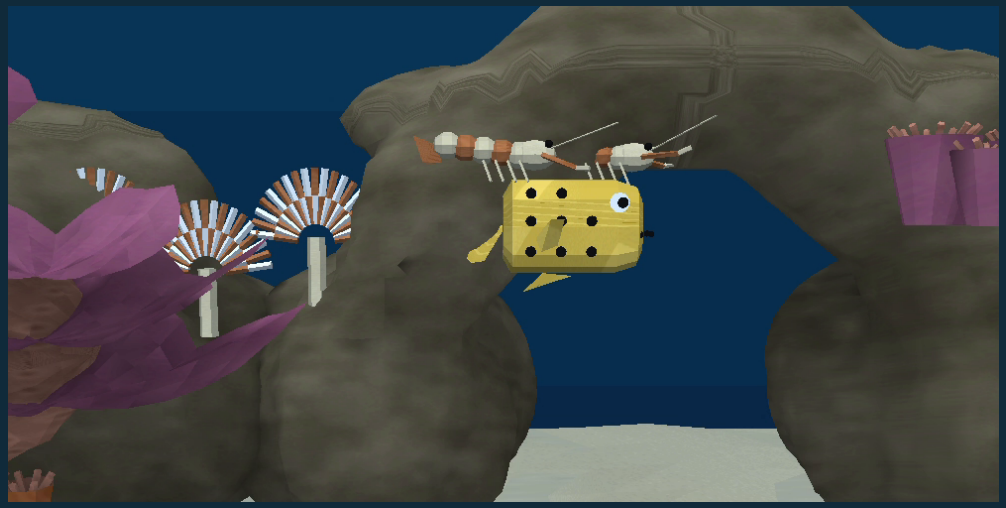
Actual Pixel captures: the shell now has visible striped color variation.

Diagnostic camera and fish placement; both views use the game atlas and lighting.

# Shrimp cleaning: placement

**BEFORE**

**AFTER**



The shrimp moved through the fish; raising their cleaning position places them on its back.

Pixel gameplay, realistic style: controlled delayed arrival; the after view is a diagnostic build.

Other fish orientations and the anime style still need gameplay review.